

Chapter 1 – Physical Quantities, Scalars, and Vectors

1. scalar vs. vector

vector: $\vec{u}$ $\bar{u}$ $\underline{u}$ $\mathbf{u}$

scalar: defined by a single number

+ or -

only compatible units allowed as arguments

* or /

may involve quantities with different units

$\sin(\dots)$, $\ln(\dots)$, $\exp(\dots)$, ... only dimensionless arguments allowed

Vector have both a magnitude and a direction

scalar (dot) product $\bar{u} \cdot \bar{w} = 0 \Leftrightarrow \bar{u} \perp \bar{w}$

orthogonal projection

右手螺旋定則 $\bar{u} \times \bar{w} = -\bar{w} \times \bar{u}$ 方向性

Cross product $\bar{u} \times \bar{w} = \vec{0} \Leftrightarrow \bar{u} \parallel \bar{w}$

2. Vectors in the Cartesian Coordinate System

▲ position vector



▲ displacement $\Delta \vec{r} = \vec{r}(t_2) - \vec{r}(t_1)$

3. Cartesian Coordinate System

$\vec{r} = x\hat{n}_x + y\hat{n}_y + z\hat{n}_z$ $x, y, z \Rightarrow$ Cartesian coordinates

$\vec{r} = (x, y, z)$. Note. The Cartesian unit vectors are also often denoted

as $\hat{i}, \hat{j}, \hat{k}$ or $\bar{e}_x, \bar{e}_y, \bar{e}_z$.

For $\bar{u} = (u_x, u_y, u_z)$ and $\bar{w} = (w_x, w_y, w_z)$, the dot product

$$\bar{u} \cdot \bar{w} = u_x w_x + u_y w_y + u_z w_z$$

- For a right-handed system (note the cyclic permutation)

$$\begin{aligned}\hat{n}_x \times \hat{n}_y &= \hat{n}_z \\ \hat{n}_y \times \hat{n}_z &= \hat{n}_x \\ \hat{n}_z \times \hat{n}_x &= \hat{n}_y\end{aligned}$$

*注意, $\hat{n}_y \times \hat{n}_x = -\hat{n}_z$.

$$\bar{u} \times \bar{w} = (u_y w_z - u_z w_y) \hat{n}_x + (u_z w_x - u_x w_z) \hat{n}_y + (u_x w_y - u_y w_x) \hat{n}_z$$

$$= \begin{vmatrix} \hat{n}_x & \hat{n}_y & \hat{n}_z \\ u_x & u_y & u_z \\ w_x & w_y & w_z \end{vmatrix}$$

determinant.

(The determinant is calculated by taking the dot product of the first row with the cross product of the second and third rows. Red arrows point from the first row to the cross product of the second and third rows, and from the second and third rows to the cross product.)

Chapter 2 – Kinematics in 1D

average velocity $V_{av,x} \stackrel{\text{def}}{=} \frac{x(t+\Delta t) - x(t)}{\Delta t}$ [m/s]

instantaneous velocity

average acceleration

instantaneous acceleration

$$\text{average speed} = \frac{\text{distance traveled}}{\text{time interval}}$$

$$\text{average velocity} = \frac{\text{displacement}}{\text{time interval}}$$

Galilean transformation $x = v_0' t + x'$

2nd assumption $\begin{cases} \textcircled{1} v_0' = \text{const} \\ \textcircled{2} t' = 0 \text{ at } t = 0 \end{cases}$

Chapter 3 – Kinematics in 3D

1. trajectory

parametric form $\vec{r}(t) = x(t)\hat{n}_x + y(t)\hat{n}_y + z(t)\hat{n}_z$

$\hat{n}_x, \hat{n}_y, \hat{n}_z$ are time-independent.

$$\dot{\hat{n}}_x = \dot{\hat{n}}_y = \dot{\hat{n}}_z = 0$$

$$x^2 + y^2 = R^2 \text{ 与 time 无关}$$

implicit form $F(x, y) = 0$

tangential component

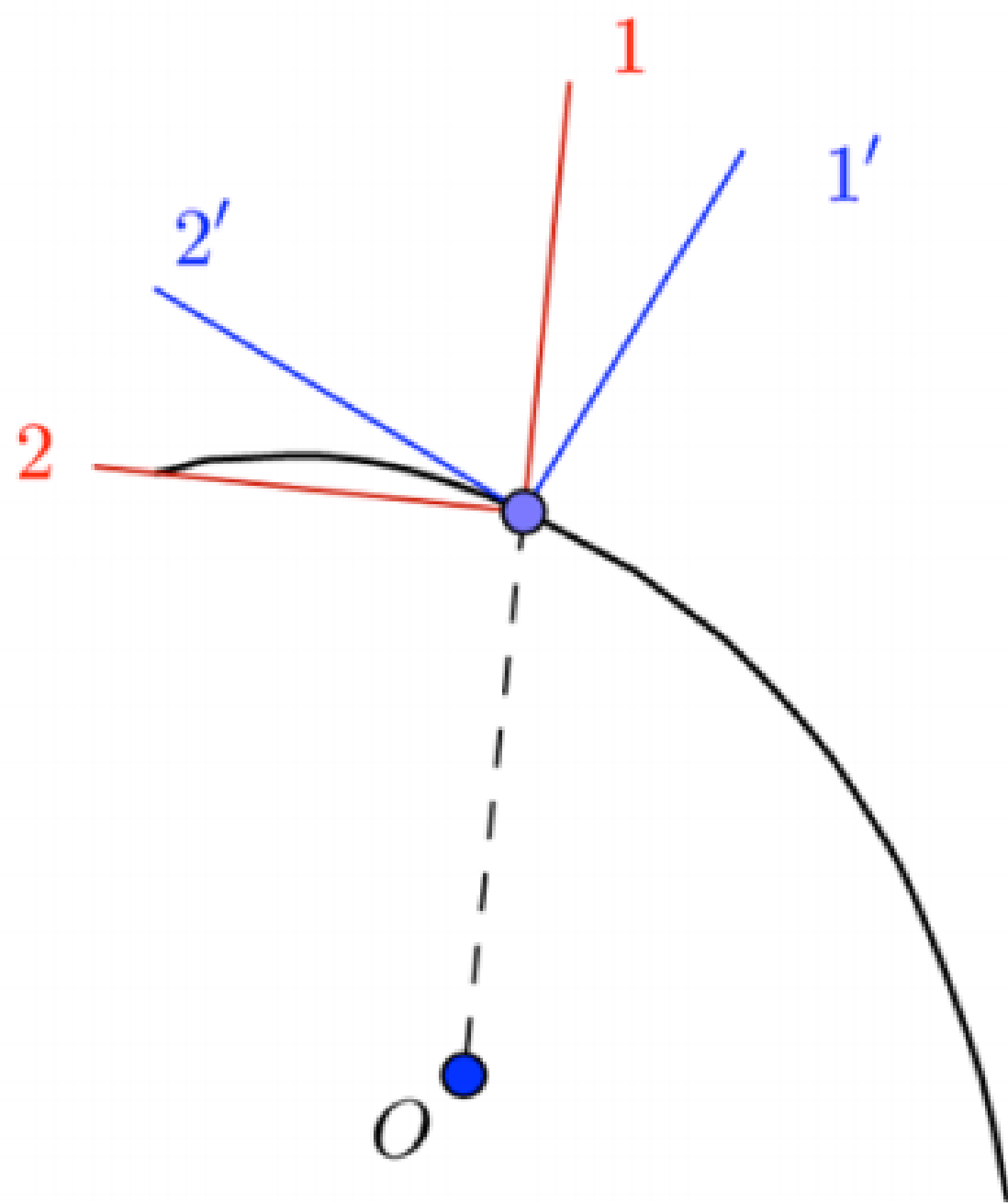
2. 切线方向 加速度改变速度大小

$$\vec{a} \parallel \vec{v}$$

3. normal component

法线方向 加速度改变速度方向

In general, radial $\neq$ normal, transverse $\neq$ tangential! (Though, it holds for uniform circular motion).

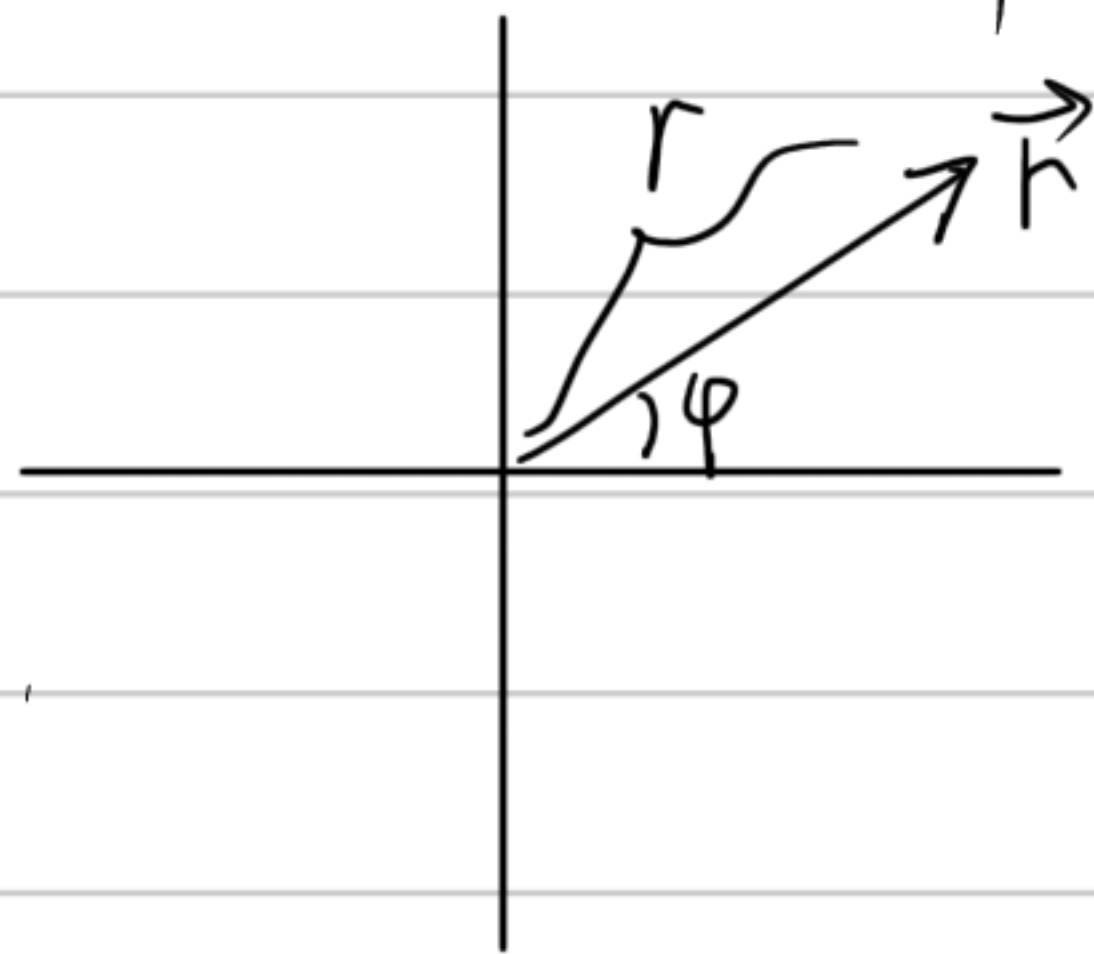


- 1' normal direction
- 2' tangential direction
- 1 radial direction
- 2 transverse direction

相较瞬时速度方向

相较原始方向

极坐标系下求解速度



① $\vec{r} = r \cdot \hat{n}_r$

② $\vec{r}$ 变 $\hookrightarrow$



$|\delta \hat{n}_r| = \Delta \varphi |\hat{n}_r| = \Delta \varphi$

$\Delta \hat{n}_r = \Delta \varphi \cdot \hat{n}_\varphi$

$\downarrow$ 同阶 Δt

$\dot{\hat{n}}_r = \dot{\varphi} \hat{n}_\varphi$

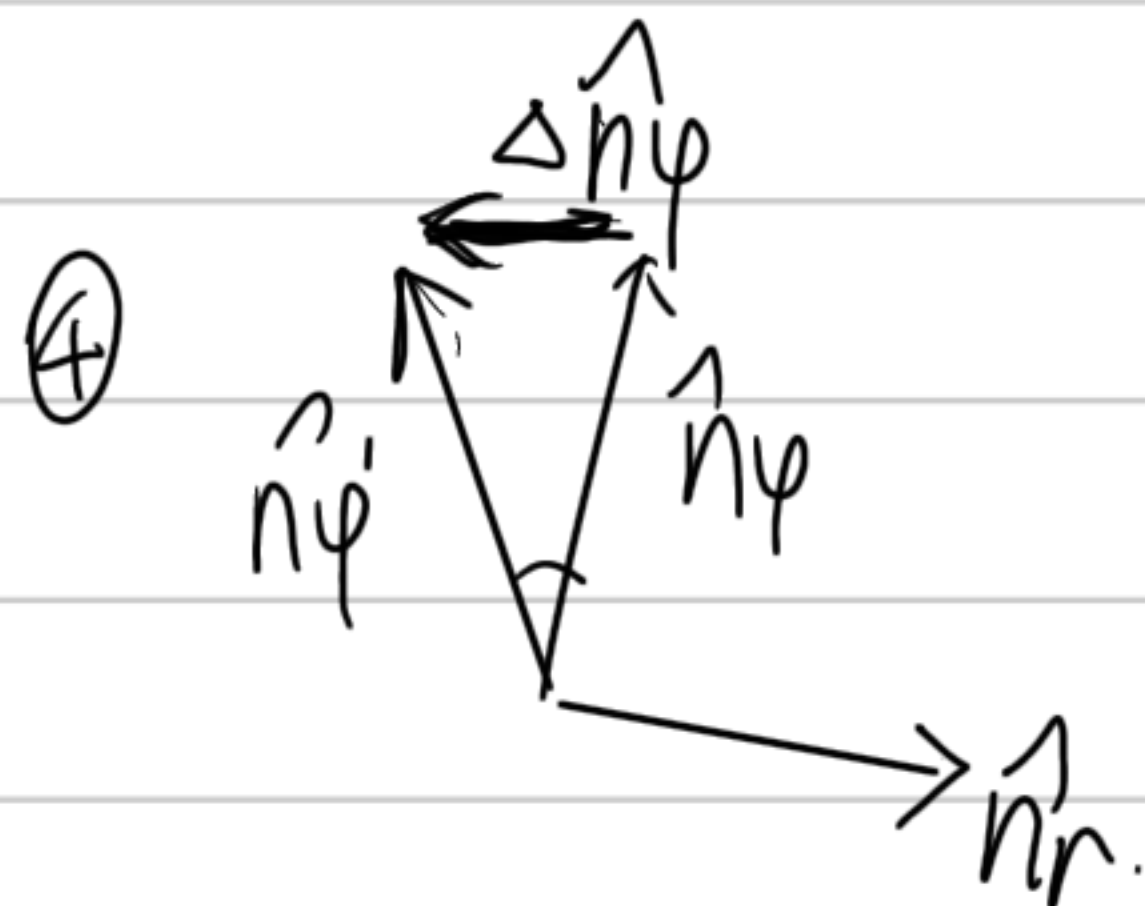
$\dot{\hat{n}}_r = \dot{\varphi} \hat{n}_\varphi$

$\Delta \hat{n}_\varphi$ 与 $\hat{n}_r$ 方向相反.

$\Delta \hat{n}_\varphi = -\Delta \varphi \hat{n}_r$

$\downarrow$

$\dot{\hat{n}}_\varphi = -\dot{\varphi} \hat{n}_r$



⑤ 现求 $\vec{v} = \frac{d\vec{r}}{dt}$

$\vec{r} = r \cdot \hat{n}_r$

求导 $\rightarrow$

$\dot{\vec{r}} = \dot{r} \hat{n}_r + r \dot{\hat{n}}_r$

$= \dot{r} \hat{n}_r + r \dot{\varphi} \hat{n}_\varphi$

radial component

transverse component

注意这里 r 也会随 t 变化.

⑥ $\vec{a} = \ddot{\vec{r}} = \ddot{r} \hat{n}_r + \dot{r} \dot{\hat{n}}_r + \dot{r} \dot{\varphi} \hat{n}_\varphi + r (\ddot{\varphi} \hat{n}_\varphi - \dot{\varphi}^2 \hat{n}_r)$

$= (\ddot{r} - r \dot{\varphi}^2) \hat{n}_r + (2\dot{r} \dot{\varphi} + r \ddot{\varphi}) \hat{n}_\varphi$

trajectory (parametric form)

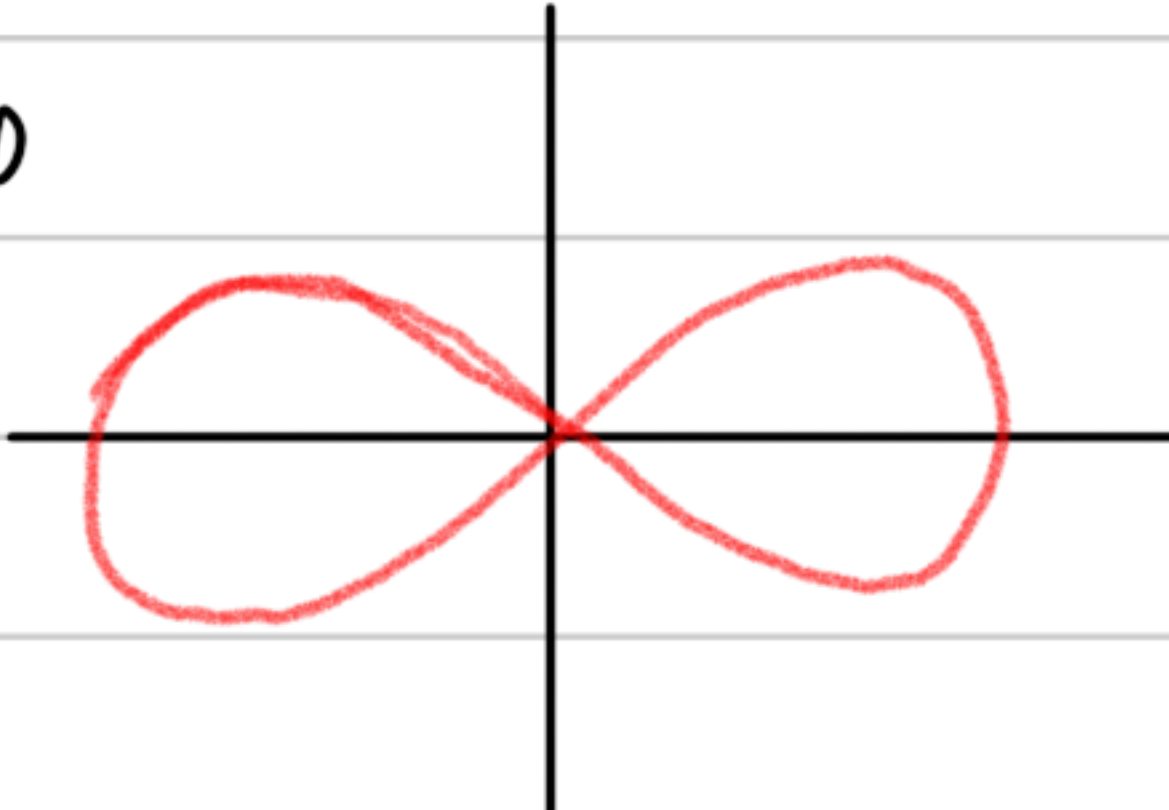
$r = r(t)$

$\varphi = \varphi(t)$

implicit form

$r = r(\varphi)$ or $\varphi = \varphi(r)$ or $f(r, \varphi) = 0$

Example : Lemniscate $r^2 = 2A^2 \cos 2\varphi$



顺理成章推导匀速圆周运动:

$$\vec{v} = \underbrace{\dot{r}}_0 \hat{n}_r + \underbrace{r\dot{\varphi}}_{|\vec{v}|} \hat{n}_\varphi$$

$$r\dot{\varphi} = |\vec{v}|$$

$$\dot{\varphi} = \frac{|\vec{v}|}{r}$$

$$\varphi = \int_0^t \frac{|\vec{v}|}{r} dt = \frac{vt}{r} = \underbrace{\omega t}_{\text{in general a vector}}$$

$$\begin{aligned} \dot{\vec{v}} &= \ddot{r} \hat{n}_r + \dot{r} \dot{\hat{n}}_r + \dot{r} \dot{\varphi} \hat{n}_\varphi + r \ddot{\varphi} \hat{n}_\varphi + r \dot{\varphi} \dot{\hat{n}}_\varphi \\ &= \ddot{r} \hat{n}_r + \dot{r} \dot{\varphi} \hat{n}_\varphi + \dot{r} \dot{\varphi} \hat{n}_\varphi + r \ddot{\varphi} \hat{n}_\varphi - \underbrace{r \dot{\varphi}^2 \hat{n}_r}_{\text{in general a vector}} \end{aligned}$$

$$= (\underbrace{\ddot{r}}_0 - r \dot{\varphi}^2) \hat{n}_r + (\underbrace{2\dot{r}\dot{\varphi}}_0 + \underbrace{r\ddot{\varphi}}_0) \hat{n}_\varphi$$

$$\vec{a} = -\underbrace{R\dot{\varphi}^2}_{\omega^2} \hat{n}_r = -R\omega^2 \hat{n}_r$$

$$\vec{v} = R\omega \hat{n}_\varphi$$

(此时 ω 不是 vector)

如果 φ 是关于 t 的函数呢? ($\dot{\varphi} \neq 0$)

$$\varphi(t) = \omega(t)$$

$$\ddot{\varphi}(t) = \ddot{\omega}(t) = \varepsilon(t)$$

angular acceleration 角加速度, vector.

$$\vec{a}_{\text{总}} = -R\omega^2 \hat{n}_r + R\varepsilon(t) \hat{n}_\varphi \quad \Leftarrow \quad \vec{v} = R\omega(t) \hat{n}_\varphi$$

蜜蜂爬转盘问题

$$t = \frac{v_0}{\omega} \varphi$$

$$\vec{v} = \dot{r} \hat{n}_r + r \dot{\varphi} \hat{n}_\varphi = v_0 \hat{n}_r + v_0 t \omega \hat{n}_\varphi$$

$$\dot{r} = v_0 \quad \ddot{r} = 0$$

时刻记作 t 与 t 有关

$$\vec{a} = (\ddot{r} - r \dot{\varphi}^2) \hat{n}_r + (2\dot{r}\dot{\varphi} + r\ddot{\varphi}) \hat{n}_\varphi$$

Natural Coordinate System

1. 定义

$\hat{n}_T$

$\hat{n}_n$

$\hat{n}_b$

$$\hat{n}_n = \frac{\dot{\hat{n}}_T}{|\dot{\hat{n}}_T|}$$

$$\hat{n}_T \times \hat{n}_n \text{ 右手}$$

构成 system

曲率半径 $R_c \stackrel{\text{def}}{=} \frac{v}{|\dot{\hat{n}}_T|}$

区别:

$$n_n \quad n_T \quad n_r \quad n_z.$$

$$\vec{a} = \underbrace{\dot{v} \hat{n}_T}_{\text{tangential component}} + \underbrace{\frac{v^2}{R_c} \hat{n}_n}_{\text{normal component}}$$

匀圆中 $R_c = R$.

直线的 $R_c \rightarrow \infty$

Relative Motion & Galilean Transformation

$$\vec{V} = \vec{V}_0' + \vec{V}'$$

$$\vec{r}_0' = \vec{V}_0' t + \vec{r}_0'_{\text{init}}$$

$$\Rightarrow \boxed{\vec{r} = \vec{V}_0' t + \vec{r}'}$$

(前提: $t=0$ A与A'原点重合)

Newton's Laws of Motion

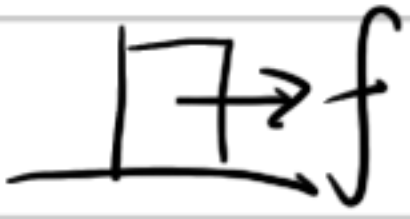
Interaction	Particles Involved	Relative Strength	Range
Gravitational always attractive holds planets in their orbits around Sun	any massive particle	$\sim 10^{-38}$ 最小	infinite
Electromagnetic attractive/repulsive fundamental in optics, chemistry, biology; source of friction	electric charge	$\sim 10^{-2}$	infinite
Weak necessary for buildup of heavy nuclei; responsible for radioactive decay (beta decay)	quarks, leptons	$\sim 10^{-6}$	short $\sim 10^{-18}$ m (0.1% of the diameter of the proton)
Strong holds protons and neutrons together in the nucleus	hadrons (protons, neutrons, mesons)	1	short $\sim 10^{-15}$ m (diameter of a medium sized nucleus)

SI unit: Newton $[1N = 1kg \cdot m/s^2]$

2. The vector sum of all forces acting upon an object is called the **resultant** of the forces or the **net force**.

3. 接触力是场力 (electromagnetic in their nature)

contact force

- Normal Force 
- Frictional force 
- Tension force 

weight long-range force.

Sun's gravitational force on the earth	$3.5 \times 10^{22} \text{ N}$
Thrust of a space shuttle during launch	$3.1 \times 10^7 \text{ N}$
Weight of a large blue whale	$1.9 \times 10^6 \text{ N}$
Maximum pulling force of a locomotive	$8.9 \times 10^5 \text{ N}$
Weight of a 250-lb linebacker	$1.1 \times 10^3 \text{ N}$
Weight of a medium apple	1 N
Weight of smallest insect eggs	$2 \times 10^{-6} \text{ N}$
Electric attraction between the proton and the electron in a hydrogen atom	$8.2 \times 10^{-8} \text{ N}$
Weight of a very small bacterium	$1 \times 10^{-18} \text{ N}$
Weight of a hydrogen atom	$1.6 \times 10^{-26} \text{ N}$
Weight of an electron	$8.9 \times 10^{-30} \text{ N}$
Gravitational attraction between the proton and the electron in a hydrogen atom	$3.6 \times 10^{-47} \text{ N}$

Inertial Frame of Reference 惯性参考系

a frame where Newton's Law holds true.
no external force $\begin{cases} \rightarrow \text{rest} \\ \rightarrow \text{uniform motion} \end{cases}$

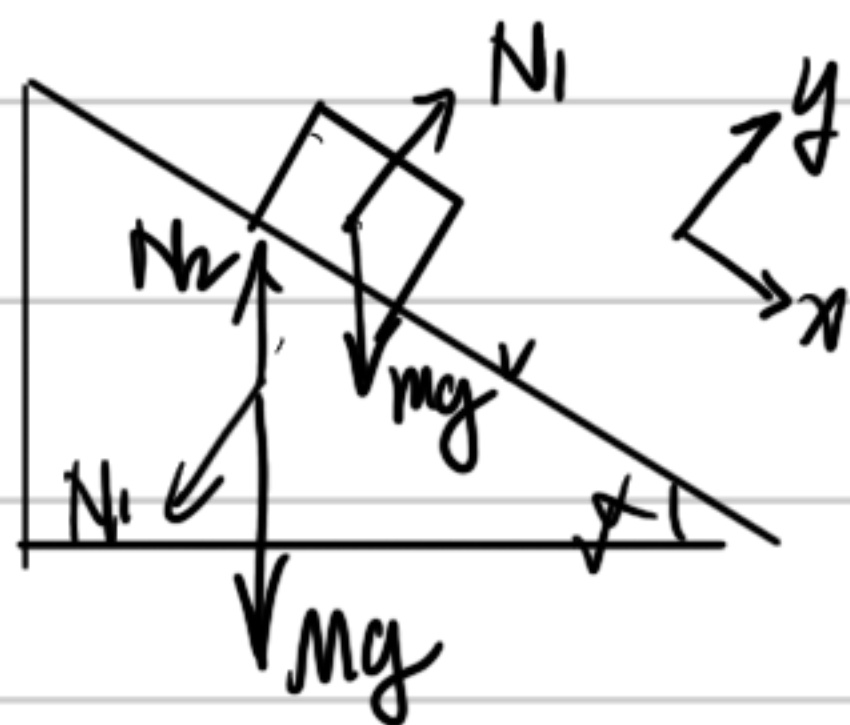
总结: ① the body accelerates $\neq v \uparrow$
↳ 表示存在 a .

② 牛一: 合力为0, 速度不变. 加速度为0
大小方向

③ $F' = F$ (according to Newton's 3rd Law)

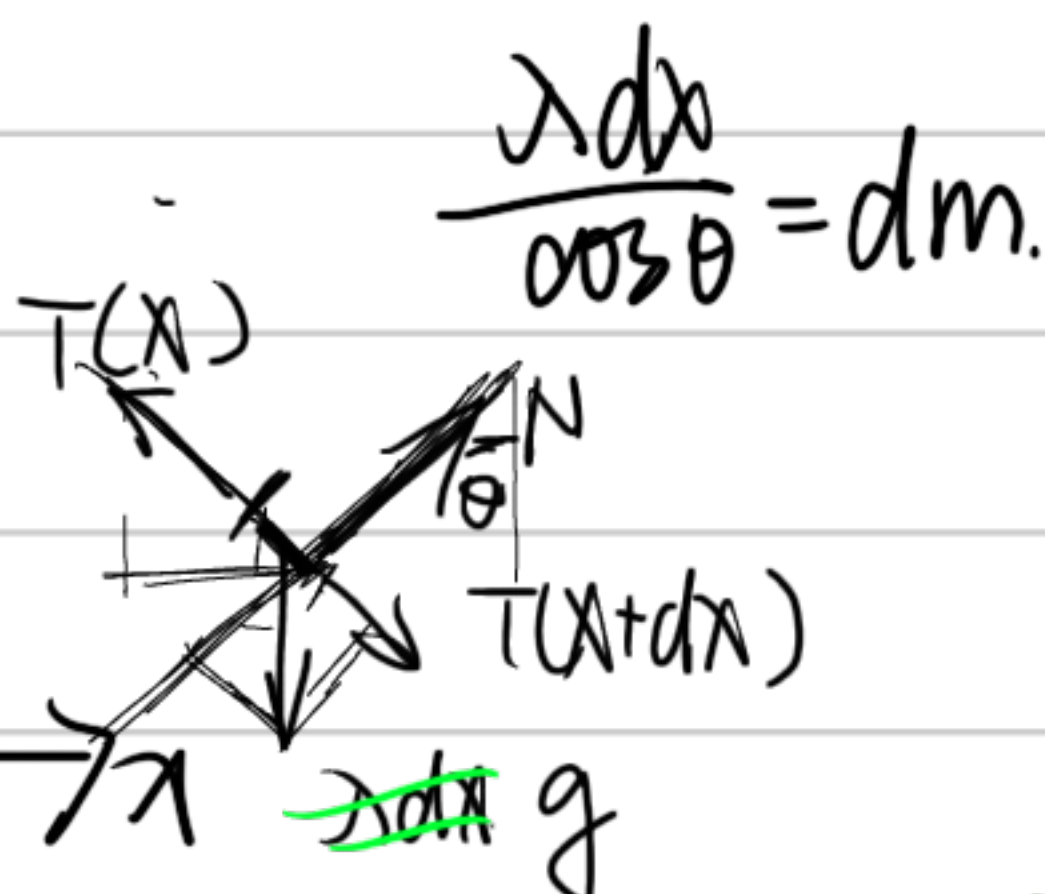
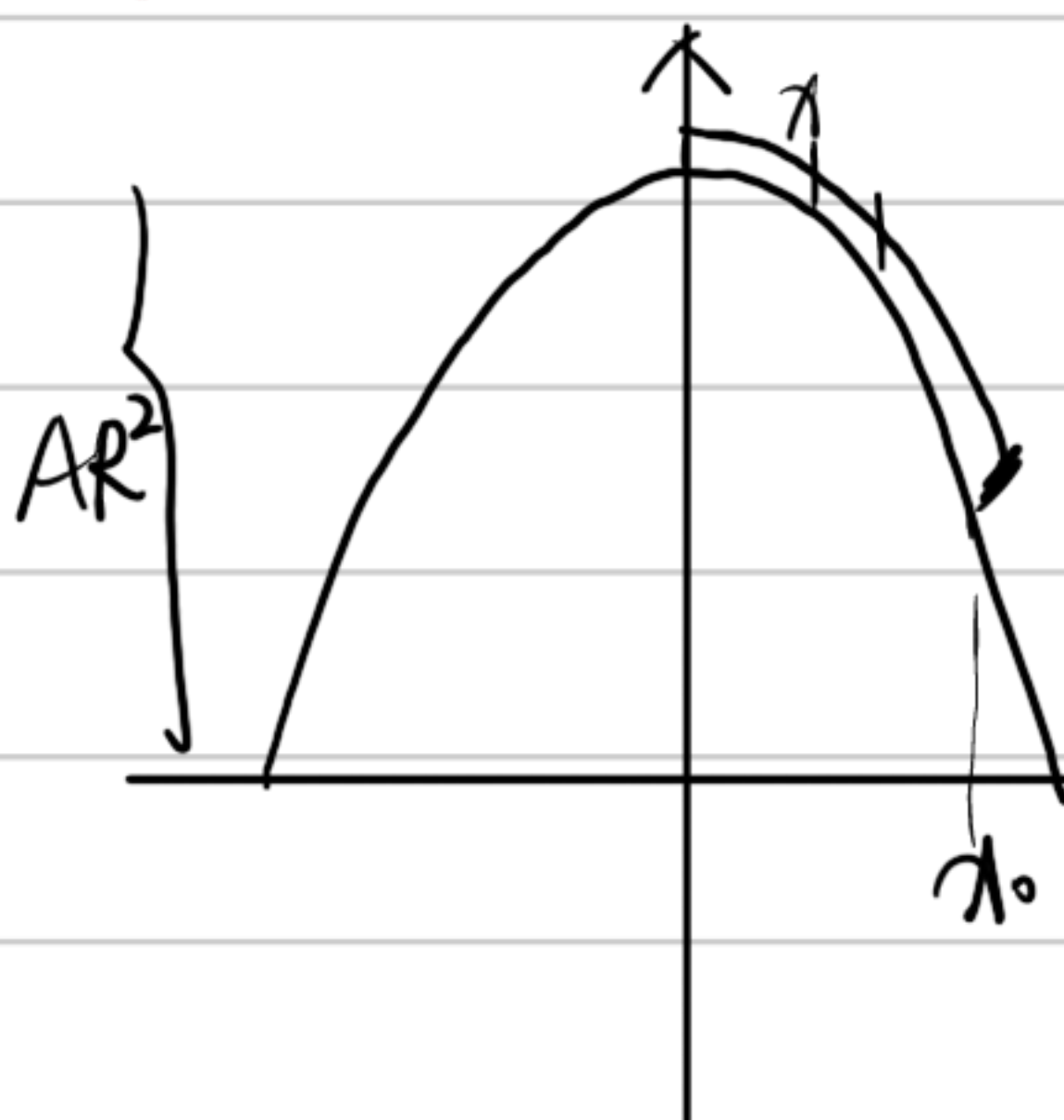
④ action 作用力
reaction 反作用力

⑤ equivalence of all inertial FORs



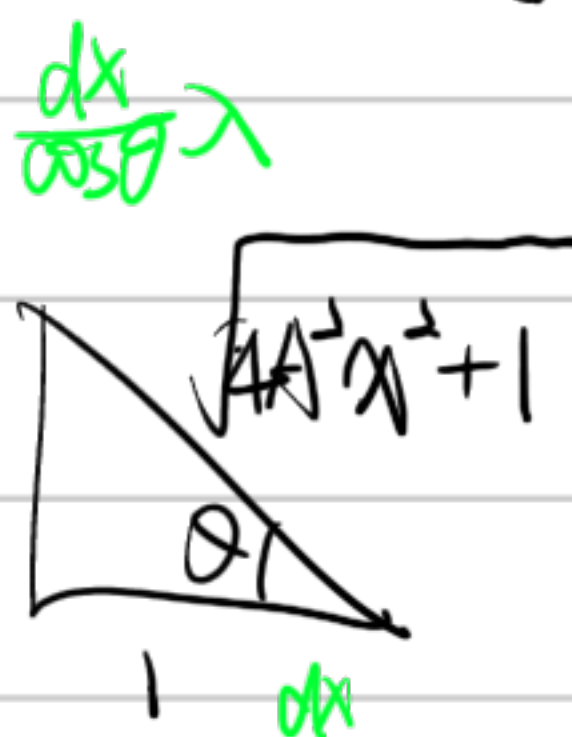
$$\frac{N_1}{m} = \frac{mg \cos \alpha - N_1}{m}$$

易错题



$$\frac{\lambda dx}{\cos \theta} = dm$$

$$\tan \theta = 2Ax$$



$$\leftarrow 2R \rightarrow$$

$$y = -A(x+R)(x-R)$$

$$y' = -2Ax$$

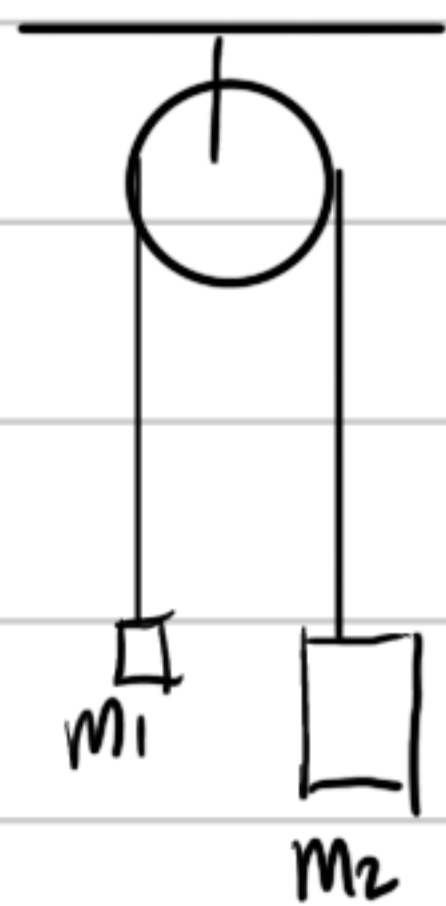
$$\lambda dx g \cos \theta = N$$

$$\lambda dx g \tan \theta + T(x+dx) = T(x)$$

$$\frac{T(x+dx) - T(x)}{dx} = \lambda g 2Ax$$

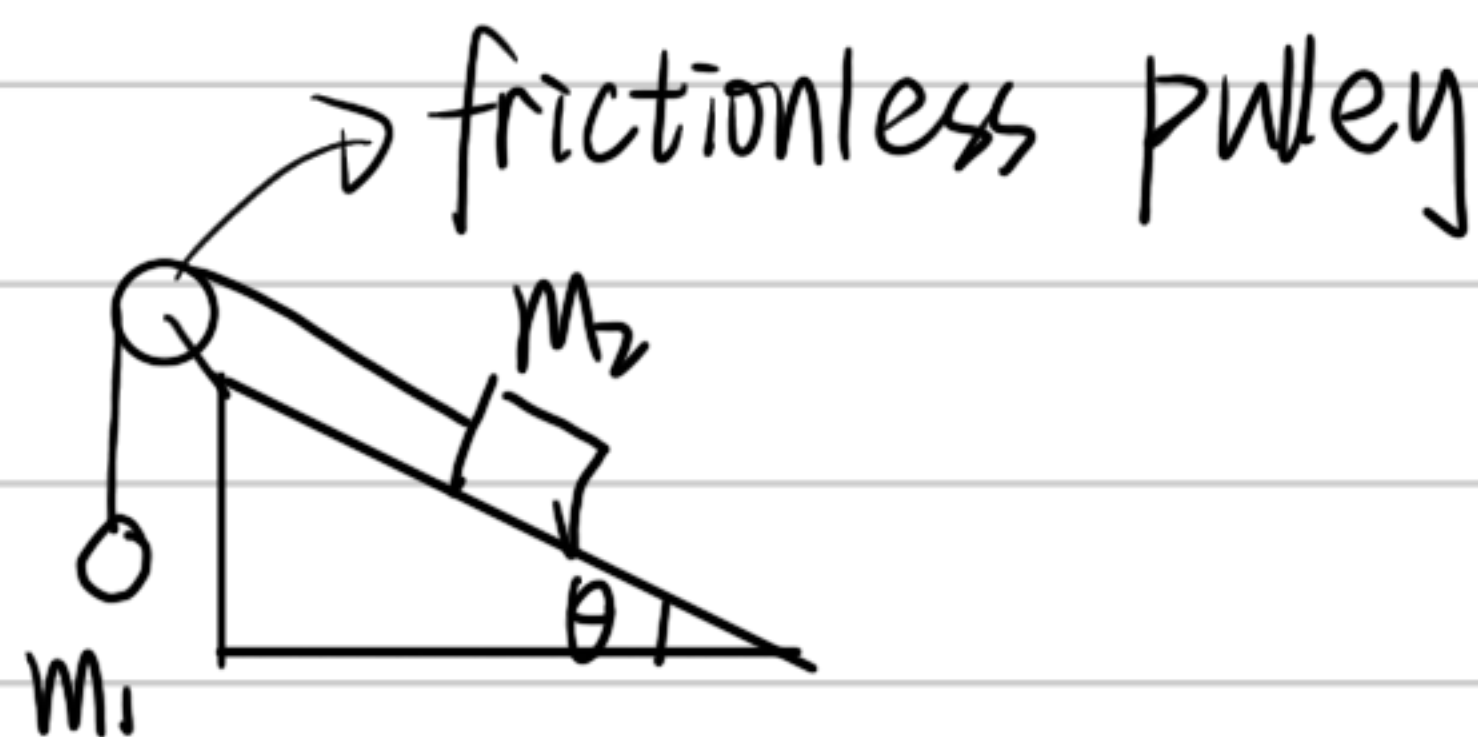
易错题:

turn to its left
≈



$$a = \frac{m_2 - m_1}{m_1 + m_2} g$$

$$T = \frac{2m_1 m_2}{m_1 + m_2} g$$



$$a = \frac{m_2 \sin \theta - m_1}{m_1 + m_2} g$$

$$T = \frac{m_1 m_2}{m_1 + m_2} (1 + \sin \theta) g$$

★ State Coordinate.

★ State the domain.

Prefix of unit.

10^9 giga G

10^6 mega M

10^3 kilo K

10^{-1} deci d

10^{-2} centi c

10^{-3} milli m

10^{-6} micro μ

10^{-9} nano n

Tangential & Normal Components of $\vec{a}$, $\vec{v}$

$$\vec{v} = \dot{r} \hat{n}_r + r \dot{\phi} \hat{n}_\phi$$

$$\vec{a} = (\ddot{r} - r \dot{\phi}^2) \hat{n}_r + (r \ddot{\phi} + 2 \dot{r} \dot{\phi}) \hat{n}_\phi$$

大RC Midterm

期中复习

单位向量 ⊥ 单位向量求导

$$\hat{n}_n = \frac{\dot{\hat{n}}_T}{|\dot{\hat{n}}_T|}$$

normal unit vector
用来描述 tangential unit vector 的导数的 unit vector

$$\hat{n}_b = \hat{n}_T \times \hat{n}_n$$

$$\vec{a} = \underbrace{\dot{\vec{v}} \cdot \hat{n}_T}_{\text{tangential component}} + \underbrace{\frac{v^2}{R_c} \hat{n}_n}_{\text{normal component}}$$

$$\vec{a} = \dot{\vec{v}} = (\dot{v} \hat{n}_T)' = \dot{v} \hat{n}_T + v \dot{\hat{n}}_T$$

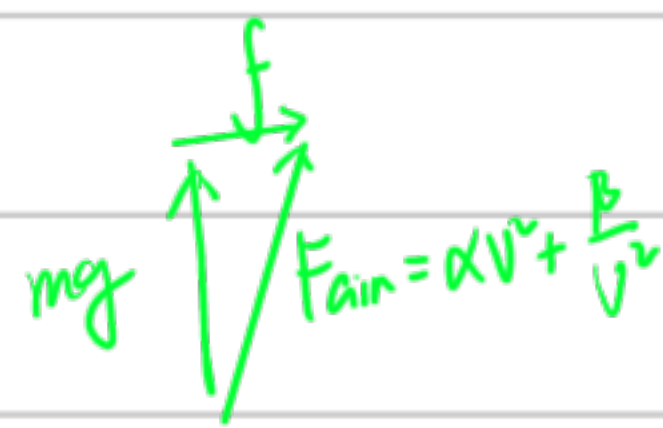
$$= \dot{v} \hat{n}_T + v \frac{|\dot{\hat{n}}_T|}{v} \hat{n}_n$$

$$R_c = \frac{v}{|\dot{\hat{n}}_T|} = \frac{v^2}{R_c}$$

★ v 是速率
 $\dot{v}$ 是速度

Galilean Transformation.

$$T = T_0 t + T'$$



$$E = f \cdot x$$

$$= f \cdot v \cdot t = \underbrace{\left(\alpha v^2 + \frac{B}{v^2} \right)}_{\downarrow} \cdot t$$

$$V^0 = \frac{B}{\alpha}$$

$$V = \left(\frac{3.5 \times 10^5}{3 \times 10^{-29}} \right)^{\frac{1}{3}} =$$

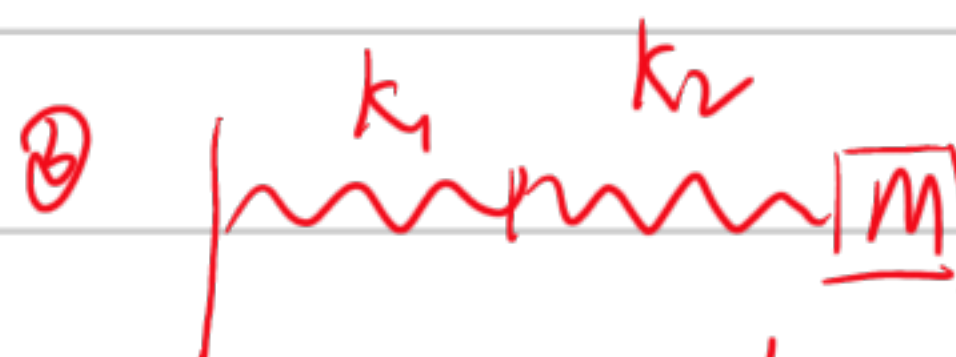
$$y' = 3\alpha v^2 - \frac{B}{v^2}$$

$$V^2 = \frac{B}{3\alpha}$$

同种弹簧模型.



$$k = k_1 + k_2$$



$$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$$



$$U_k = \frac{mg(d+x)\sin\theta - \frac{1}{2}Cx^3}{mg(d+x)\cos\theta}$$

↑

$$\frac{1}{2}Cx^3 = mg(d+x)\sin\theta - U_k mg(d+x)\cos\theta$$

$$mg\sin\theta = U_s mg\cos\theta$$

$$U_s = \tan\theta$$



$$\text{balance } mg\sin\theta = kx^3$$

$$\frac{G(m_0 + m)^3}{m} - mg\sin\theta = m\ddot{x}$$

$$= -\frac{k}{m}x$$

$$Cx^3 \leq U_s mg\cos\theta + mg\sin\theta$$

$$Cx^3 \leq 2mg\sin\theta$$

Quizzes 精彩集锦

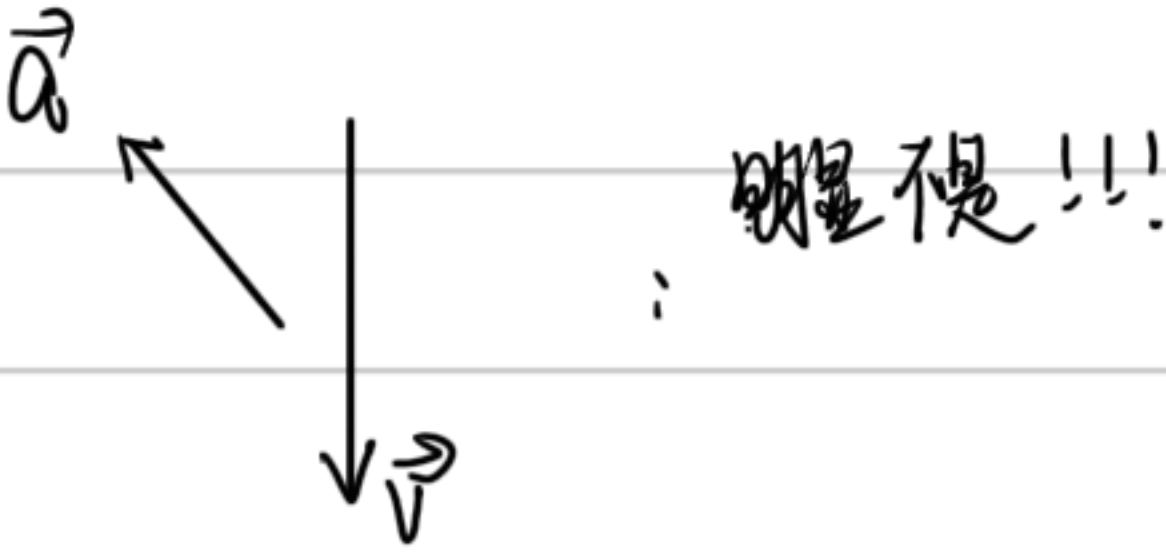
① N 不是 SI ^{base} unit

② You Answered ☒ The magnitude of a vector can be zero even though one of its Cartesian components is not zero.

$|v|=0 \Rightarrow$ every component is 0

③ ☒ In the Galilean transformation, time is ~~not~~ relative to the frame of reference.

④ Shown below are the velocity and acceleration vectors for a car in several different types of motion. In which case is the car slowing down and turning to its left?



明显慢!!!

Materials	Coefficient of Static Friction, μ_s	Coefficient of Kinetic Friction, μ_k
Steel on steel	0.74	0.57
Aluminum on steel	0.61	0.47
Copper on steel	0.53	0.36
Brass on steel	0.51	0.44
Zinc on cast iron	0.85	0.21
Copper on cast iron	1.05	0.29
Glass on glass	0.94	0.40
Copper on glass	0.68	0.53
Teflon on Teflon	0.04	0.04
Teflon on steel	0.04	0.04
Rubber on concrete (dry)	1.0	0.8
Rubber on concrete (wet)	0.30	0.25

The coefficient of static friction between two teflon-covered surfaces is smaller than the coefficient of static friction between a rubber car tyre and a concrete road (in wet conditions).

☐ True ☒ False

$r'' - r\omega^2 = C_1$
 $r = C_2 t$ $\dot{r} = C_2$

$\dot{\varphi} = \omega$

$\vec{v} = \dot{r} \hat{n}_r + r \dot{\varphi} \hat{n}_\varphi$

$\vec{a} = (\ddot{r} - r\omega^2) \hat{n}_r + (r\ddot{\varphi} + 2\dot{r}\omega) \hat{n}_\varphi$



$\vec{a} = (\ddot{r} - r\dot{\varphi}^2) \hat{n}_r + (r\ddot{\varphi} + 2\dot{r}\dot{\varphi}) \hat{n}_\varphi$

理解：
圆周运动时
 $\ddot{r} = 0$

匀圆: $\vec{a} = -r\omega^2 \hat{n}_r$

变速圆 $\vec{a} = -r\omega^2 \hat{n}_r + r\ddot{\varphi} \hat{n}_\varphi$
 $\omega, \dot{\varphi}$ 角速度 $\uparrow$ $\omega, \ddot{\varphi}$ 角度加速度 $\downarrow$
 法向加速度 切向加速度

For a system that exhibits deterministic chaos, the word "chaos" refers to the fact that

☐ The evolution of the system is extremely sensitive to initial conditions.

■ Kinematics 2D & 3D

- Circular Motion (Tangential & Normal Components of $\vec{a}$, $\vec{v}$)
- Projectile Motion

■ Polar Coordinates

- $\vec{v}$, $\vec{a}$ Expression
- Radial & Transverse Components

$$\vec{v} = \dot{r} \hat{n}_r + r \dot{\varphi} \hat{n}_\varphi$$

$$\vec{a} = (\ddot{r} - r \dot{\varphi}^2) \hat{n}_r + (r \ddot{\varphi} + 2\dot{r} \dot{\varphi}) \hat{n}_\varphi$$

A frictionless simple pendulum of length 1 m is released from an initial position, 65 degrees with the vertical. Take the acceleration due to gravity equal to 10 m/s².

True or False? The period of oscillations of this pendulum is equal to $2\pi\sqrt{0.1}$ seconds.

☒ True it's false

Consider a particle in a liquid that is subject to a restoring force and linear drag. The system is in the overdamped regime.

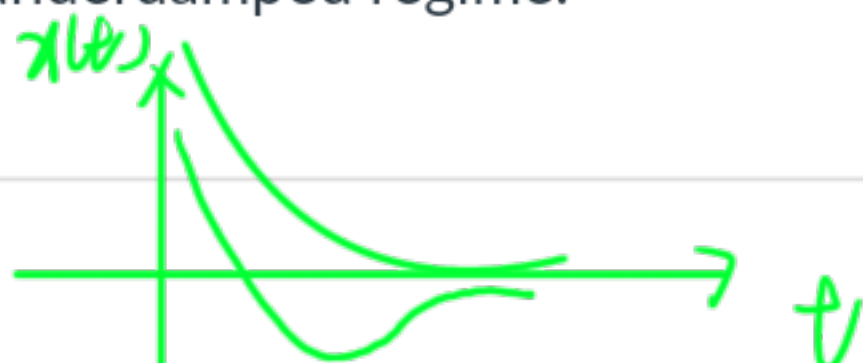
Mark all the statements that are **True**.

☒ $x(t) = Ae^{-\frac{b}{2m}t} \cos(\sqrt{\omega_0^2 - (\frac{b}{2m})^2}t + \phi_0)$
The resulting oscillations are with exponentially decreasing amplitude and their frequency is lower than the natural angular frequency of the system,

$\Rightarrow$ 指 underdamped

☒ The equilibrium position is visited ~~multiple~~ at most once times, with regular time intervals.

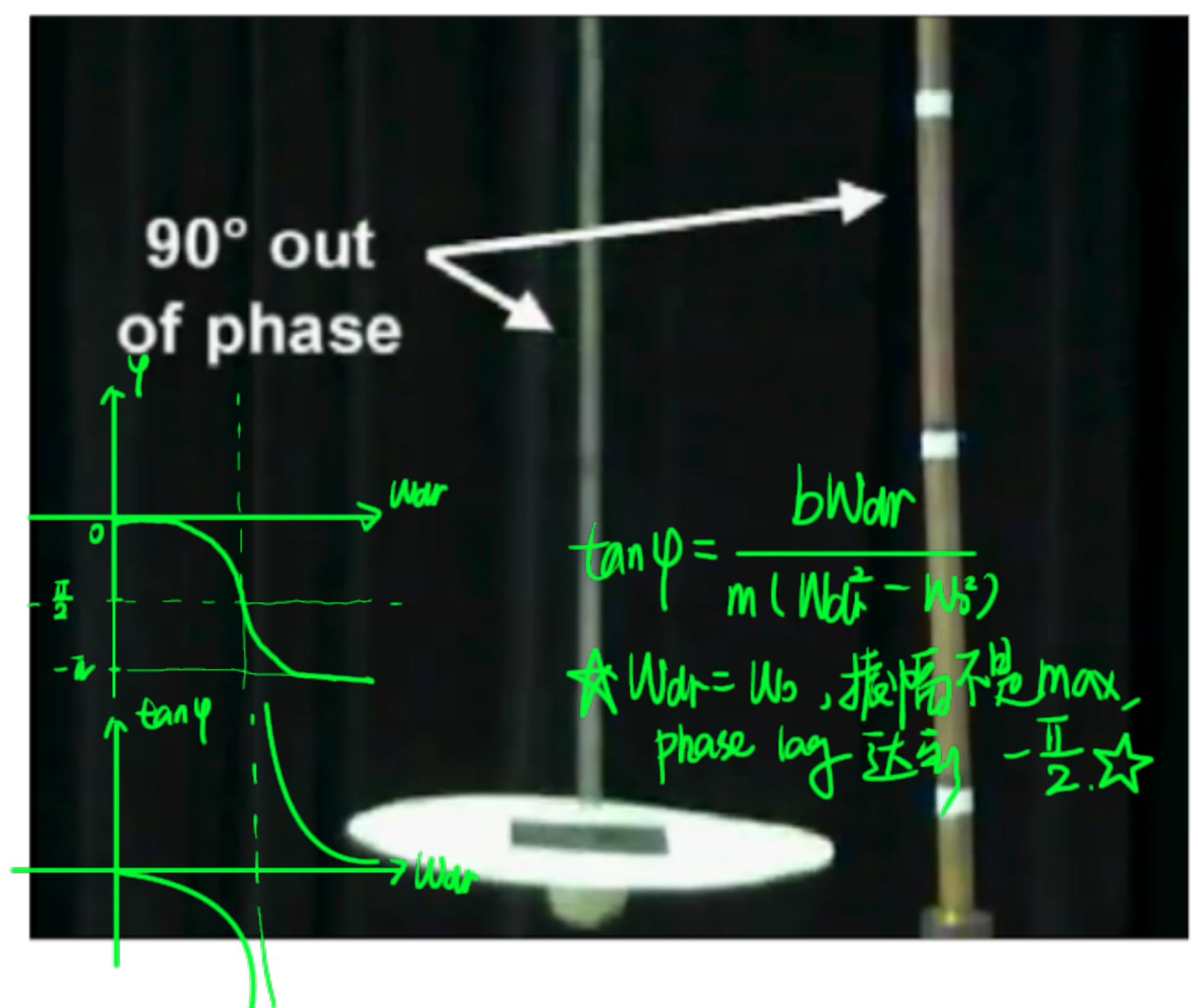
☒ If we replace the liquid with another one, of a much smaller viscosity (while leaving other parameters of the system unchanged), it may be possible that the system crosses over to the underdamped regime.



☒ After a long time, the system returns to the equilibrium position.

Shown below is a still frame from the MIT video which we watched in class, showing a 90-degree phase shift between the displacement and the driving force for a harmonic oscillator driven by a sinusoidal driving force.

Mark all the statements below that are **True**.



☐ The angular frequency $\omega_{dr} = \omega_0$ of the driving force is well above the resonance frequency. $\sqrt{\omega_0^2 - \frac{b^2}{2m^2}}$

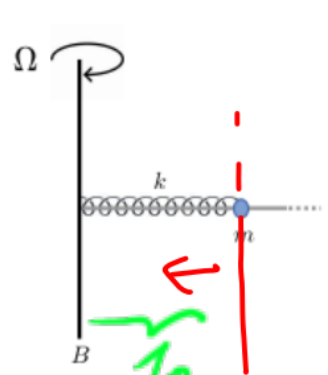
☐ The angular frequency of the driving force is well below the resonance frequency,

☒ $A(\omega_{dr})$
The amplitude, as a function of the angular frequency of the driving force, is close to maximum. Q

☒ The response of the system lags the drive by a half of the cycle. ✗

☒ The angular frequency of the driving force is close to the resonance frequency. Q

7 points) [Note. There is no gravitational force in this question.]
 A linear spring with spring constant k and unstretched/uncompressed length is x_0 is coiled around a light long horizontal rod (see the figure).
 The rod is fixed to a vertical axle B , which spins always with the same angular speed $\Omega = \sqrt{k/m}$. One end of the spring is fixed at the axle. At the other end of the spring there is bead with mass m that can slide along the rod, but the rod is rough so that the bead moves with friction.
 What should the value of the coefficient of kinetic friction μ be, so that the bead slides away from the axle with constant speed v_0 ? Comment briefly on whether the scenario described in this problem is sustainable?



$$\dot{\phi} = \sqrt{\frac{k}{m}} \quad r = r_0 + v_0 t \quad \dot{r} = v_0 \quad \ddot{r} = 0$$

$$k v_0 t + \mu m g = m a$$

$$\vec{a} = (\ddot{r} - r\dot{\phi}^2)\hat{r} + (r\ddot{\phi} + 2\dot{r}\dot{\phi})\hat{\phi}$$

$$= -(\dot{r} + v_0 t)\frac{k}{m}\hat{r} + 2v_0\sqrt{\frac{k}{m}}\hat{\phi}$$

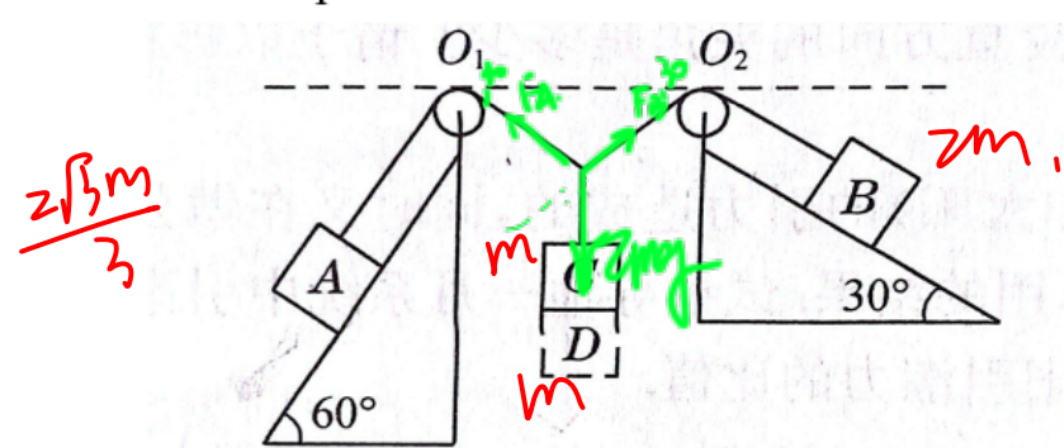
$$a = \sqrt{(\dot{r} + v_0 t)^2 \frac{k^2}{m^2} + 4v_0^2 \frac{k}{m}}$$

轴向加速度由轴向力造成

$$(\dot{r} + v_0 t)\frac{k}{m} = \frac{v_0 t k + \mu m g}{m}$$

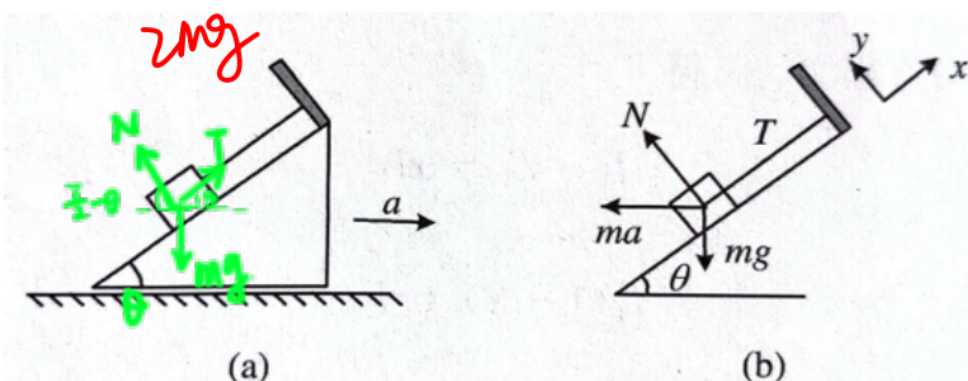
$$\mu = \frac{v_0 k}{m g}$$

O_1 and O_2 are of the same height and they are frictionless. The mass of A, B and C are $\frac{2\sqrt{3}}{3}m$, $2m$ and m . The slopes are frictionless. When equilibrium has reached, we put D (mass: m) on C with 0 velocity and CD stick together. What is the acceleration of A and B at the instant we put D?



$$\begin{cases} F_A - mg = \frac{2}{3}ma_A \\ F_B - mg = 2ma_B \\ \frac{1}{2}F_A + mg - F_B = 2ma_B \\ \frac{1}{2}F_B + mg - F_A = 2ma_A \end{cases}$$

An object of mass m is placed on a frictionless slope. A light sting connects the object to a plank on the top of the slope. The maximum force the string can stand is $2mg$. In order to make sure that m doesn't separate from slope, what is the range of the acceleration of the slope?



$$\begin{cases} T \cos \theta - N \sin \theta = ma \\ mg = T \sin \theta + N \cos \theta \end{cases}$$

$$N = \frac{mg - T \sin \theta}{\cos \theta}$$

$$N \geq 0$$

Write the velocity and displacement formula with regard to time in the following condition: 1) $f = -kv$ 2) $f = -kv$ during the process of a free fall 3) $f = -kv^2$
 Mass: m initial velocity: v_0 for 1) & 3)

A ship of mass M sails with constant velocity, while its engine provides a constant propelling force of magnitude F_{prop} . There is a linear drag force acting upon the ship, $F_{drag} = -\beta v$, where $\beta > 0$ is constant. At the instant $t=0$ s, the magnitude of the ship's propelling force suddenly drops to 25% of the original value F_{prop} . Find

- the speed of the ship before the reduction of the propelling force, and
- the dependence of ship's speed on time after the propelling force has been reduced.

$$\frac{1}{4}F_p - \beta v = m \frac{dv}{dt}$$

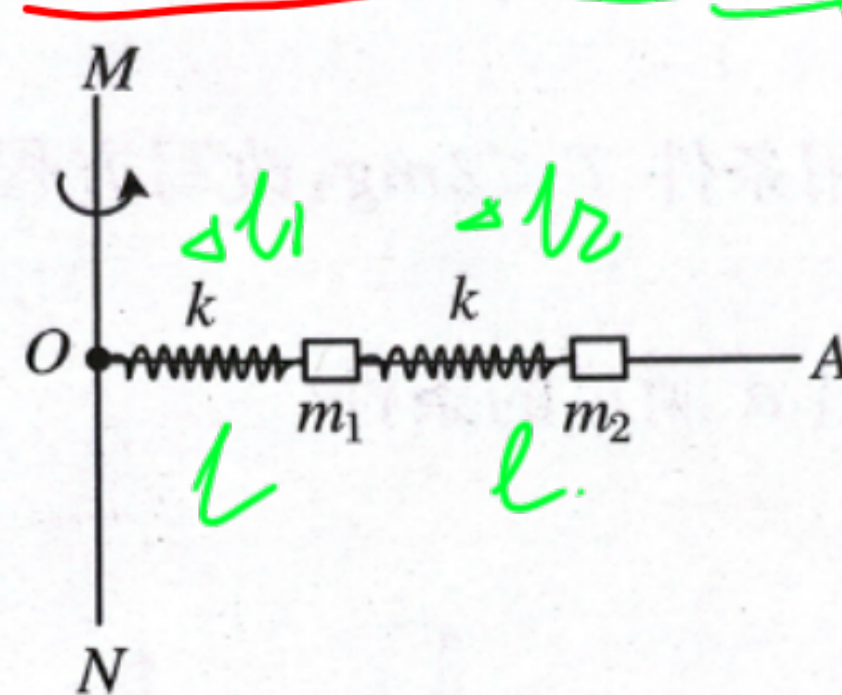
$$-\beta \left(v + \frac{F_p}{4\beta} \right) = m \frac{dv}{dt}$$

$$\int_0^t -\frac{\beta}{m} dt = \int_{v_0}^{v(t)} \frac{1}{v + \frac{F_p}{4\beta}} dv$$

$$-\frac{\beta}{m} t = \ln \left(v(t) + \frac{F_p}{4\beta} \right) - \ln \left(v_0 + \frac{F_p}{4\beta} \right)$$

$$\ln \left(v_0 + \frac{F_p}{4\beta} \right) - \frac{\beta}{m} t = \ln \left(v(t) + \frac{F_p}{4\beta} \right)$$

A smooth rod OA, can be rotated horizontally around point O, there are two identical springs (k), and two identical objects (m), if the system around the MN axis does stable uniform rotation, find the range of angular velocity of rotation.



$$l = \frac{-4\delta l_2^2 - \delta l_1^2 + 3\delta l_1 \delta l_2}{2\delta l_1 - 6\delta l_2}$$

$$l + \delta l_1 = \frac{-4\delta l_2^2 + \delta l_1^2 - 3\delta l_1 \delta l_2}{2\delta l_1 - 6\delta l_2}$$

$$= \frac{(\delta l_1 + \delta l_2)(\delta l_1 - 4\delta l_2)}{2\delta l_1 - 6\delta l_2}$$

$$(\delta l_1 - \delta l_2)k = m \omega^2 (l + \delta l_1)$$

$$\delta l_2 k = m \omega^2 (l_1 + l_2 + \delta l_1 + \delta l_2)$$

$$\delta l_1 - \delta l_2 = \left(\frac{m \omega^2}{k} \right) (l + \delta l_1)$$

$$\textcircled{1} (1-A)\delta l_1 = A l + \delta l_2 = A l + \frac{2A l + A \delta l_1}{1-A}$$

$$\delta l_2 = A (2l + \delta l_1 + \delta l_2)$$

$$(1-A)\delta l_2 = 2A l + A \delta l_1$$

$$(1-A)^2 \delta l_1 = (1-A)A l + 2A l + A \delta l_1$$

$$[(1-A)^2 - A] \delta l_1 = 3A l - A^2 l$$

$$\delta l_1 = \frac{3A l - A^2 l}{1 - 3A + A^2}$$

$$\frac{\delta l_1 - \delta l_2}{\delta l_1} = \frac{1 - \delta l_2 + \delta l_1}{2l_1 + \delta l_1 + \delta l_2}$$

$$\frac{\delta l_1 - \delta l_2}{\delta l_1} = \frac{\delta l_1 - \delta l_2}{2l_1 + \delta l_1 + \delta l_2}$$

$$\delta l_1 (2l_1 + \delta l_1 + \delta l_2) = \delta l_1 (2l_1 + \delta l_1 + \delta l_2)$$

$$- (\delta l_1 - \delta l_2)(\delta l_1 + \delta l_2) = (\delta l_1 - \delta l_2)(\delta l_1 + \delta l_2)$$

$$- \delta l_1^2 + \delta l_2^2 = - \delta l_1^2 + \delta l_2^2$$

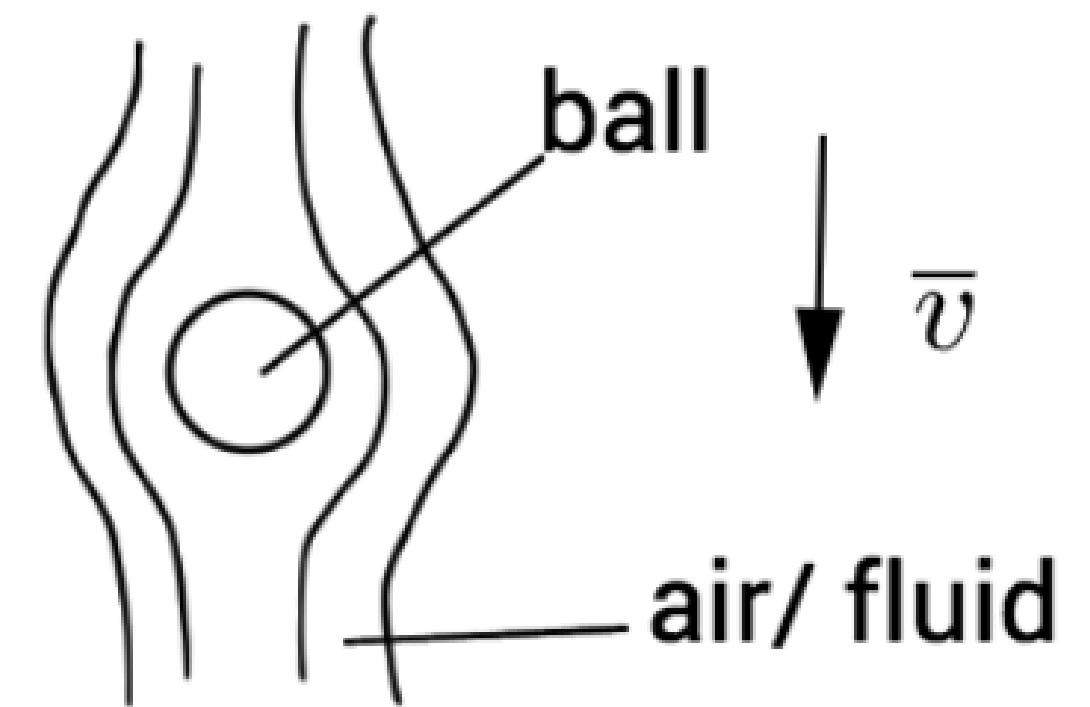
Chapter 5a – Motion with Fluid/Air Resistance

Stokes (or linear) drag [small objects/low speeds]

$$\bar{F}_{\text{drag}} = -k\bar{v} = -k\nu\left(\frac{\bar{v}}{\nu}\right)$$

where $k = 6\pi\eta R_s = \text{const}$, η is the fluid viscosity, R_s is the Stokes' radius (property of the object)

垂直于运动方向的面



Quadratic drag [large objects/high speeds]

2次

$$\bar{F}_{\text{drag}} = -b\nu^2\left(\frac{\bar{v}}{\nu}\right)$$

where $b = \frac{1}{2}\rho C_d A = \text{const}$, ρ is the fluid density, C_d is a drag coefficient (e.g. for cars 0.25-0.5), A is the cross-sectional area of the object perpendicular to the direction of motion

Observation: $m \uparrow$ terminal speed $\uparrow$

General Problem: how to deal with $F(v_y)$

step 1: $\Sigma = a = \frac{F(v_y)}{m}$

step 2: $\frac{dv_y}{dt} = \frac{F(v_y)}{m}$

$\int \frac{m}{F(v_y)} dv_y = \int dt$

这一步需要 separate variable

Eg. 1. Linear Drag.

$$m a_y = mg - k v_y \Rightarrow \frac{dv_y}{dt} = \frac{mg - k v_y}{m} = g - \frac{k}{m} v_y$$

$$\frac{dv_y}{g - \frac{k}{m} v_y} = dt \quad = -\frac{k}{m} \left(v_y - \frac{m}{k} g \right)$$

$$\int_0^{v_y(t)} \frac{dv_y}{v_y - \frac{m}{k} g} = -\frac{k}{m} \int_0^t dt$$

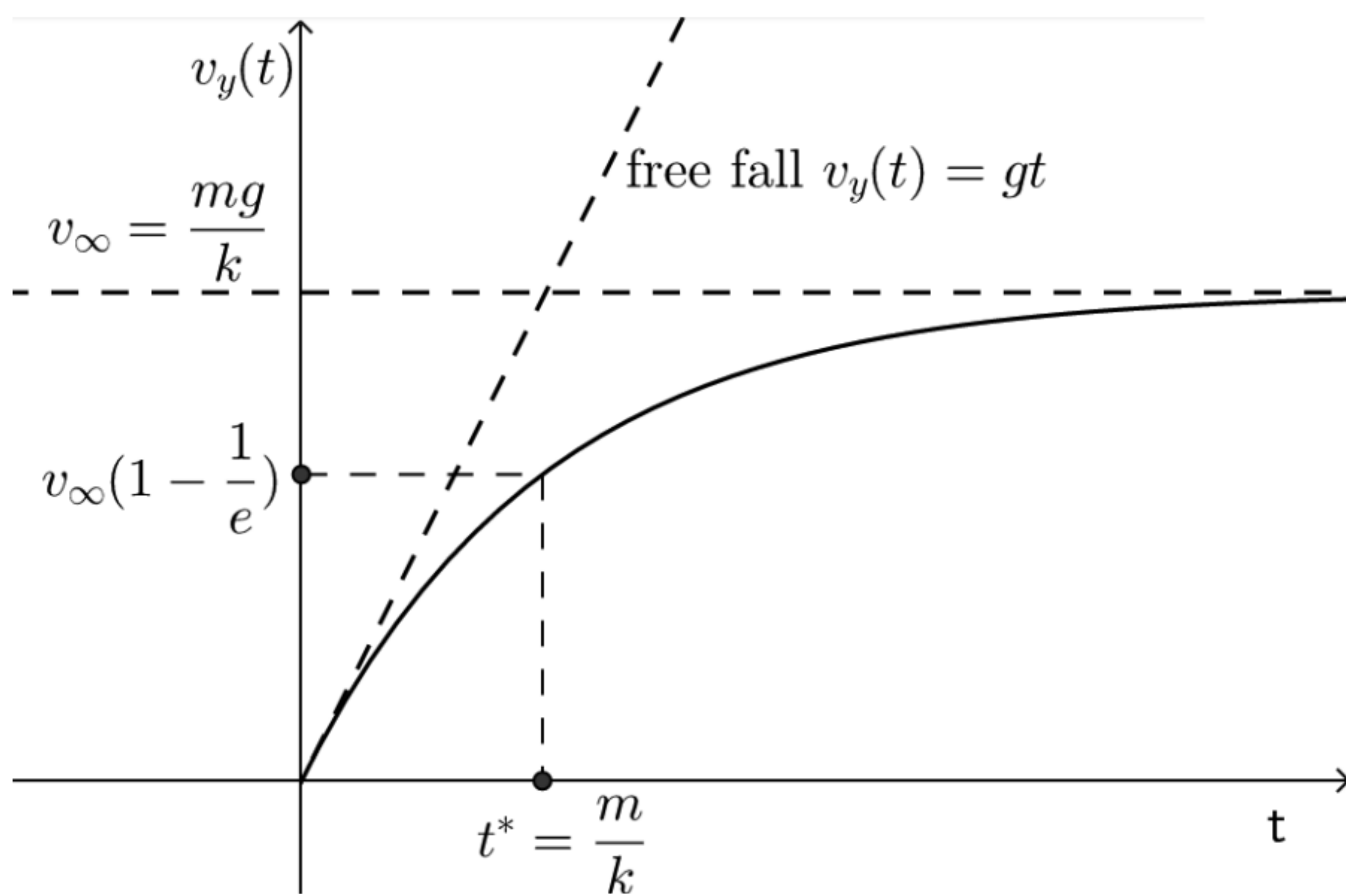
$$\ln \left[v_y(t) - \frac{m}{k} g \right] - \ln \left(0 - \frac{m}{k} g \right) = -\frac{k}{m} t$$

记住 $v_y(t) \rightarrow \frac{mg}{k}$ (terminal speed)

$$\int_0^{v_y(t)} \frac{dv_y}{v_y - \frac{mg}{k}} = -\frac{k}{m} \int_0^t dt \Rightarrow \ln \left| \frac{v_y(t) - \frac{mg}{k}}{-\frac{mg}{k}} \right| = -\frac{k}{m} t$$

$$\Rightarrow v_y(t) = \frac{mg}{k} (1 - e^{-\frac{k}{m} t})$$

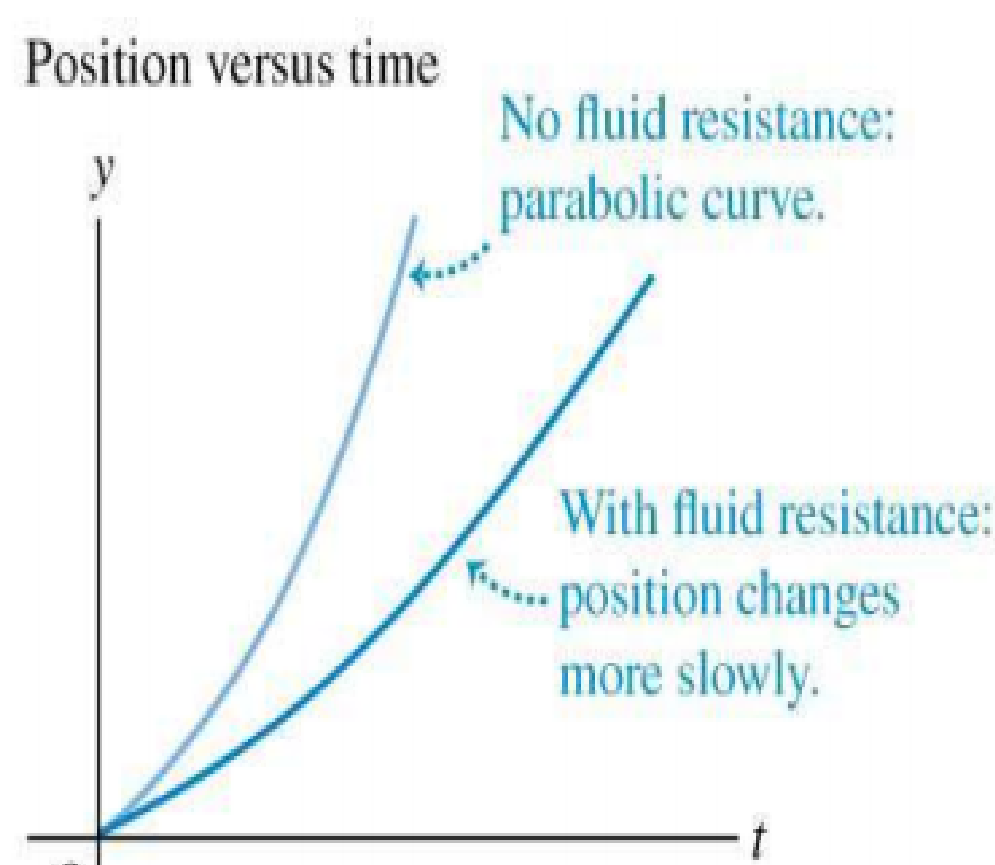
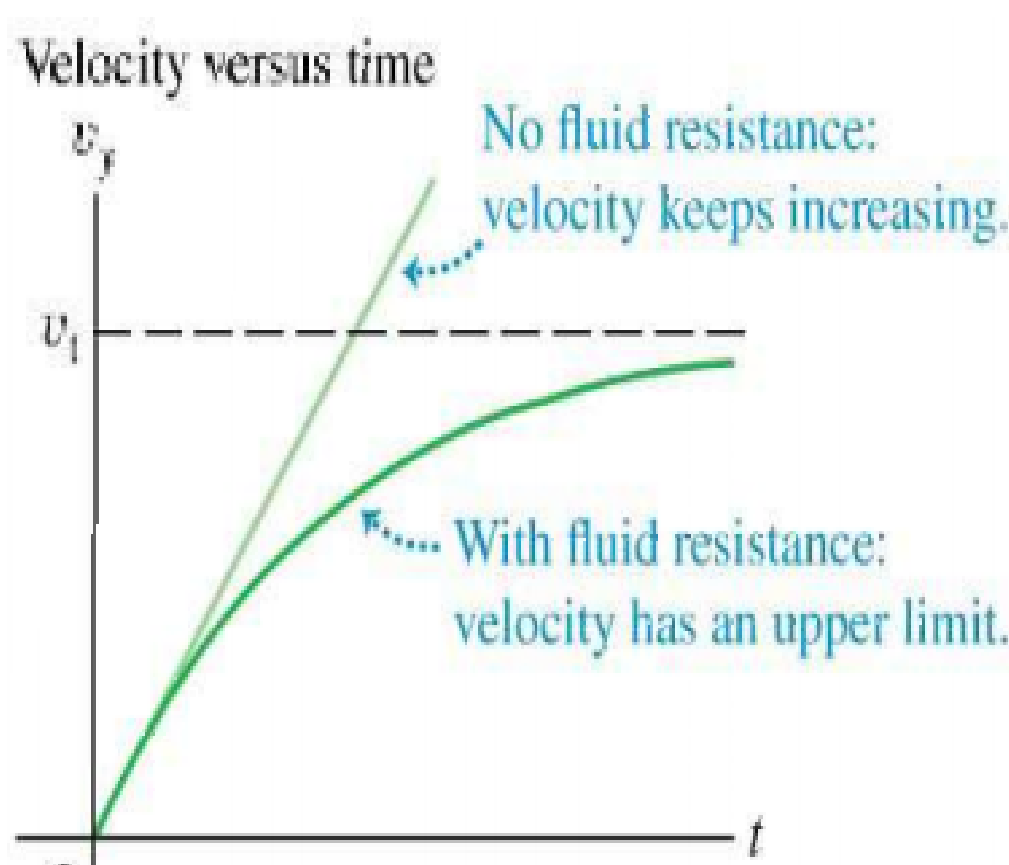
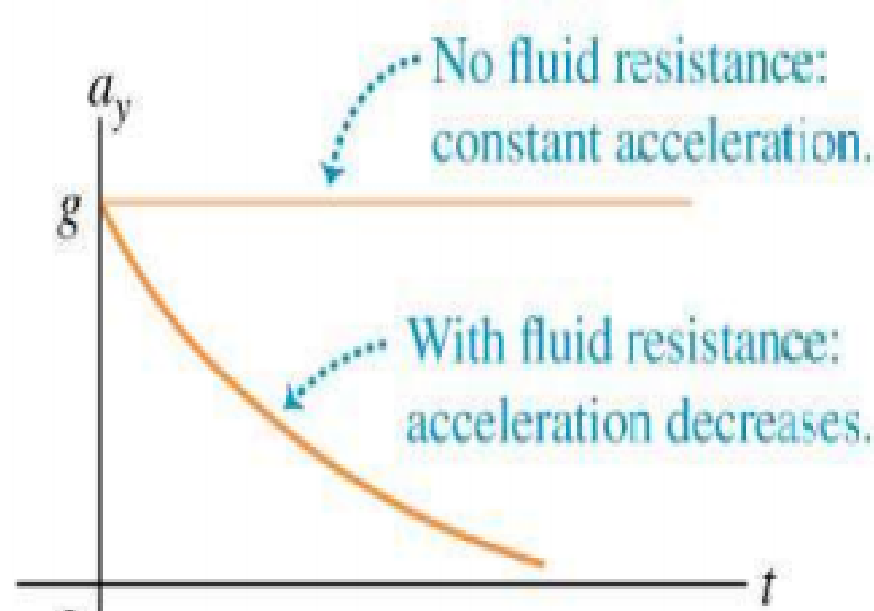
graph:



① $t \ll \frac{m}{k} \quad \frac{k}{m} t \ll 1$
 $v_y(t) \approx gt$

② $t \gg \frac{m}{k}$

$$v_{\infty} = \frac{mg}{k} = \text{const.}$$



so $a_y(t) = g e^{-\frac{k}{m}t}$

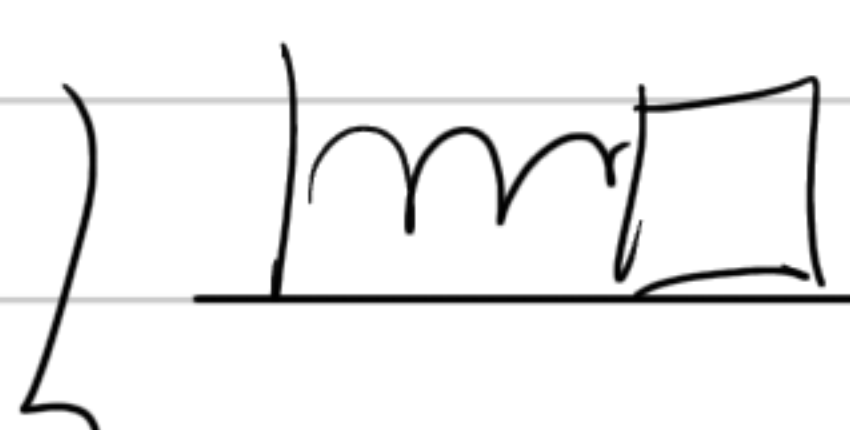
$$y(t) = \frac{mg}{k} \left[t + \frac{m}{k} (e^{-\frac{k}{m}t} - 1) \right]$$

short time $e^{-\frac{k}{m}t} = 1 + \frac{-\frac{k}{m}t}{1!} + \frac{(\frac{k}{m}t)^2}{2!} \dots$

$$y(t) \approx \frac{1}{2}gt^2$$

Long time $e^{-\frac{k}{m}t} \approx 0$ $y(t) = \frac{mg}{k} (t - \frac{m}{k})$

Chapter 6 – Harmonic Oscillator and Mechanical Resonance

 $ma_x = F_x \Rightarrow \ddot{x} + \frac{k}{m}x = 0$

 $ma_x = -kx \Rightarrow \ddot{x} + \frac{k}{m}x = 0$



$$\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$$

$ma_\theta = F_\theta \Leftrightarrow m l \ddot{\theta} \approx -mg\theta$

$T = 2\pi\sqrt{\frac{l}{g}}$ 前提条件: θ 很小!!! $\Rightarrow \ddot{\theta} + \frac{g}{l}\theta = 0$

Observation:

In all three cases, the equation of motion is of the same form

$$\ddot{x} + \omega_0^2 x = 0$$

[with $\omega_0^2 = k/m$ for the mass-spring systems and $\omega_0^2 = g/l$ for the simple pendulum]

Any particle (system) with the equation of motion of the above form is called a **simple harmonic oscillator** (SHO).

to find the most general form of $x(t)$

$$x(t) = A \cos(\omega_0 t + \phi)$$

amplitude initial phase

$\updownarrow$

$$\dot{x} = -\omega_0 A \sin(\omega_0 t + \phi)$$

$$\ddot{x} = -\omega_0^2 A \cos(\omega_0 t + \phi)$$

$$x(t) = B \cos \omega_0 t + C \sin \omega_0 t$$

$$A \cos \omega_0 t \cos \phi - A \sin \omega_0 t \sin \phi$$

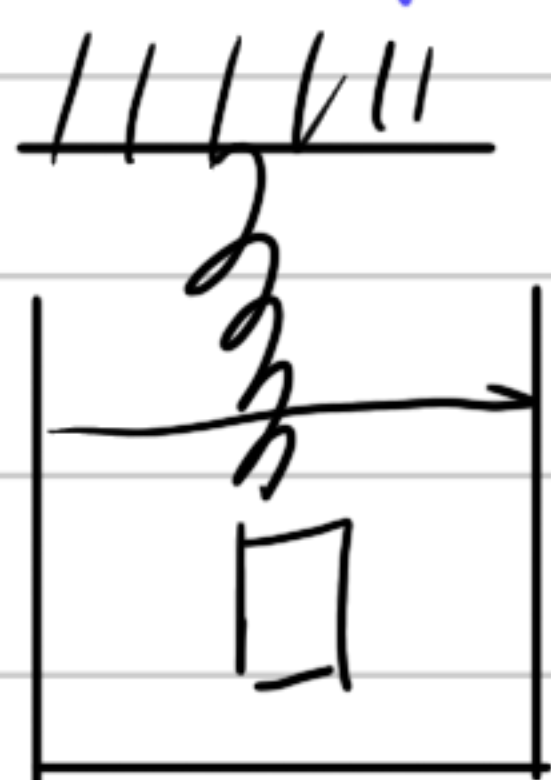
$$B = A \cos \phi \quad C = -A \sin \phi$$

note: $x-t$ $\rightarrow$ v_x-t $\rightarrow$ a_x-t 图.

向左移动本周期

向左移动本周期

Damped Oscillation



$$m \frac{d^2 x}{dt^2} = -b \frac{dx}{dt} - kx$$

$$\Rightarrow \ddot{x} + \frac{b}{m} \dot{x} + \omega_0^2 x = 0$$

设 $x(t) = e^{\lambda t}$ 解 $\lambda^2 + \frac{b}{m} \lambda + \omega_0^2 = 0$

$$\Delta = \left(\frac{b}{m}\right)^2 - 4\omega_0^2$$

underdamped.

① $\Delta < 0$; complex roots $\lambda_{1,2} = -\frac{b}{2m} \pm i \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}$

② $\Delta > 0$; real & different roots $\lambda_{1,2} = -\frac{b}{2m} \pm \sqrt{\left(\frac{b}{2m}\right)^2 - \omega_0^2}$ overdamped.

③ $\Delta = 0$; real root (repeated) $\lambda = -\frac{b}{2m}$ critical.

$$\Delta < 0 \quad \Rightarrow \quad \left(\frac{b}{m}\right)^2 - 4\omega_0^2 < 0 \quad \Rightarrow \quad \boxed{\left(\frac{b}{m}\right)^2 < 4\omega_0^2}$$

It is the case of weak damping (**underdamped regime**). Then, the general solution

不是很懂

$$\begin{aligned} x(t) &= C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} = \\ &= C_1 e^{-\frac{b}{2m}t} e^{i\sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}t} + C_2 e^{-\frac{b}{2m}t} e^{-i\sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}t} \end{aligned}$$

But $x(t)$ is a physical quantity (displacement from the equilibrium position); must be real, so

$$C_1 = \frac{1}{2}Ae^{i\phi_0} = C_2^* \quad \Rightarrow \quad C_2 = \frac{1}{2}Ae^{-i\phi_0}$$

Hence (recall the formula: $e^{iu} = \cos u + i \sin u$)...

阻尼振荡运动学公式: underdamped regime.

$$x(t) = Ae^{-\frac{b}{2m}t} \cos \left(\sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2} t + \phi_0 \right)$$

Overdamped regime

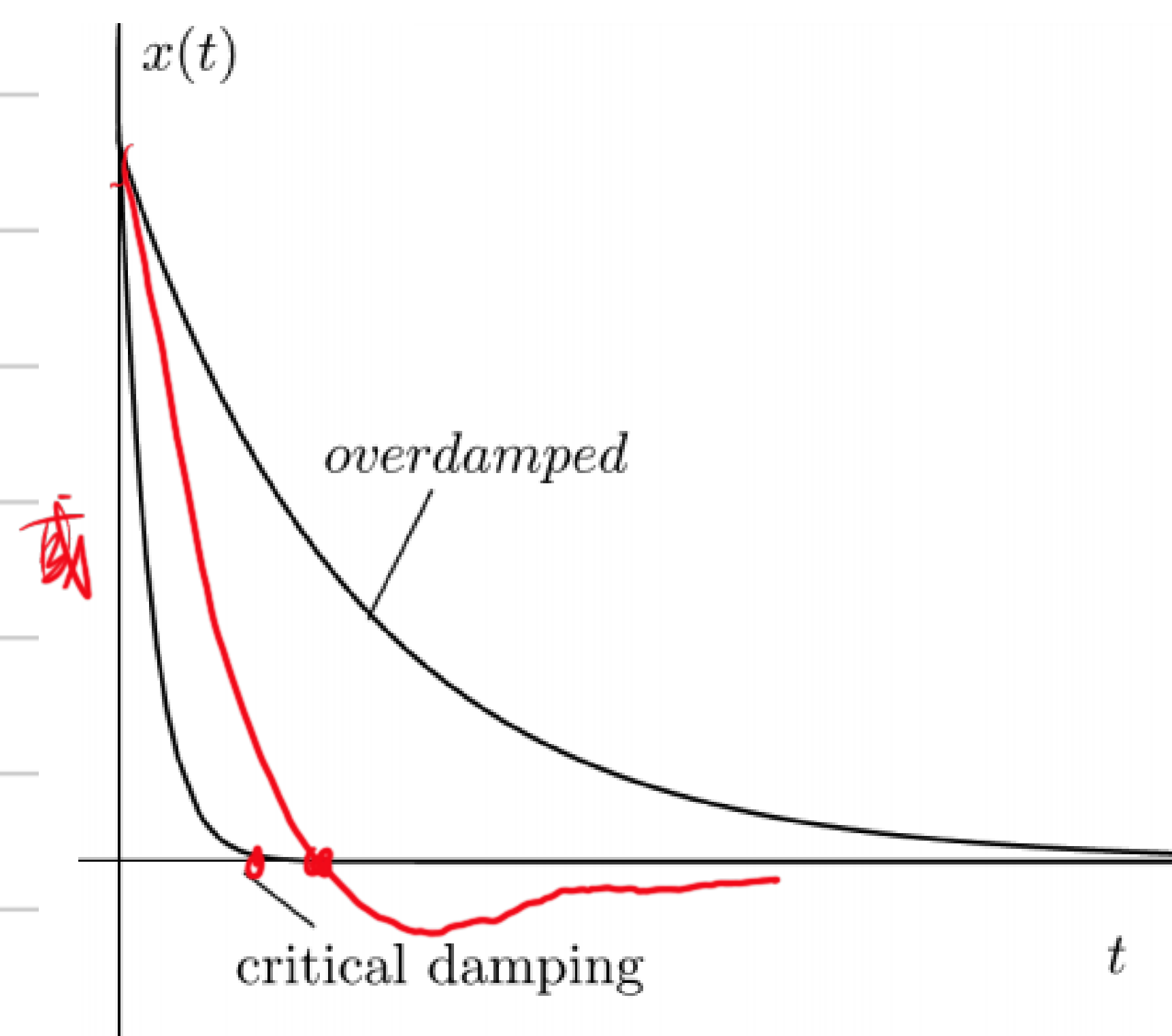
$$x(t) = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} =$$

$$= C_1 e^{-\left(\frac{b}{2m} - \sqrt{\left(\frac{b}{2m}\right)^2 - \omega_0^2}\right)t} + C_2 e^{-\left(\frac{b}{2m} + \sqrt{\left(\frac{b}{2m}\right)^2 - \omega_0^2}\right)t}$$

{ underdamped $b < 2m\omega_0$ $x(t) = Ae^{-\frac{b}{2m}t} \cos \left(\sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2} t + \phi \right)$
 overdamped $b > 2m\omega_0$ $x(t) = D_1 e^{-\frac{b}{2m}t} + D_2 t e^{-\frac{b}{2m}t}$
 critical damping $b = 2m\omega_0$

Critical damping.

$$x(t) = D_1 e^{-\frac{b}{2m}t} + D_2 t e^{-\frac{b}{2m}t}$$



Forced (or Driven) Oscillations. Mechanical Resonance

$$m\ddot{x} = F_x = -kx - b\dot{x} + F_0 \cos \omega_{dr} t.$$
$$\ddot{x} + \underbrace{\frac{b}{m}}_{\omega_0^2} \dot{x} + \frac{k}{m} x = \frac{F_0}{m} \cos \omega_{dr} t.$$

For steady-state: $x_{ss}(t) = A \cos(\omega_{dr} t + \phi)$

$$A(\omega_{dr}) = \frac{F_0}{m \sqrt{(\omega_0^2 - \omega_{dr}^2)^2 + \left(\frac{b\omega_{dr}}{m}\right)^2}}$$
$$\tan \phi = \frac{b\omega_{dr}}{m(\omega_{dr}^2 - \omega_0^2)}$$

In general, $\phi \neq 0$

Pendulum: $\omega_0 = \sqrt{\frac{g}{l}}$

Spring: $\omega_0 = \sqrt{\frac{k}{m}}$

共振频率 $\omega_{\text{dri}} = \omega_{\text{res}} = \sqrt{\omega_0^2 - \frac{b^2}{2m^2}}$
 外力驱动频率 $\omega = \sqrt{\omega_0^2 - \frac{b^2}{2m^2}}$
 $\omega_0 > \omega_{\text{dri}} > \omega_{\text{resonance}}$

Motion Equation: $\ddot{x} + \frac{b}{m}\dot{x} + \omega_0^2 x = \frac{F_0}{m} \cos \omega_{\text{dr}} t$

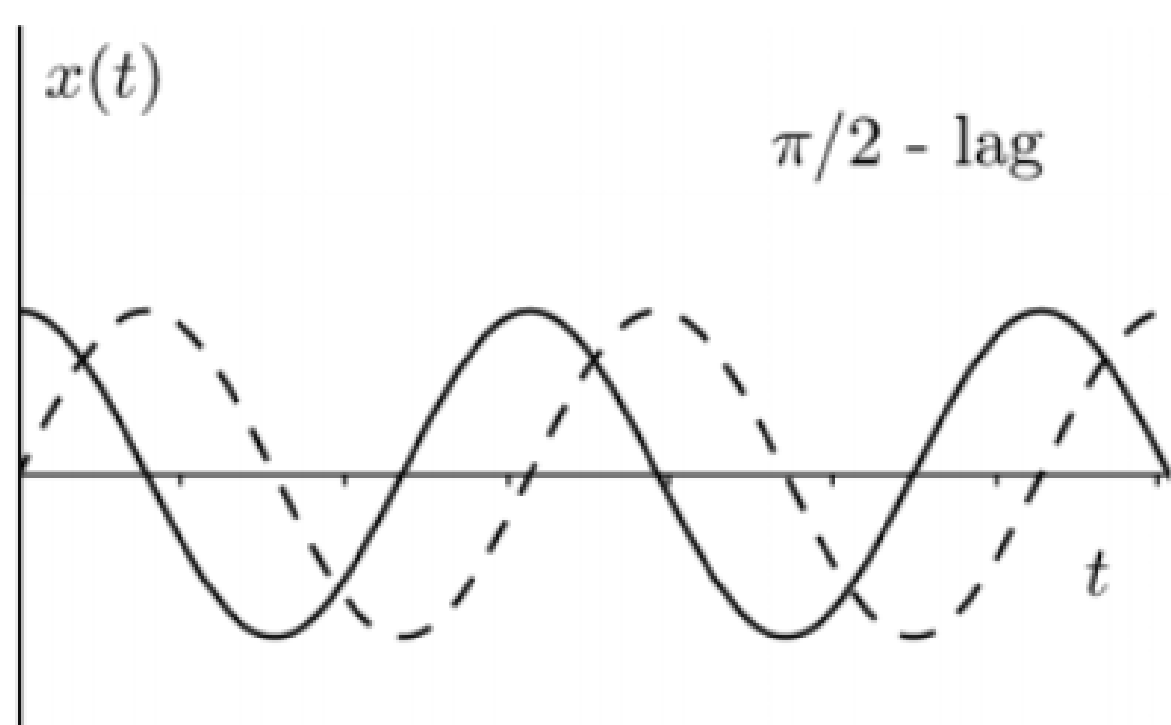
Steady State Solution: $x_s(t) = A \cos(\omega_{\text{dr}} t)$

Amplitude: $A(\omega_{\text{dr}}) = \frac{F_0}{m \sqrt{(\omega_0^2 - \omega_{\text{dr}}^2)^2 + \left(\frac{b\omega_{\text{dr}}}{m}\right)^2}}$

Phase Shift: $\tan \varphi = \frac{b\omega_{\text{dr}}}{m(\omega_{\text{dr}}^2 - \omega_0^2)}$

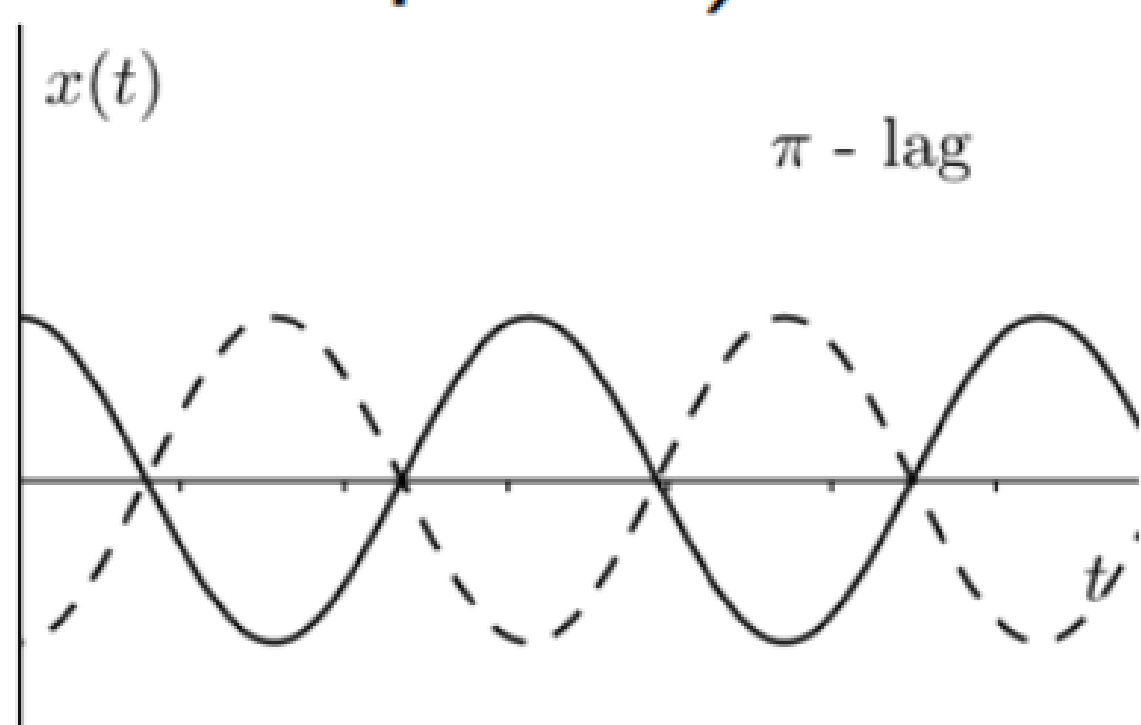
Resonance Frequency: $\omega_{\text{res}} = \sqrt{\omega_0^2 - b^2/2m^2}$

- If $\omega_{\text{dr}} \rightarrow \omega_0$ (close to resonance³), then $\phi \rightarrow -\pi/2$.



The response ($x_s(t)$) lags the drive ($F_{\text{dr}}(t)$) by 1/4 of the cycle.

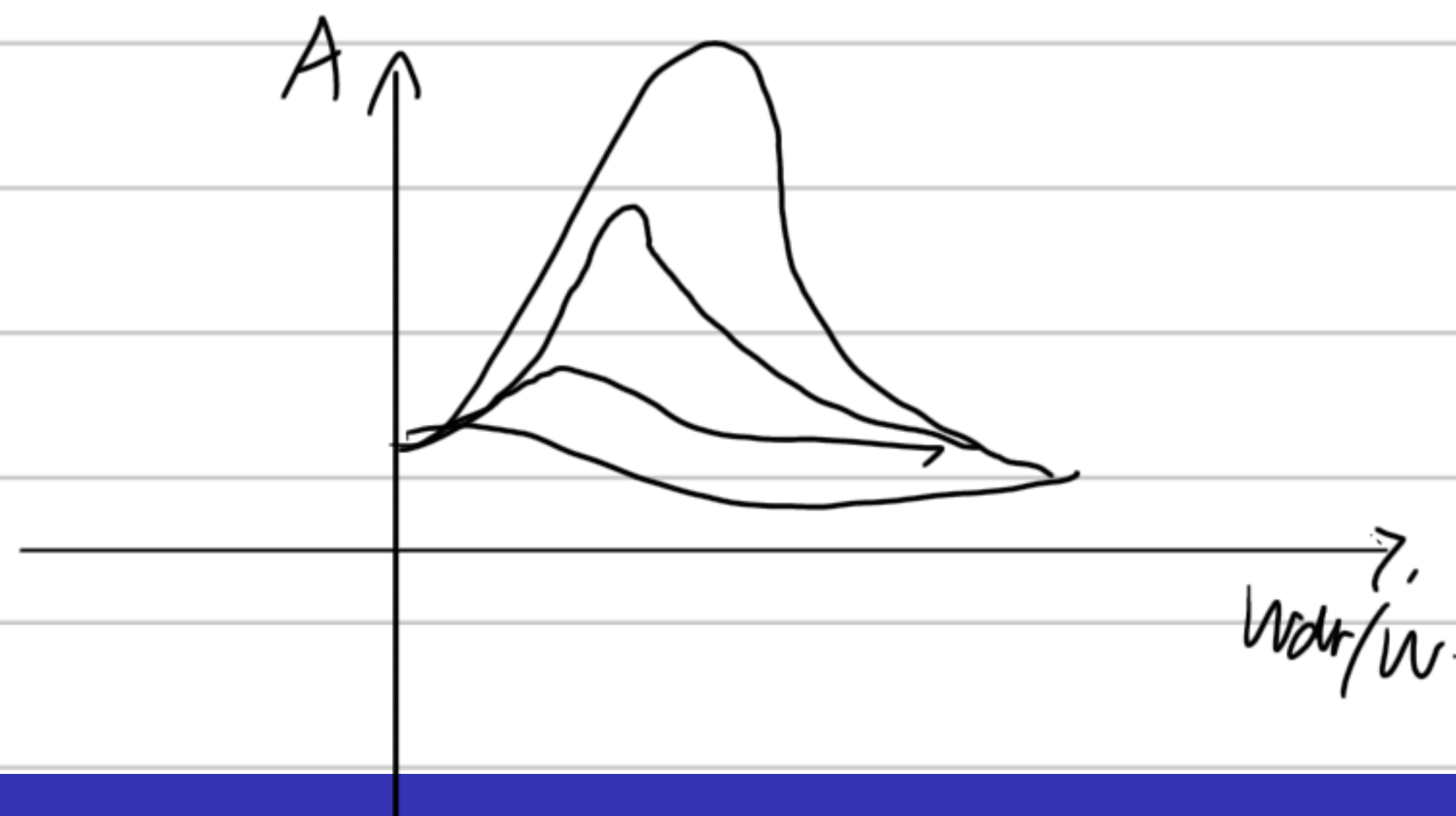
- If $\omega_{\text{dr}} \rightarrow \infty$ (high frequencies), then $\phi \rightarrow -\pi$ the response lags the drive by 1/2 of the cycle (displacement and drive are in antiphase)



The response ($x_s(t)$) lags the drive ($F_{\text{dr}}(t)$) by 1/2 of the cycle.

$A = A(\omega_{dr})$ at the resonance frequency. $\omega_{dr} = \omega_{res} = \sqrt{\omega_0^2 - \frac{b^2}{2m^2}}$
 if damping is negligible, $\omega_{dr} = \omega_0$

Increasing damping shifts the resonance frequency downwards.



Chapter 7 – Dynamics of Circular Motion Dynamics in Non-Inertial Frames of Reference

$$\vec{v} = \vec{v}_0 + \vec{v}' + (\vec{\omega} \times \vec{r}') \quad (\text{canvas})$$

$$\vec{a} = \vec{a}_0 + \vec{a}' + 2\vec{\omega} \times \vec{v}' + \frac{d\vec{\omega}}{dt} \times \vec{r}' + \vec{\omega} \times (\vec{\omega} \times \vec{r}')$$

real force

$$m\vec{a}' = \underbrace{m\vec{a}}_{\substack{\uparrow \\ \text{real force}}} - m\vec{a}_0 - \dots \Rightarrow \text{non-inertial P.O.V.}$$

$\vec{a}' = \vec{a}_{\text{rel}}$ (non-inertial reference frame acceleration)

$\vec{a}_0$ (pseudo force)

Newton's 2nd Law

$$m\vec{a}' = \underbrace{\vec{F}}_{\text{real force (of material origin)}} + \underbrace{-m\vec{a}_0}_{\text{d'Alembert force}} + \underbrace{-m \frac{d\vec{\omega}}{dt} \times \vec{r}'}_{\text{Euler}} + \underbrace{-2m(\vec{\omega} \times \vec{v}')}_{\text{Coriolis}} + \underbrace{-m\vec{\omega} \times (\vec{\omega} \times \vec{r}')}_{\text{centrifugal force}}$$

pseudo-forces (also called fictitious forces or forces of inertia); these are kinematic corrections (which have units of [N]) that **must** be included due to the fact that we describe dynamics in a non-inertial FoR (force = mass × acceleration is valid only in inertial FoRs). These "forces" **never** appear in inertial FoRs!

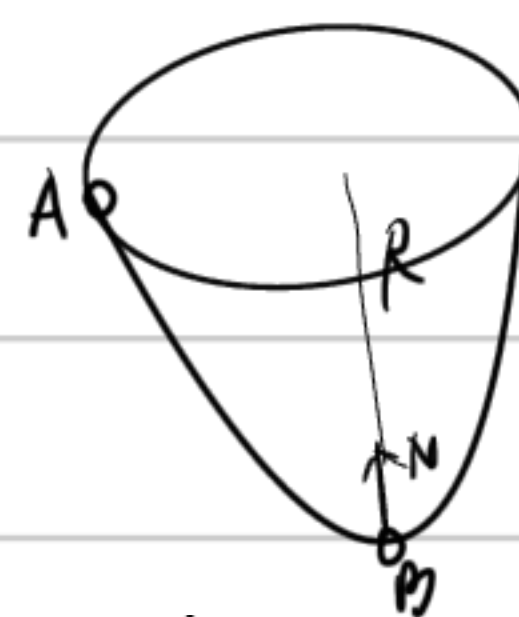
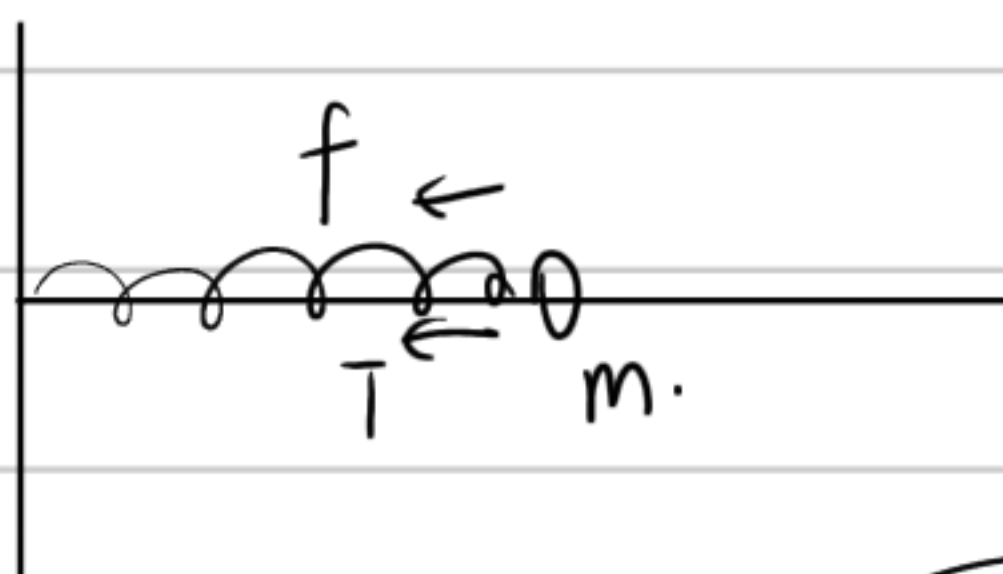
Each of the fictitious forces has its own name

北顺		$-m\vec{a}_0$	d'Alembert (or drift) "force" 非惯性系的平动加速度
南逆		$-m \frac{d\vec{\omega}}{dt} \times \vec{r}'$	Euler "force" 系统角加速度
		$-2m(\vec{\omega} \times \vec{v}')$	Coriolis "force" "北顺南逆"
		$-m\vec{\omega} \times (\vec{\omega} \times \vec{r}')$	centrifugal "force"

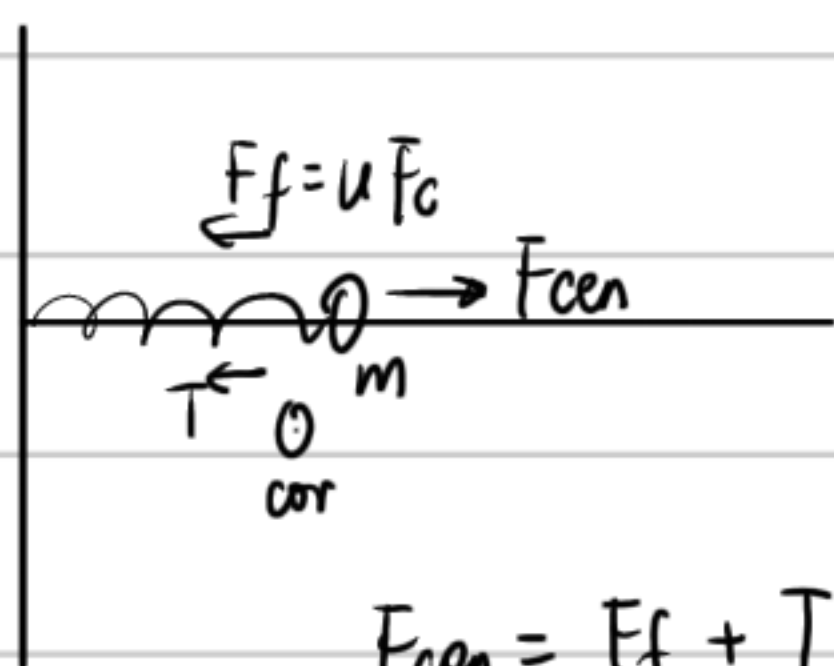
$F_{cen}, v'=0, \frac{d\omega}{dt}=0$

$f \rightarrow v_0$
 $\leftarrow 0$
 Kol

$$F_f = \mu N = \mu F_c$$



$$N - mg = m \frac{v^2}{R}$$



$$0 = ma' = F + F_{inertial}$$

$$F_{cen} = F_f + T$$

$$m\omega^2(r_0 + \delta x) = \mu \cdot 2m\omega^2 r_0 + k\delta x$$

$$\delta x = 0 \Rightarrow 2\mu m\omega^2 r_0 = m\omega^2 \delta x$$

$$\mu = \frac{r_0}{2\omega^2} \sqrt{\frac{k}{m}}$$

$$\mu = \frac{r_0}{2\omega^2}$$

$$\vec{v} = \dot{r}\hat{n}_r + r\dot{\phi}\hat{n}_\phi$$

$$\vec{a} = (\ddot{r} - r\omega^2)\hat{n}_r + (r\ddot{\omega} + 2\dot{r}\dot{\omega})\hat{n}_\phi$$

Example

B

A particle with mass m moves along the x axis in a potential force field with potential energy $U(x) = Be^{2x^2}$. The period of small oscillations of this particle about the stable equilibrium at $x_0 = 0$ is

Δ

(a) $2\pi\sqrt{\frac{m}{2B}}$

(b) $2\pi\sqrt{\frac{m}{4B}}$

(c) $\sqrt{\frac{2m}{B}}$

(d) $\sqrt{\frac{m}{2B}}$

$$U(x) = U(0) + \frac{1}{2}U''(0)x^2$$

$$U(x) = Be^{2x^2}$$

$$U' = 4Bxe^{2x^2}$$

$$U'' = 4B e^{2x^2} + 16Bx^2 e^{2x^2}$$

$$\omega = \sqrt{\frac{k}{m}}$$

$$U(x) = B +$$

$$2B x^2$$

$$U'(x) = 4Bx = -F$$

$$k = 4B$$

$$\omega = \sqrt{\frac{4B}{m}}$$

$$T = 2\pi\sqrt{\frac{m}{B}}$$

$$\star \bar{A} \times (\bar{B} \times \bar{C}) = \bar{B} (\bar{A} \circ \bar{C}) - \bar{C} (\bar{A} \circ \bar{B})$$

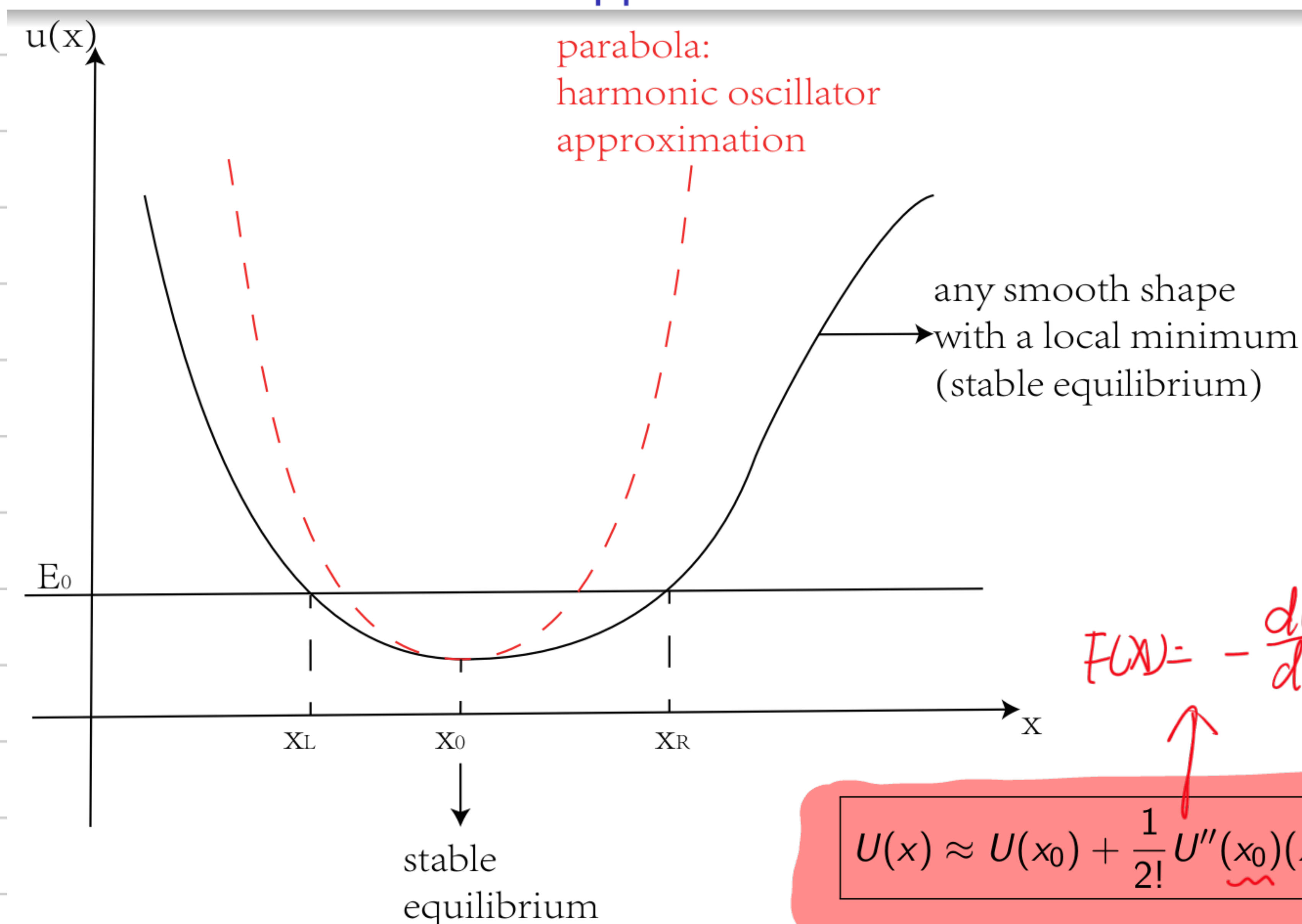
outer clock wise



$\Rightarrow$ on the southern hemisphere

Chapter 9 - potential energy - conservative force,

Motion in a Potential Well. Harmonic Approximation



$$F(x) = -\frac{du}{dx} = -k(x-x_0)$$

$$U(x) \approx U(x_0) + \frac{1}{2!} \underbrace{U''(x_0)}_{\text{spring constant}} (x - x_0)^2.$$

$\star$ 前提条件 $|x - x_0| \rightarrow \text{'small'}$

specially: for a spring = $U(x) = \frac{1}{2}k(x-x_0)^2$
 so $k = U''(x_0)$.

Conclusion. A particle moving in a potential well of *any smooth shape*, not too far away from a stable equilibrium, moves as if it was in harmonic motion with the natural frequency

$$\omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{U''(x_0)}{m}} \quad \text{前提: linear force}$$

Consequently, $x(t) = A \cos(\omega_0 t + \phi)$.

$$\star \vec{F} = \left(-\frac{\partial U}{\partial x}, -\frac{\partial U}{\partial y}, -\frac{\partial U}{\partial z} \right) = -\nabla U.$$

↓
各个方向上的偏导.

$$E_{\text{mean}} = K + U = \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 + \frac{1}{4}cx^4 = \frac{1}{4}cA^4$$

$$2m\left(\frac{dx}{dt}\right)^2 + cx^4 = cA^4$$

$$\frac{dx}{dt} = \sqrt{\frac{cA^4 - cx^4}{2m}} \Rightarrow \int_0^A \sqrt{\frac{2m}{cA^4 - cx^4}} dx = \int_0^{\frac{1}{2}T} dt = \frac{1}{2}T$$

For a certain oscillator the net force on the body with mass m is given by $F_x = -cx^3$.

One-quarter of a period is the time for the body to move from $x = 0$ to $x = A$. Calculate this time and hence the period.

Express your answer in terms of the variables A , m , and c .

$$T = \frac{7.41}{A} \sqrt{\frac{m}{c}}$$

Submit

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Answer Requested

$$F_x = -cx^3$$

$$U = \int -F_x dx = \int cx^3 dx = \frac{1}{4}cx^4$$

$$W_0 = \frac{3cx^2}{m}$$

$$U' = cx^3$$

$$U'' = 3cx^2$$

—————→ x
1/2

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \frac{\sqrt{\frac{2m}{c}} \frac{1}{A^2 - x^4}}{\frac{1}{2A}} dx &= \frac{1}{2A} \int_0^{\frac{\pi}{2}} \frac{dx}{\cos^2 \theta} = \frac{1}{2A} \int_0^{\frac{\pi}{2}} \sec^2 \theta d\theta \\ &= \frac{1}{2A} \tan \theta \Big|_0^{\frac{\pi}{2}} = \frac{1}{2A} \lim_{\theta \rightarrow \frac{\pi}{2}} \tan \theta = \frac{1}{2A} \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\sin \theta}{\cos \theta} = \frac{1}{2A} \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1}{0} = \frac{1}{2A} \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\cos \theta}{-\sin \theta} = \frac{1}{2A} \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{0}{-1} = \frac{1}{2A} \end{aligned}$$

Part C

According to the result you obtained in part (b), does the period depend on the amplitude A of the motion? Are the oscillations simple harmonic?

非线性力 T与A相关.

yes;no

Submit

[Previous Answers](#)

$$F = -3000(x + 160x^3) = -3000x - 48000x^3 \quad x=0 \quad U_a=0$$

$$U_a = -\int F dx = \int (3000x + 48000x^3) dx = 1500x^2 + 12000x^4$$

$$-m \frac{d\omega}{dt} \times \vec{r}$$

ruler

$$-2m(\vec{\omega} \times \vec{v})$$

Coriolis

$$-m\vec{\omega} \times (\vec{\omega} \times \vec{r})$$

Centrifugal force

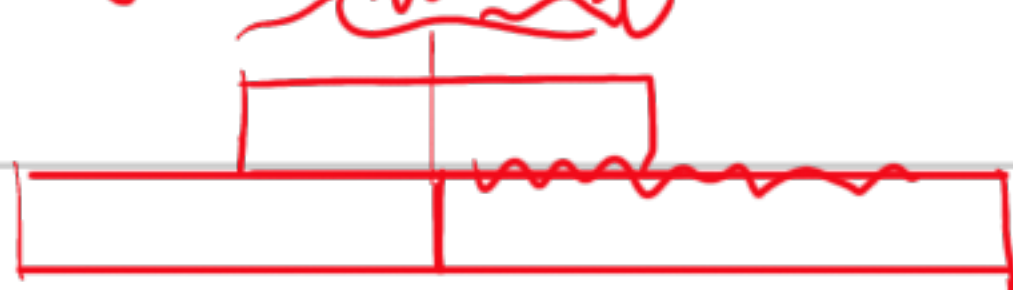
☆☆☆ 一定要记得 $F = -\nabla U = -(\frac{\partial U}{\partial x}, \frac{\partial U}{\partial y}, \frac{\partial U}{\partial z})!!!$

eg. 已知 $F = (3y^2 + z^2, 9xy^2, 2xz)$

$$U = (-3xy^2 - xz^2) \checkmark$$

$$U = (3xy^2 + xz^2 + 4) \times \times \times$$

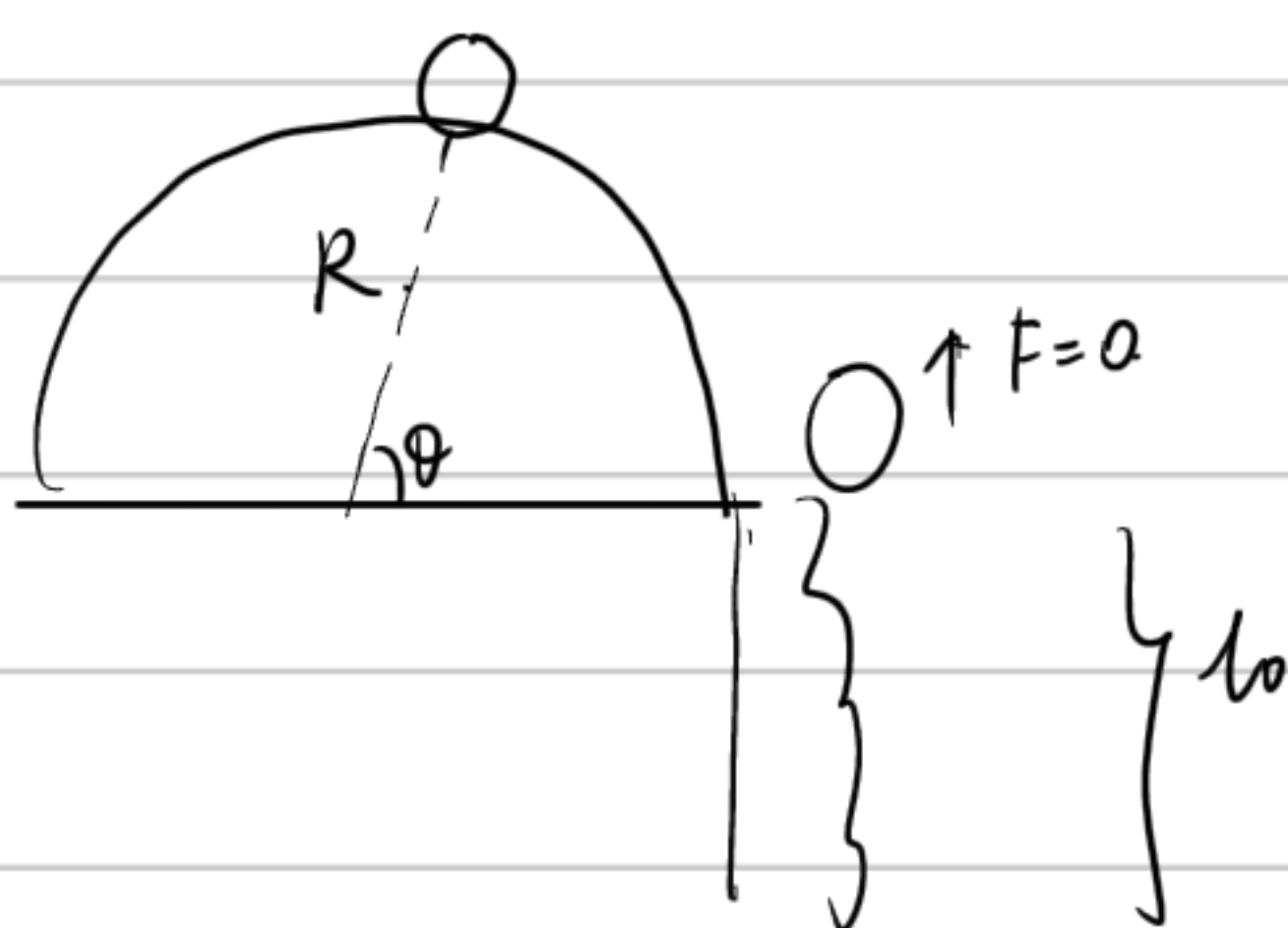
☆☆☆ 记得



$$f \propto l.$$

$$umg \frac{l}{l_0}$$

$$Q = \int umg \frac{l}{l_0} dl.$$



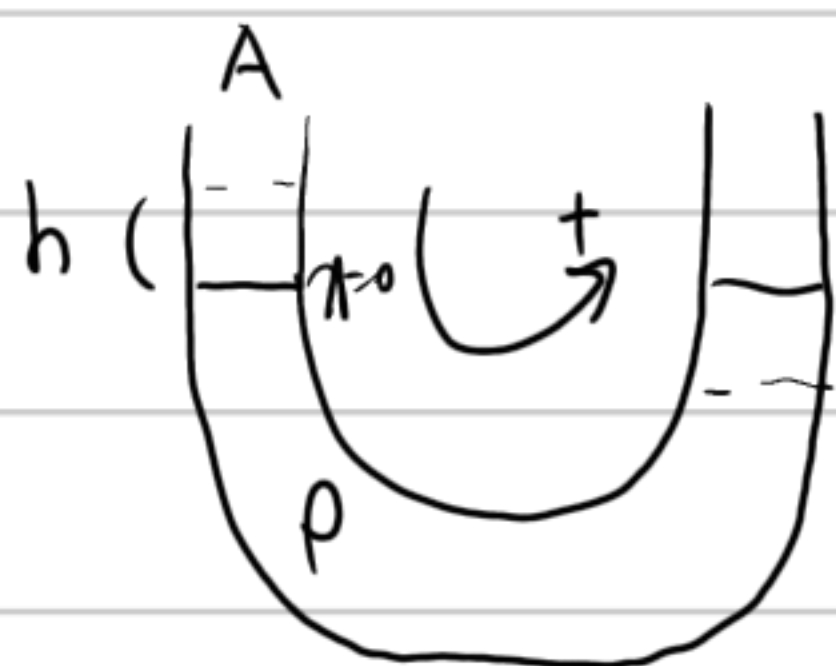
$$x = l + R\theta$$

$$W = mgR \sin \theta + \frac{1}{2}k(x - l_0)^2 - \frac{1}{2}k(l - l_0)^2$$

$$\frac{1}{2}k \left(\cancel{x^2} - \cancel{2xl_0} - \cancel{x^2} + \cancel{2xl_0} \right)$$

$$x^2 + 2R\theta l + R^2\theta^2$$

$$\frac{1}{2}k(2R\theta l + R^2\theta^2 - 2R\theta l_0)$$



$$F = 2\rho A g h.$$

$$F = -2\rho A g$$

$$\ddot{x} = \frac{2\rho A g h}{m}$$

$$\ddot{x} + \frac{k}{m}x + \omega_0^2 x = 0$$

$$\ddot{x} + \frac{2\rho A g}{m}x = 0$$

$$\omega_0 = \sqrt{\frac{2\rho A g}{m}}$$

Chapter 10 – Momentum and Conservation of Momentum

$$\vec{F} = m\vec{a} = m \frac{d\vec{v}}{dt} = \frac{d}{dt}(m\vec{v}) = \frac{d\vec{p}}{dt}$$

$$\vec{F} = \frac{d\vec{p}}{dt}$$

$$p_2 - p_1 = \underbrace{\int_{t_1}^{t_2} \text{net force } F_x dt}_{\text{impulse of the net force}}$$

N.S

the same net change in momentum can be achieved in various ways

Conservation of Momentum

Newton's 3rd law

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

Newton's 2nd law

$$\vec{F}_{AB} = \frac{d\vec{p}_A}{dt}$$

$$\frac{d\vec{p}_A}{dt} = -\frac{d\vec{p}_B}{dt}$$

$$F_{BA} = \frac{d\vec{p}_B}{dt}$$

$$\Rightarrow \frac{d\vec{p}_A}{dt} + \frac{d\vec{p}_B}{dt} = 0$$

$$\frac{d}{dt}(\vec{p}_A + \vec{p}_B) = 0$$

$$\Rightarrow \vec{p}_A + \vec{p}_B = \text{const}$$

Conservation of Momentum Law

If the sum of external forces on a system is equal to zero, then the total momentum of the system is constant (i.e. is conserved).

Collision

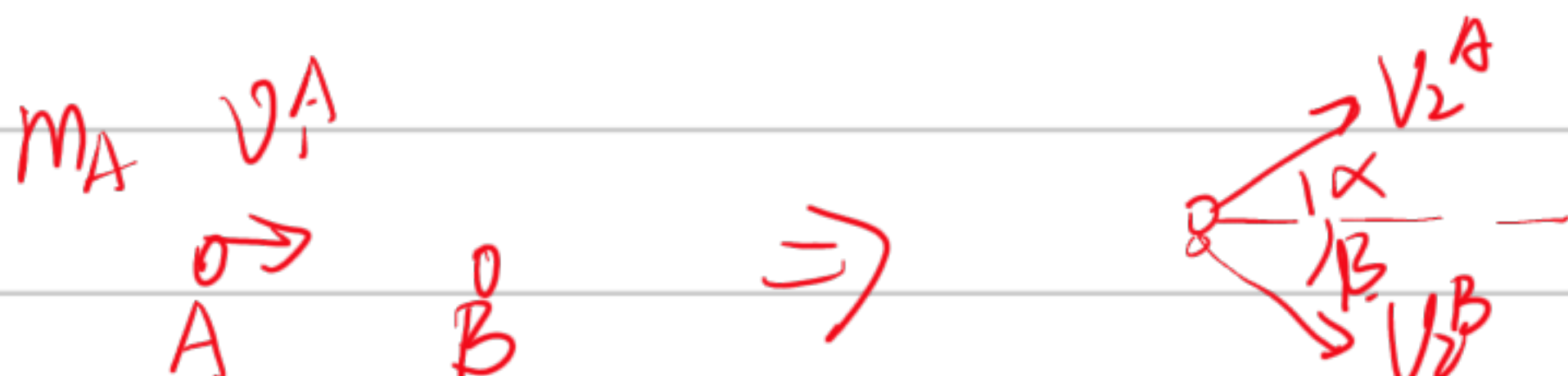
$$\tan \beta = -\frac{v_2^A \sin \alpha}{v_1^A - v_2^A \cos \alpha}$$

$$v_2^B = -\frac{m_A v_2^A \sin \alpha}{m_B \sin \beta}$$

$$\begin{matrix} v_1 & v_2 & v_1' & v_2' \\ m_1 & m_2 & m_1 & m_2 \\ \boxed{\rightarrow} & \boxed{\rightarrow} & \Rightarrow \boxed{\rightarrow} & \boxed{\rightarrow} \end{matrix}$$

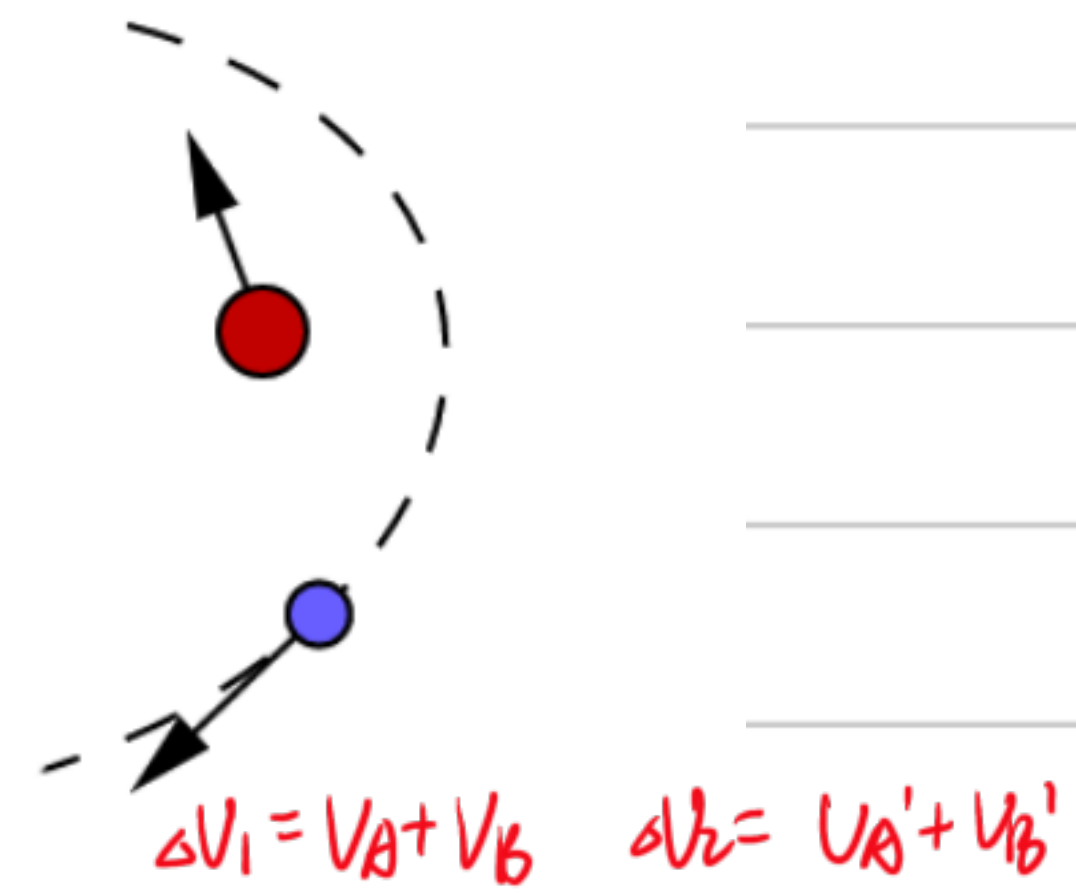
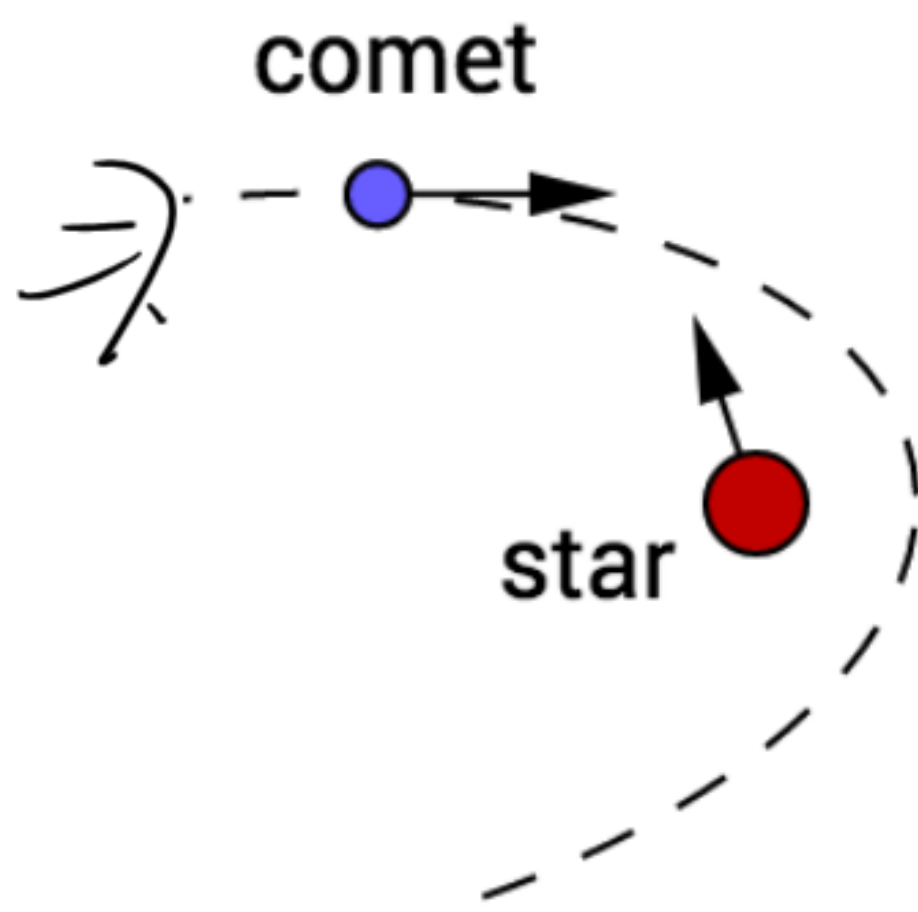
$$v_1' = \frac{(m_1 - m_2)v_1 + 2m_2 v_2}{m_1 + m_2}$$

$$v_2' = \frac{(m_2 - m_1)v_2 + 2m_1 v_1}{m_1 + m_2}$$



Collision – two objects interact (directly or indirectly) over a finite time-interval.

这个也是
collision



恢复系数 coefficient of restitution

$$e = \frac{|\Delta v_2|}{|\Delta v_1|}$$

$$m_A v_A^2 + m_B v_B^2 = m_A v_A'^2 + m_B v_B'^2$$

$$m_A v_A + m_B v_B = m_A v_A' + m_B v_B'$$

$$K_{\text{final}} = K_{\text{initial}} + Q$$

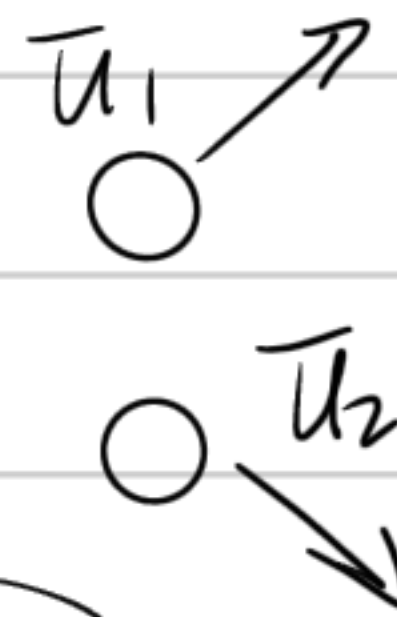
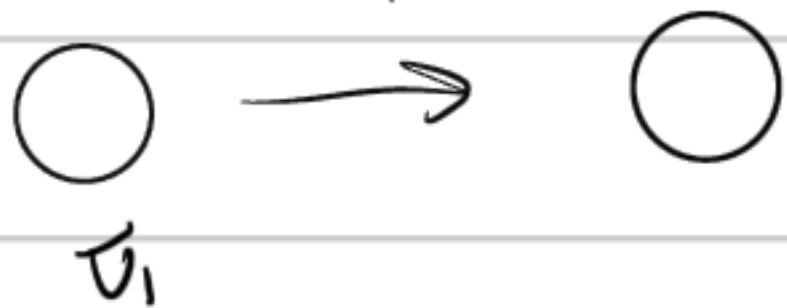
with Q

$$\begin{cases} = 0 & \text{(elastic)} & e=1 \\ > 0 & \text{(superelastic)} & e>1 \\ < 0 & \text{(inelastic)} & 0<e<1 \end{cases}$$

$$v_A + v_A' = v_B + v_B'$$

$$v_A - v_B = v_B' - v_A'$$

两个球相撞



$$\begin{cases} v_1 = u_1 + u_2 \\ v^2 = u_1^2 + u_2^2 \end{cases}$$



$$v^2 = u_1^2 + u_2^2 + 2u_1 \cdot u_2$$

release
[m]

Center of Mass

$$\bar{r}_{cm} = \frac{\sum_{i=1}^N m_i \bar{r}_i}{\sum_{i=1}^N m_i} \rightarrow \text{discrete distribution of mass}$$

Note. The sum is replaced by an integral for a continuous distribution of mass.

$$\rho(\bar{r}) = \alpha \bar{r}$$

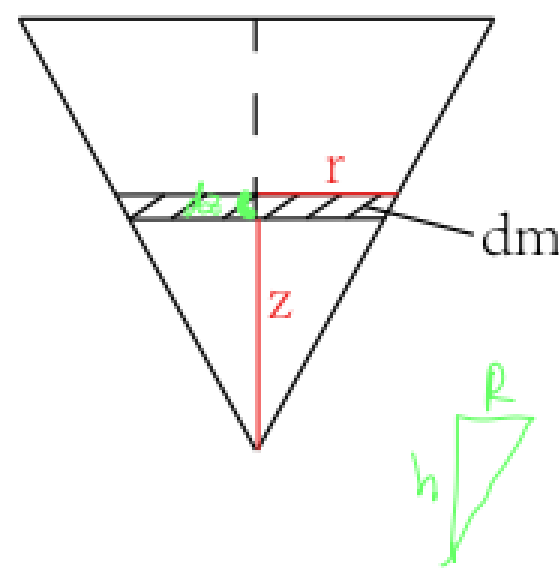
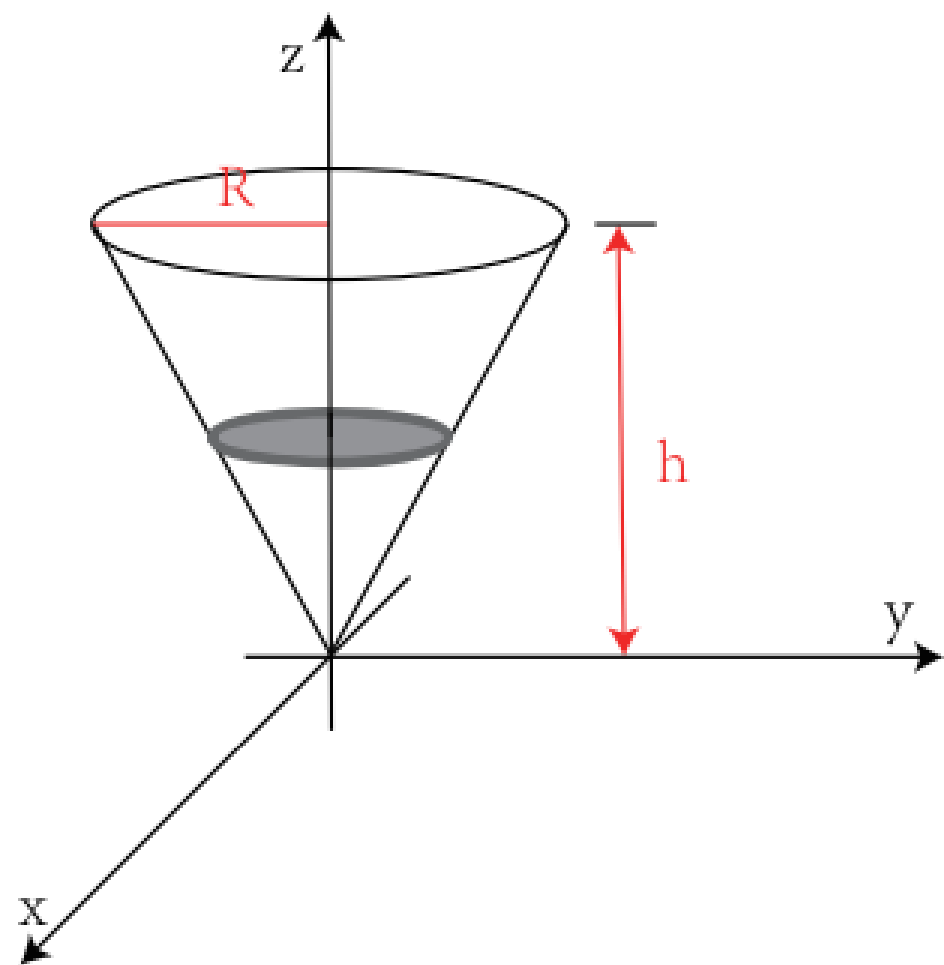
$$dm = \alpha \bar{r} dv$$

$$= \alpha \bar{r} d$$

$$dv = \frac{1}{2} \pi r^2 (\bar{r} + d\bar{r}) - \frac{1}{2} \pi \left(\frac{\bar{r}}{h} \right)^2 \bar{r}$$

$$= \frac{1}{3}$$

Find the center of mass of a non-uniform cone with $\rho(\vec{r}) = \alpha z$.



From symmetry: $x_{cm} = 0$, $y_{cm} = 0$. To find z_{cm} , divide the cone into infinitesimal slices of mass $dm = \rho(z) \pi r^2 dz$.

But $\frac{r}{R} = \frac{z}{h}$, hence $r = R \frac{z}{h}$ and $dm = \alpha z \pi R^2 \frac{z^2}{h^2} dz$. Hence

$$\int_{\text{cone}} dm = \int_0^h \alpha \pi \frac{R^2}{h^2} z^3 dz = \frac{1}{4} \alpha \pi R^2 h^2$$

$$\int_{\text{cone}} z dm = \int_0^h \alpha \pi \frac{R^2}{h^2} z^4 dz = \frac{1}{5} \alpha \pi R^2 h^3$$

Eventually

$$z_{cm} = \frac{\int_{\text{cone}} z dm}{\int_{\text{cone}} dm} = \frac{4}{5} h$$

$$M \bar{v}_{cm} = \sum_{i=1}^N \bar{p}_i = \bar{p}$$

$$\text{where } M = \sum_{i=1}^N m_i$$

Motion with varying mass

(ground as FoR)

$$\leftarrow \begin{matrix} V-u \\ 0 \\ -dm \end{matrix} \quad \bigtriangleright$$

$$dm < 0 \quad -dm > 0$$

$$mV = (m+dm)(V+dv) - dm(V-u)$$

$$\cancel{mV} = \cancel{mV} + m dv + \cancel{dmV} + \cancel{dm dv} - \cancel{V dm} + u dm$$

$$m \frac{dv}{dt} = -u \frac{dm}{dt}$$

$$a = -\frac{u}{m(t)} \frac{dm}{dt}$$

$$V(t) = V_0 + u \ln \frac{m_0}{m(t)}$$

u 要大

$\frac{m_0}{m(t)}$ 尽可能大

$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$|\vec{v}| = |\vec{\omega} \times \vec{r}| = |\vec{\omega}| \cdot |\vec{r}_\perp|$$

$$\vec{a}_{\text{tangential}} = \vec{\omega} \times \vec{r}_\perp \quad |\vec{a}_t| = |\vec{\omega}| |\vec{r}_\perp|$$

$$\vec{a}_{\text{centripetal}} = \vec{\omega} \times (\vec{\omega} \times \vec{r}) = \vec{\omega} \times (\underbrace{\vec{\omega} \times \vec{r}}_{\perp}) =$$

$$|\vec{a}_{\text{centri}}| = |\vec{\omega}|^2 |\vec{r}_\perp|$$

kinetic energy of a Rotating Rigid Body

$$K = \sum_{i=1}^N K_i = \sum_{i=1}^N \frac{1}{2} m_i v_i^2 \rightarrow \text{discrete}$$

$$K = \sum_{i=1}^N \frac{1}{2} m_i \omega^2 r_{i\perp}^2 = \frac{1}{2} \left(\sum_{i=1}^N m_i r_{i\perp}^2 \right) \omega^2 = \boxed{\frac{1}{2} I \omega^2}$$

where $I = \sum_{i=1}^N m_i r_{i\perp}^2$ is the **moment of inertia** about the (fixed) axis of rotation A.

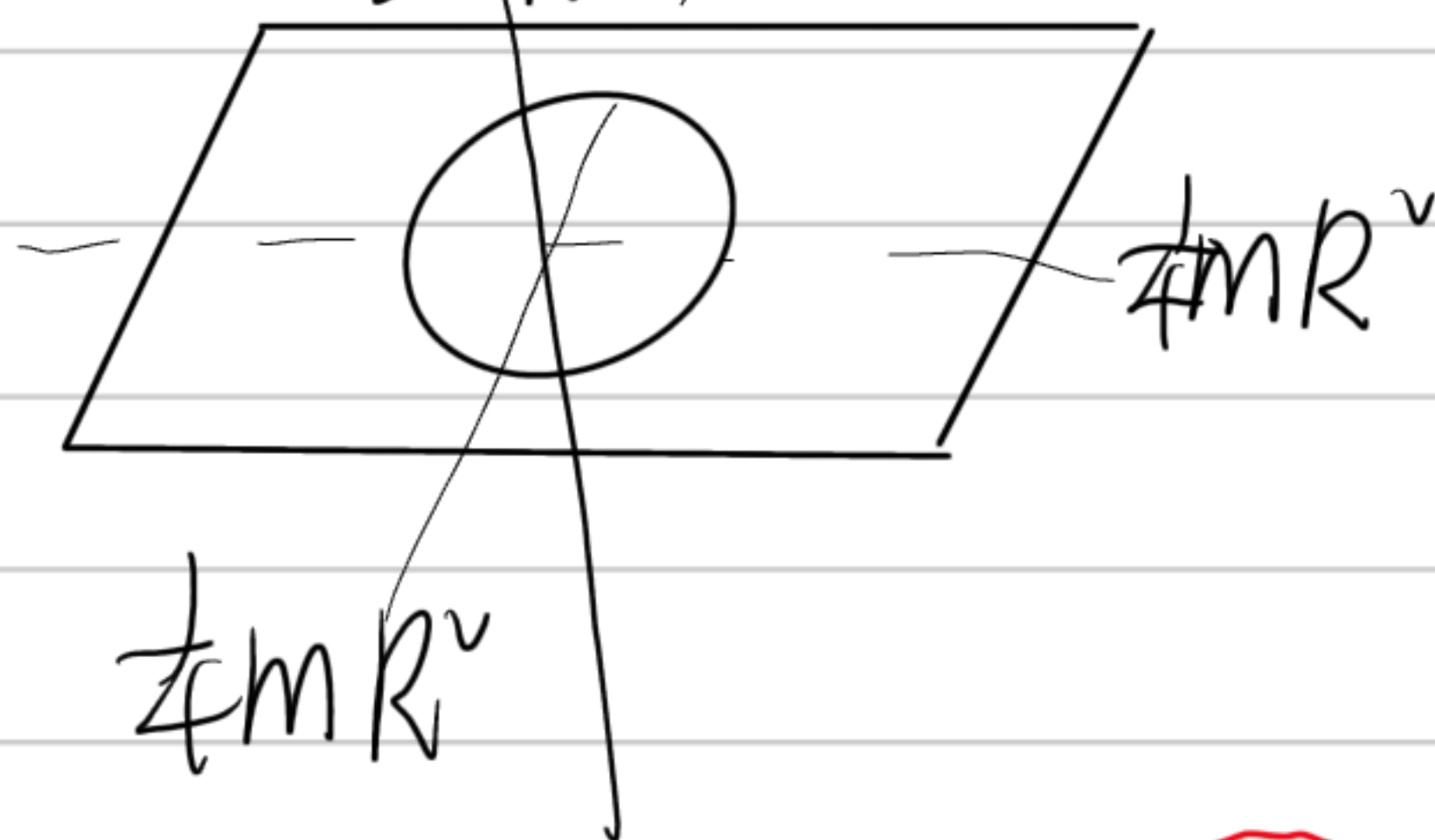
$$K = \int_{\text{object}} dk = \int_{\text{object}} \frac{1}{2} \omega^2 r_\perp^2 dm = \frac{1}{2} \left(\int_{\text{object}} r_\perp^2 dm \right) \omega^2$$

$$\boxed{I_A = \int_{\text{object}} r_\perp^2 dm}$$

Perpendicular

Axis

Theorem

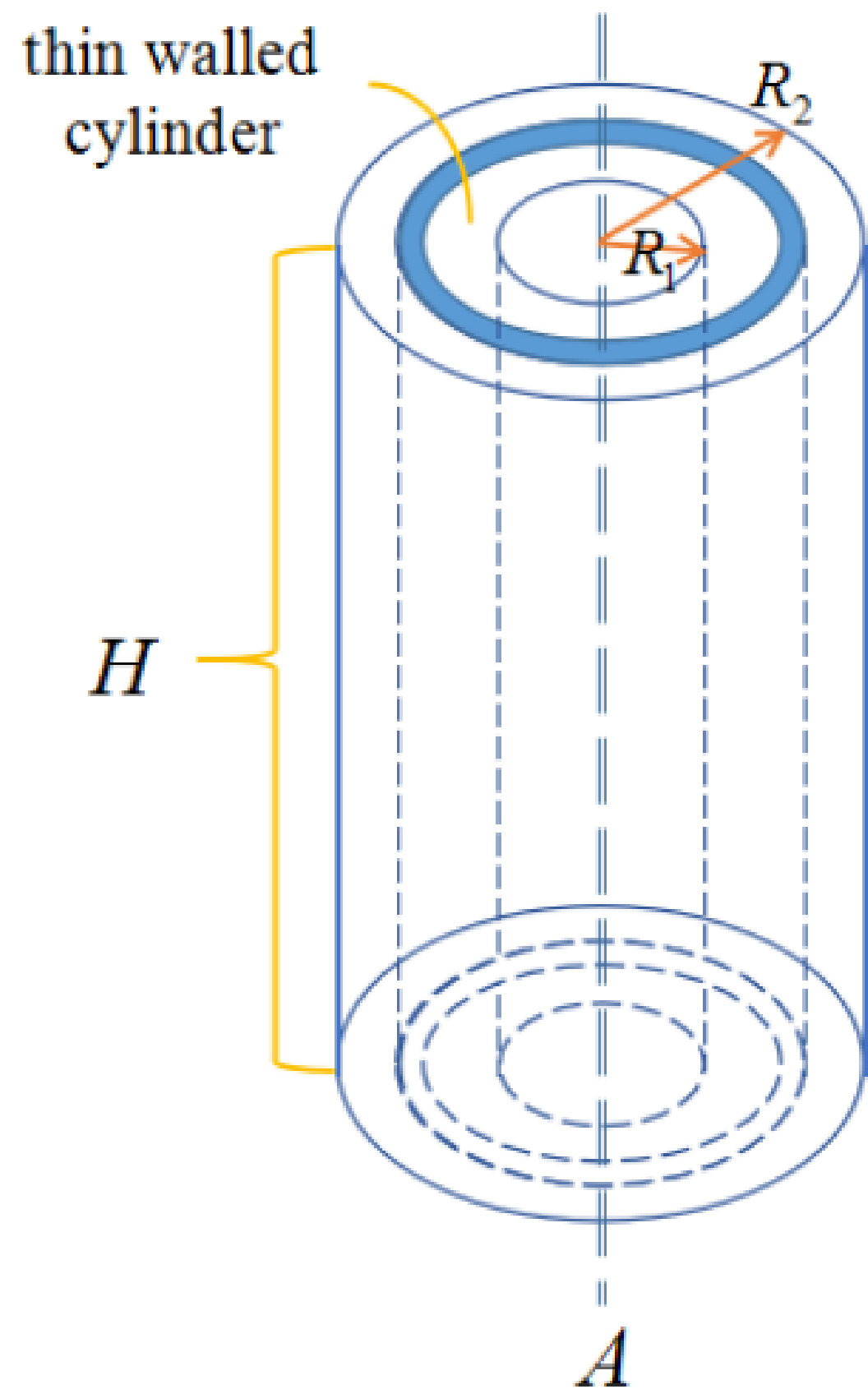


$$I_z = I_x + I_y$$



$$= \frac{1}{2} m R^2$$

for uniform hollow cylinder

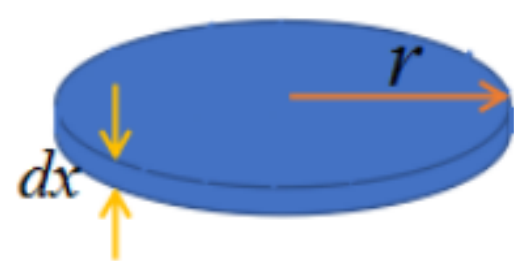
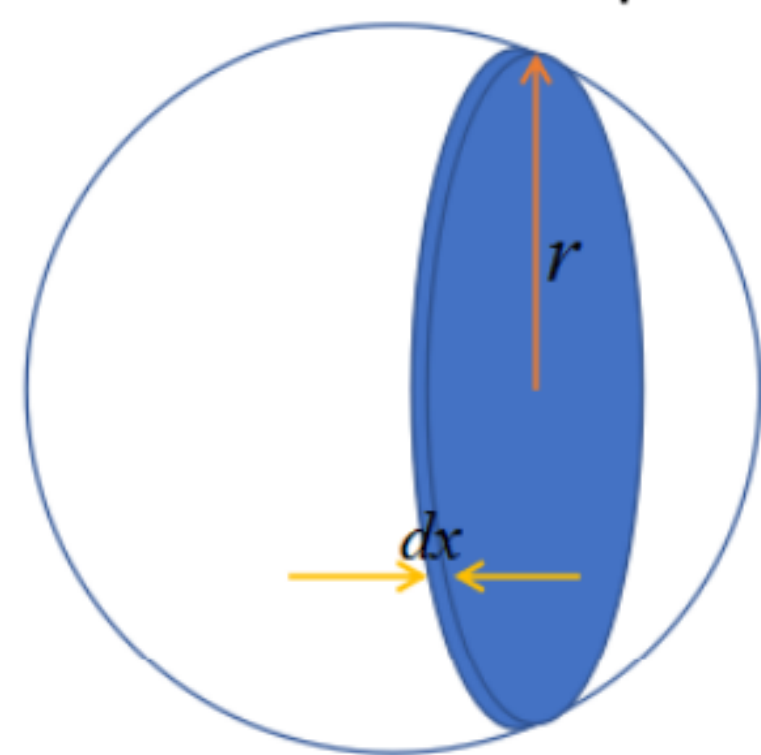


$$I_A = \frac{\pi \rho H}{2} (R_2^2 - R_1^2)(R_2^2 + R_1^2) = \frac{1}{2} m (R_2^2 + R_1^2)$$

转动惯量: additive and quantitative.

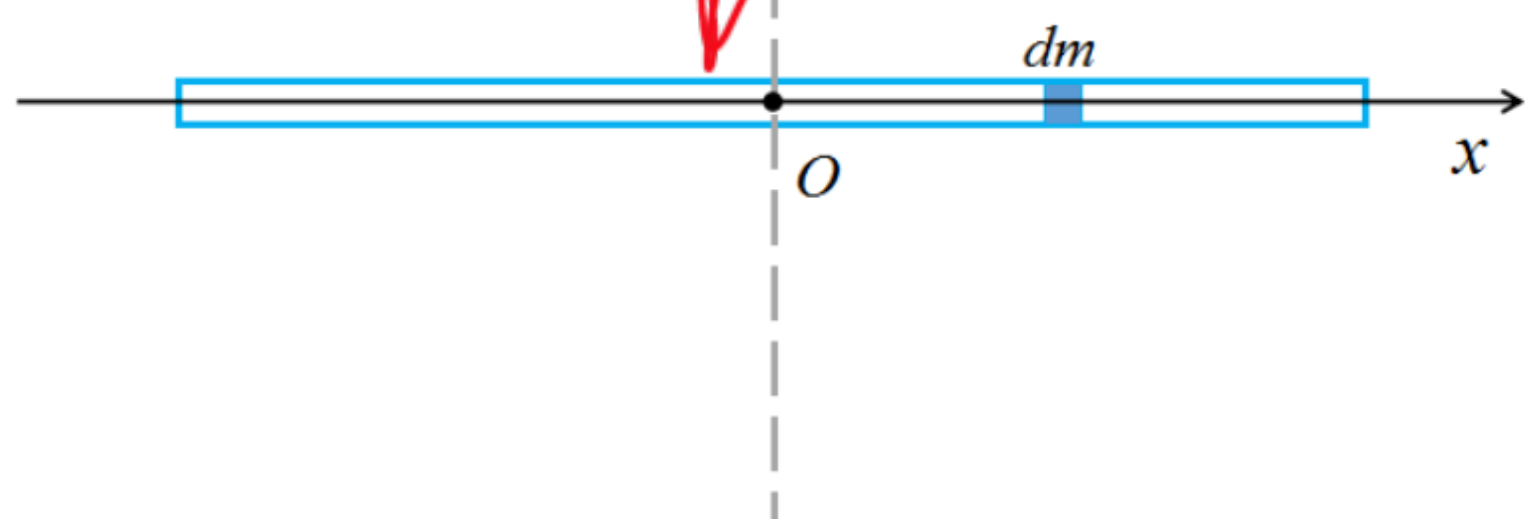
$$\int_{\text{object}} r^2 dm = \int_{\text{object}} r^2 dm + \int_{\text{object}_2} r^2 dm$$

关于球体:

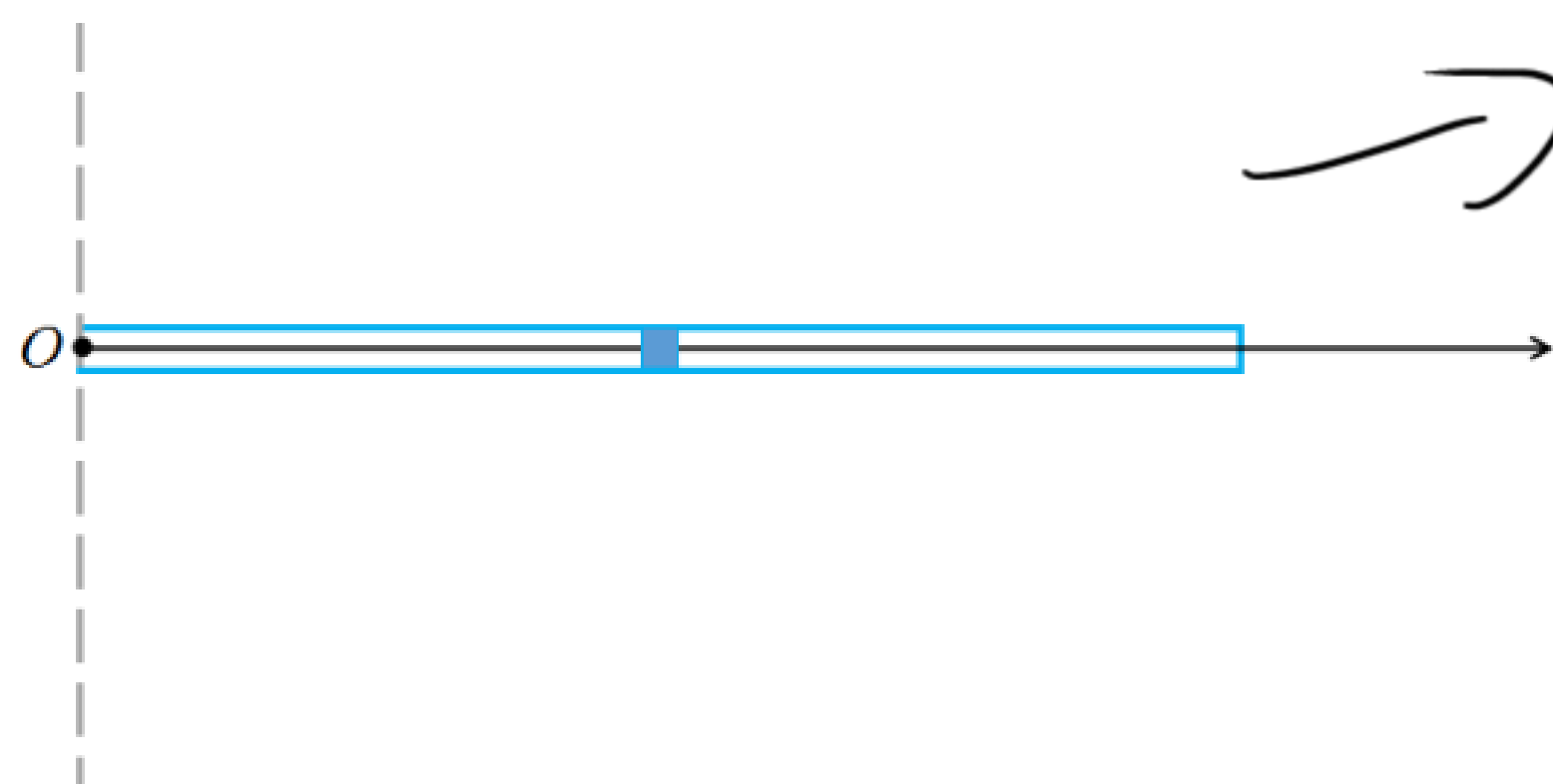


$$I_A = \frac{2}{5} m R^2$$

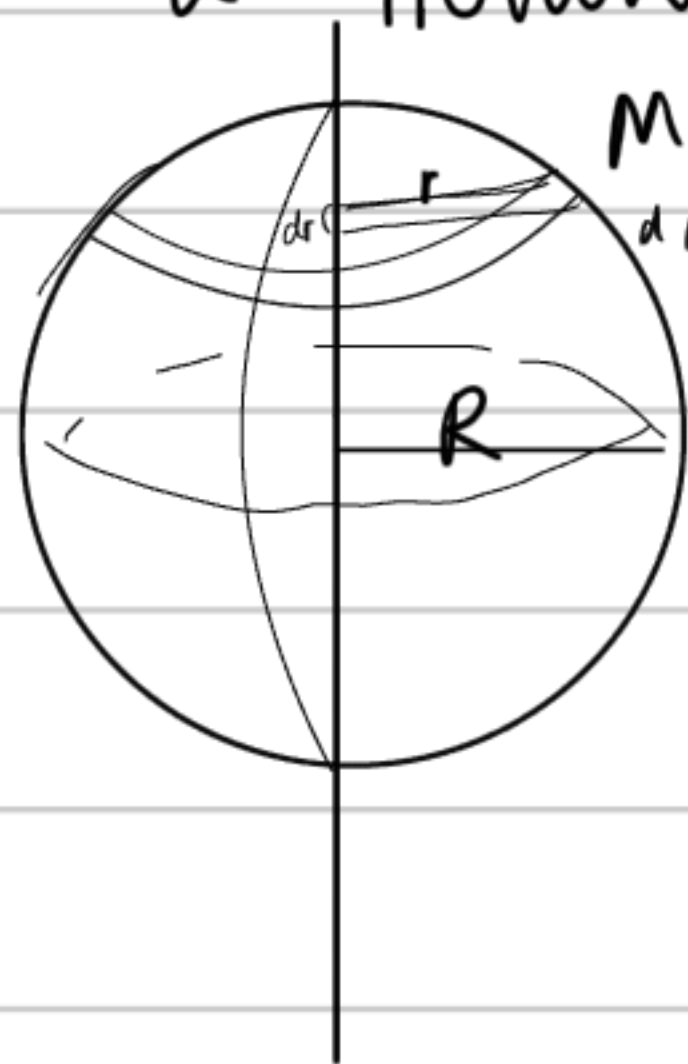
center of mass $\rightarrow I_A = \frac{1}{12} m L^2$



$$I_{A'} = \frac{1}{3} m L^2 = I_A + m \left(\frac{L}{2}\right)^2$$



for a hollow spherical shell



$$dM = \lambda 2\pi r dr$$

$$\lambda = \frac{M}{4\pi R^2}$$

$$\int r^2 dm$$

$$= \lambda 2\pi \int_0^R r^3 dr$$

$$= \lambda 2\pi \left(\frac{1}{4} r^4 \right)_0^R$$

$$= \frac{1}{4} R^4 \lambda 2\pi = \frac{1}{4} R^4 \frac{M}{4\pi R^2}$$

$$= \frac{RM}{4}$$

薄圆盘和圆柱体： $\frac{1}{2}MR^2$

Assume you get a summer job as a crane operator. On the first day you are lifting a heavy piece of machinery. Even though you have the arm at 70° above the horizontal, the crane begins to tip slowly forward. Consider the following possible actions:

1. Release the brake on the lifting cable so that the load accelerates downward.
2. Release the lifting arm so that θ decreases rapidly and the load accelerates downward.
3. Increase θ while simultaneously letting out the lifting cable so that the load accelerates downward.
4. Put the crane wheels in gear and accelerate the crane forward.

None of these solutions is ideal, but which will have the short-term effect of restoring contact of the crane's rear wheels with the ground?

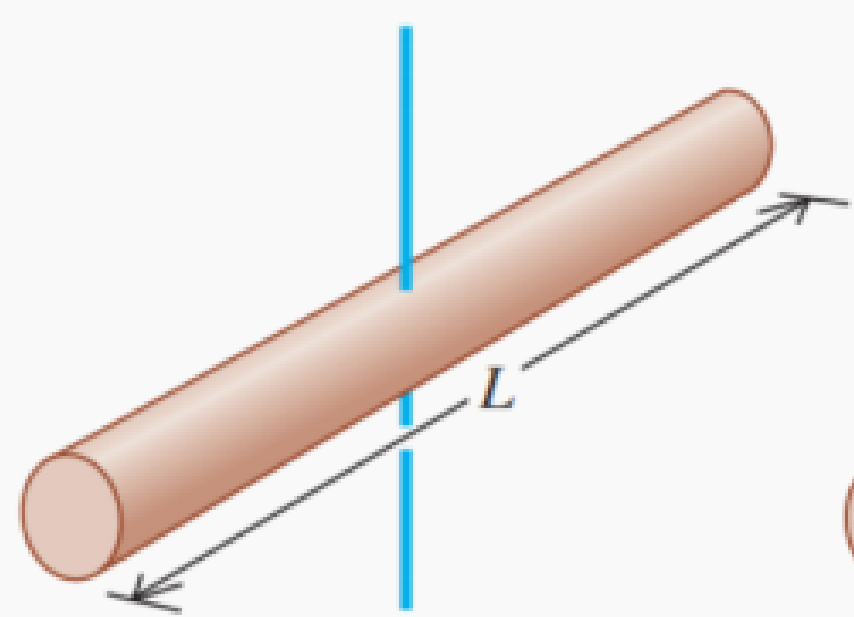
- ☐ all but 1
- ☐ all but 2
- ☐ all but 3
- ☐ all but 4
- ☒ all of them

都正确

TABLE 9.2 Moments of Inertia of Various Bodies

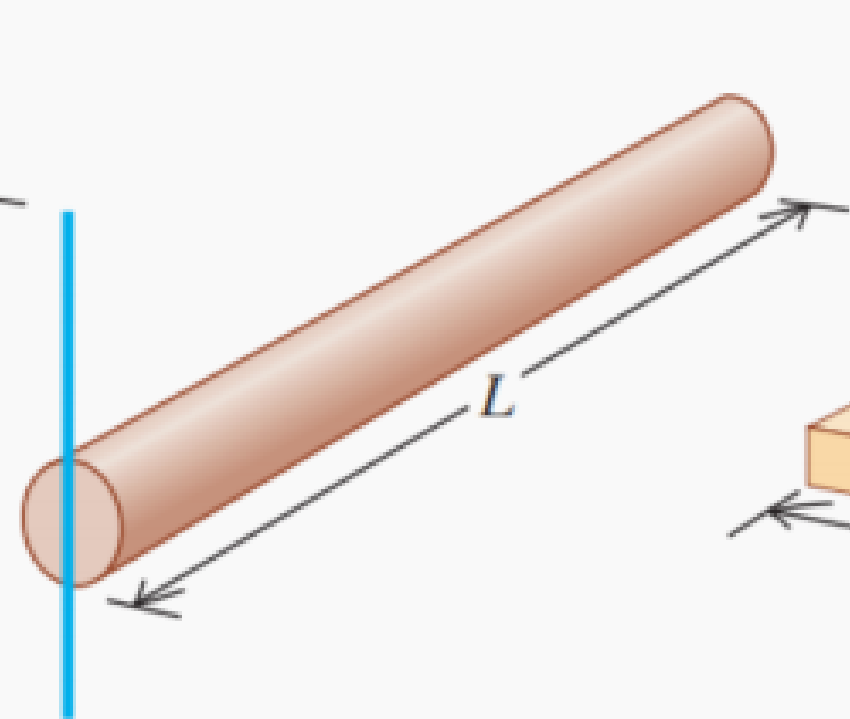
(a) Slender rod,
axis through center

$$I = \frac{1}{12}ML^2$$



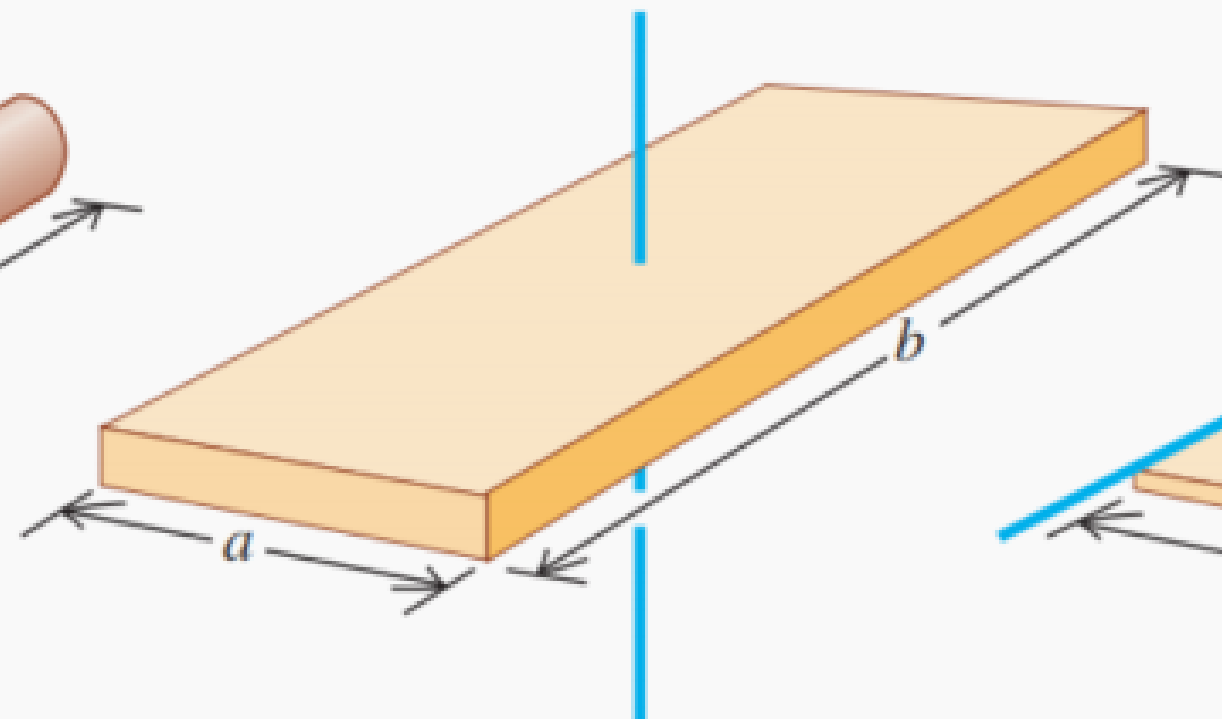
(b) Slender rod,
axis through one end

$$I = \frac{1}{3}ML^2$$



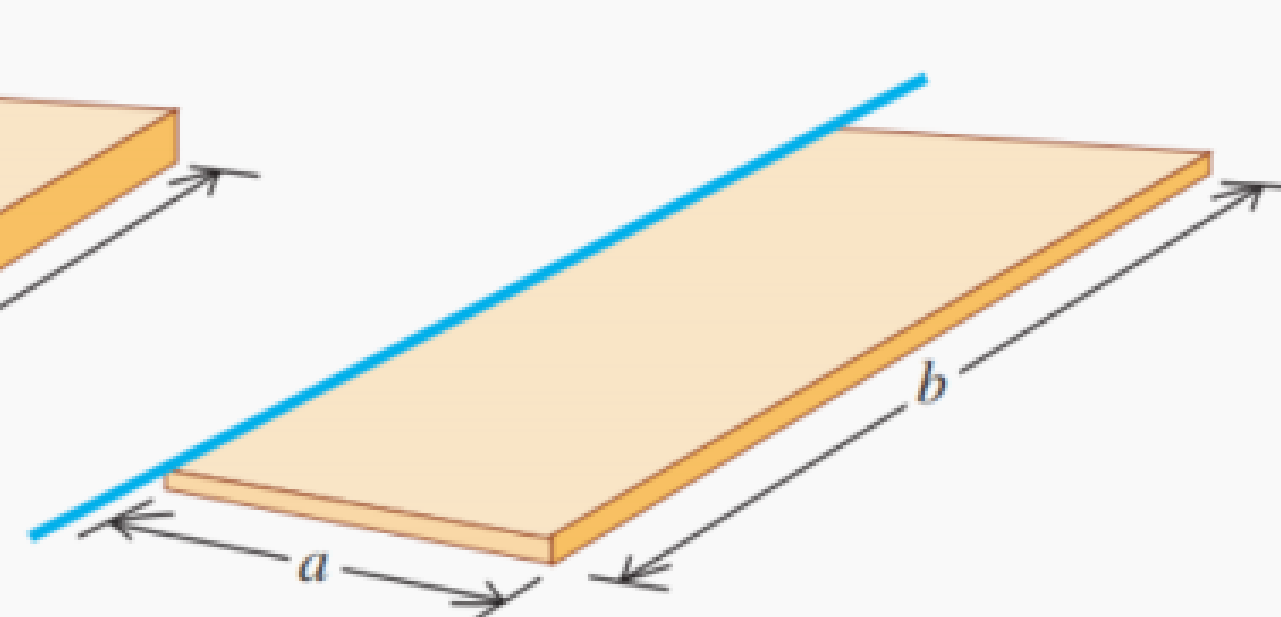
(c) Rectangular plate,
axis through center

$$I = \frac{1}{12}M(a^2 + b^2)$$



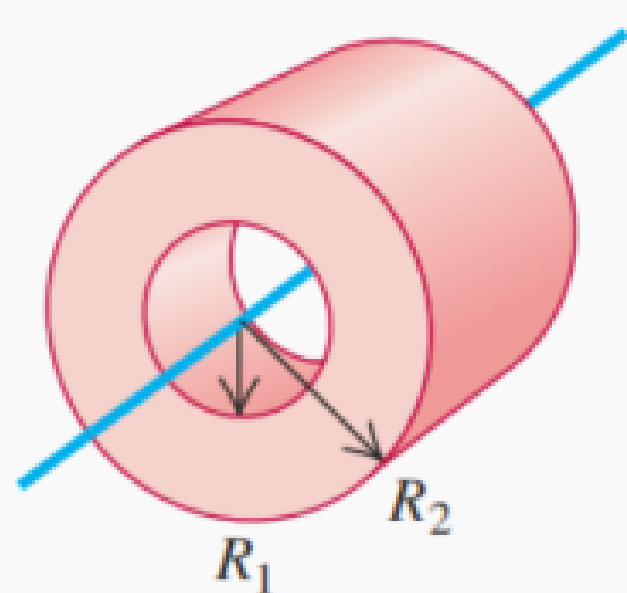
(d) Thin rectangular plate,
axis along edge

$$I = \frac{1}{3}Ma^2$$



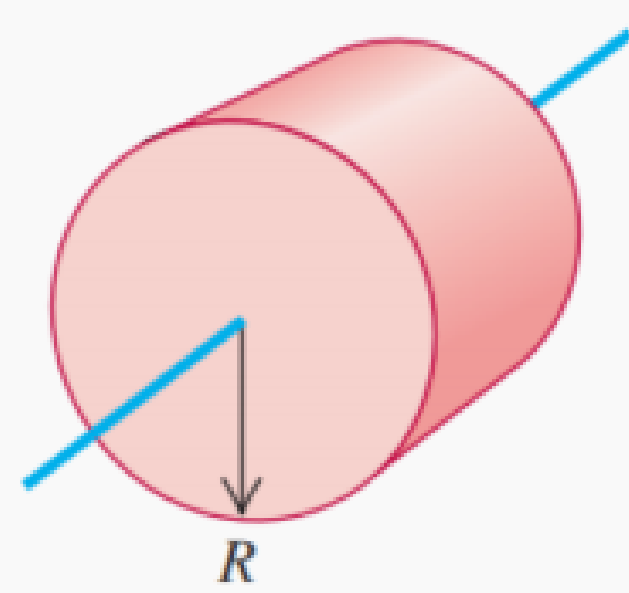
(e) Hollow cylinder

$$I = \frac{1}{2}M(R_1^2 + R_2^2)$$



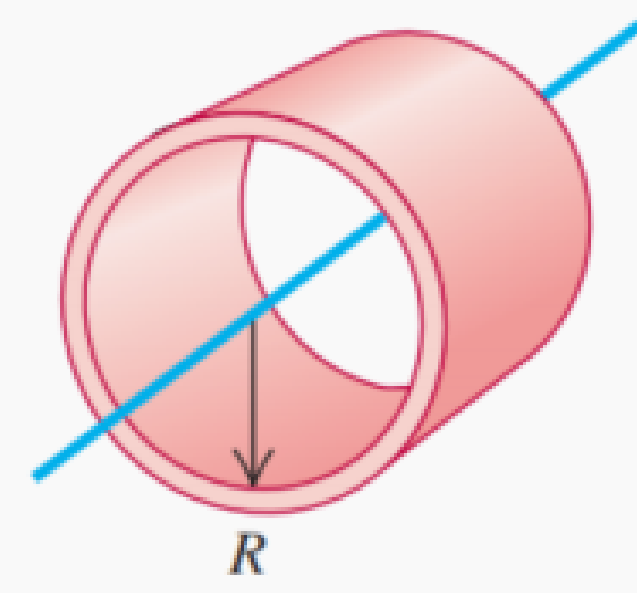
(f) Solid cylinder

$$I = \frac{1}{2}MR^2$$



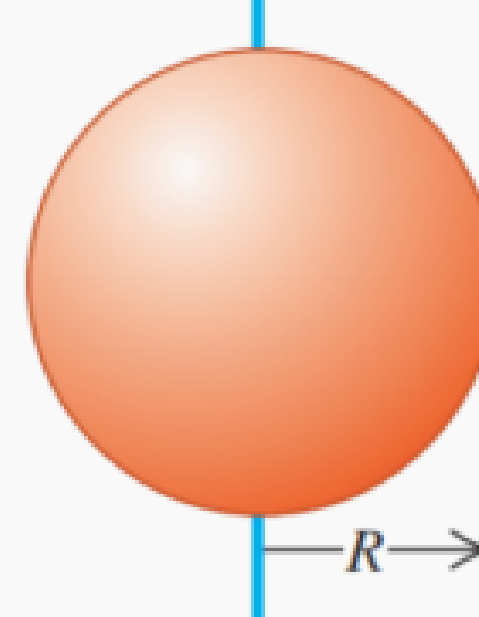
(g) Thin-walled hollow
cylinder

$$I = MR^2$$



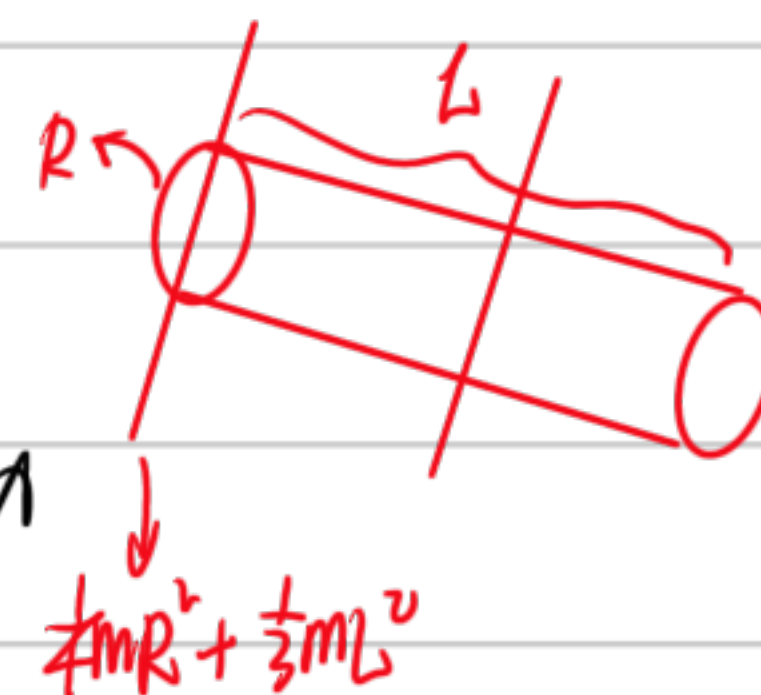
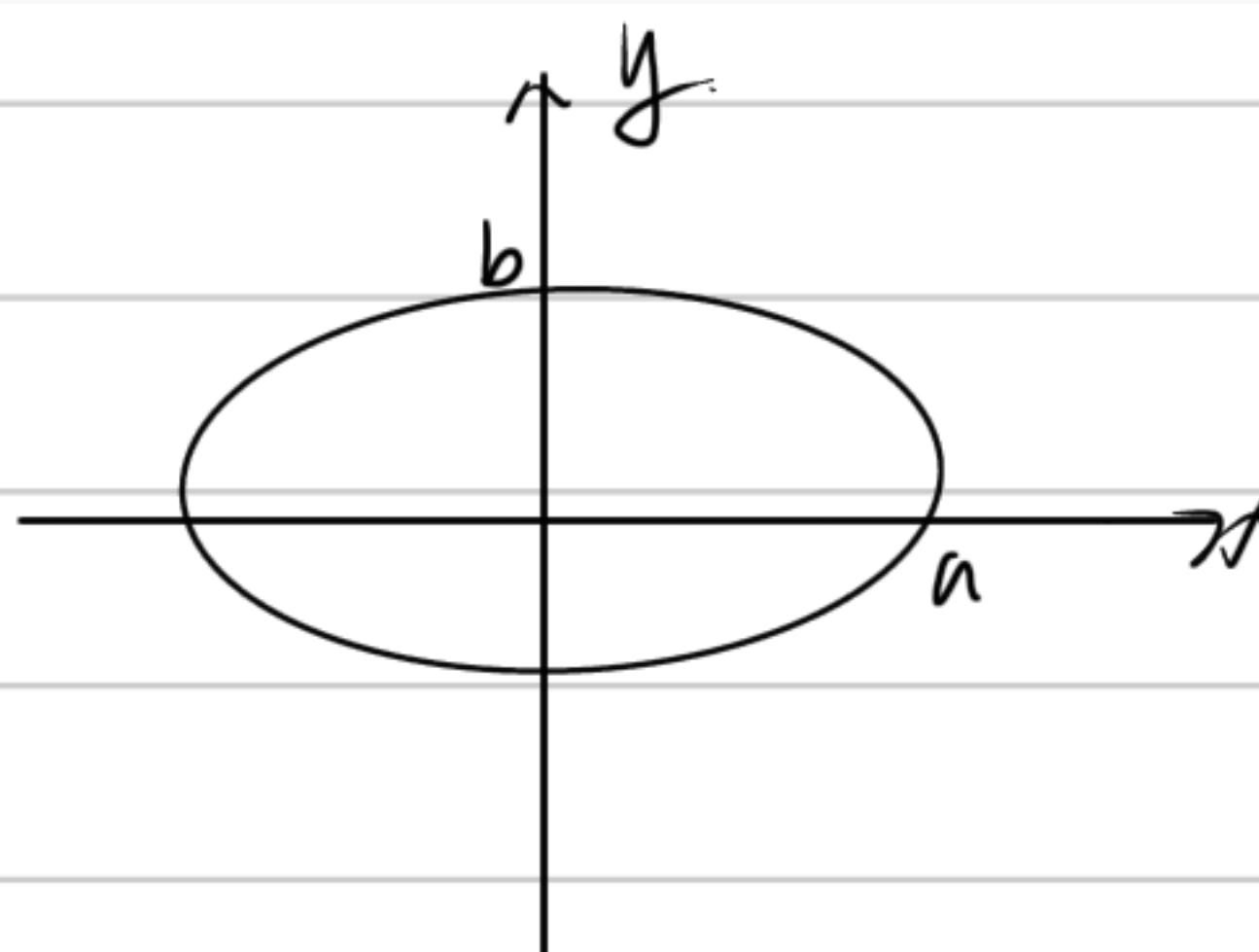
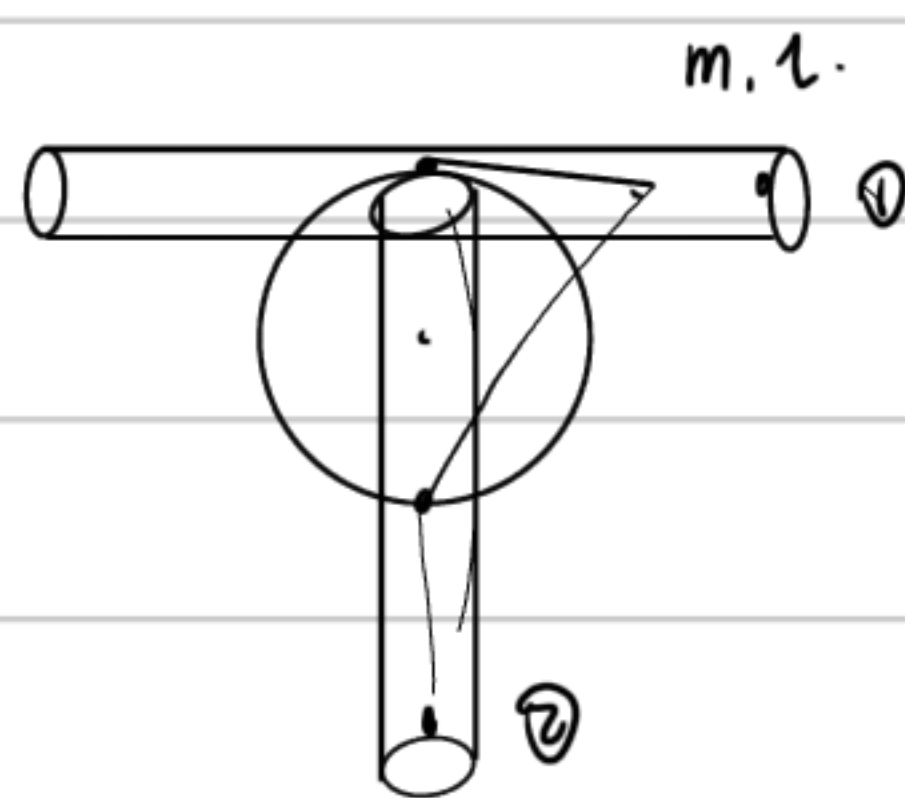
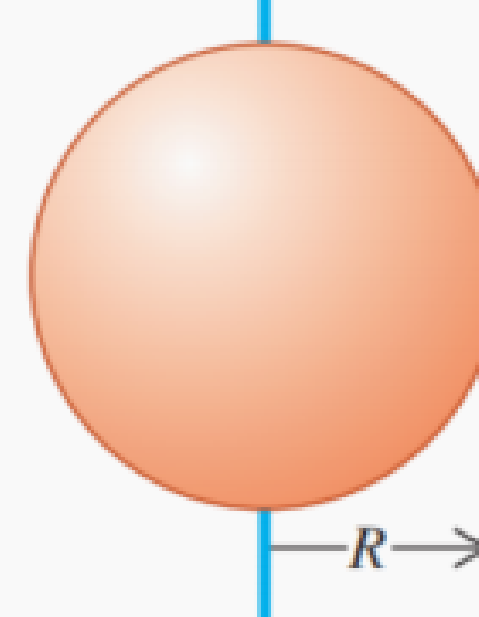
球体 $\frac{2}{5}MR^2$
(h) Solid sphere

$$I = \frac{2}{5}MR^2$$



球壳 $\frac{2}{3}MR^2$
(i) Thin-walled hollow
sphere

$$I = \frac{2}{3}MR^2$$

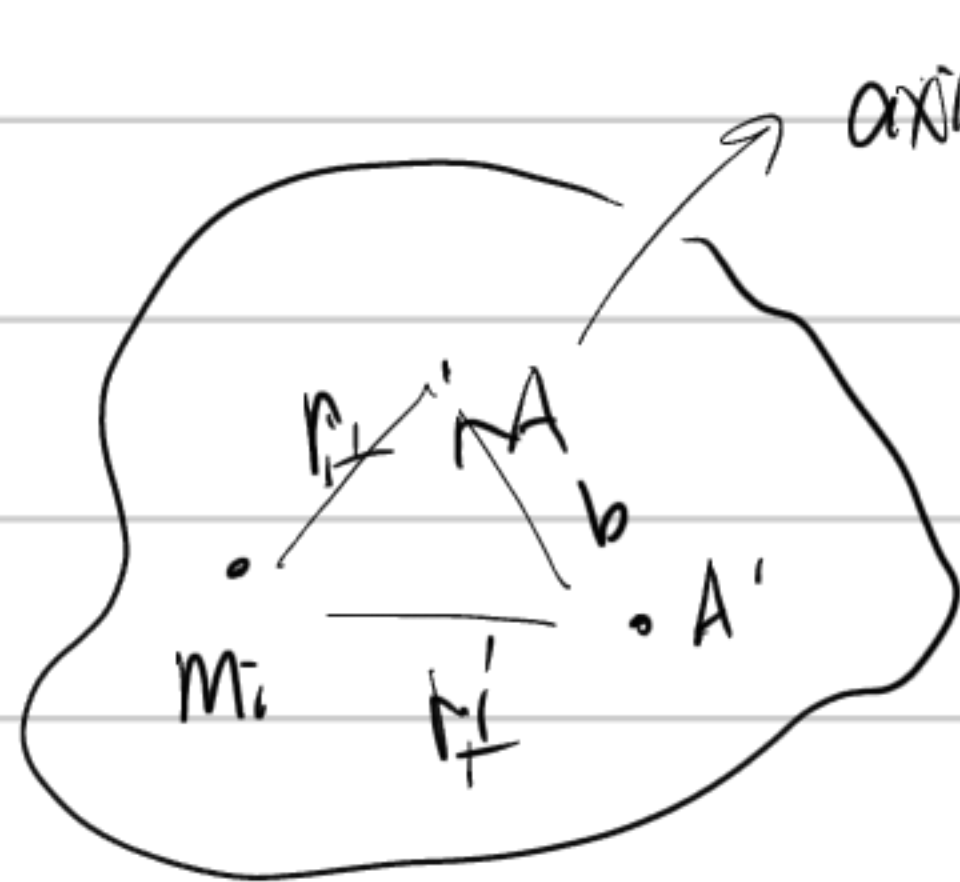


$$I_a = I_x = \int dm y^2$$

$$I_b = I_y = \int dm x^2$$

$$\frac{I_a}{a^2} + \frac{I_b}{b^2} = \frac{\int dm y^2}{a^2} + \frac{\int dm x^2}{b^2} = \int dm \cdot 1 = \int dm = m.$$

Parallel Axis Theorem (Steiner's Theorem)



axis through center

$$I_{A'} = I_A + mb^2$$

$$I_{A'} = \sum m_i r_{iA'}^2 = \sum m_i (r_{iA} + \vec{b})^2 = I_A + mb^2 + 2 \sum m_i \vec{r}_{iA} \cdot \vec{b}$$

$$\Downarrow$$

$$= m \vec{r}_{cm} = 0$$

Gravitational Potential Energy of a Rigid Body

$$U_{grav} = \sum_{i=1}^N m_i g z_i = g \sum_{i=1}^N m_i z_i = Mg z_{cm}$$

$$z_{cm} = \frac{\sum_{i=1}^N m_i z_i}{\sum_{i=1}^N m_i}$$

Rigid Body Mechanics (II) Dynamics

$$\vec{\tau} = \vec{r} \times \vec{F}$$

Second Law of Dynamics

$$\tau_z = I_z \epsilon_z$$

(ϵ_z is axis)

$$\Downarrow$$

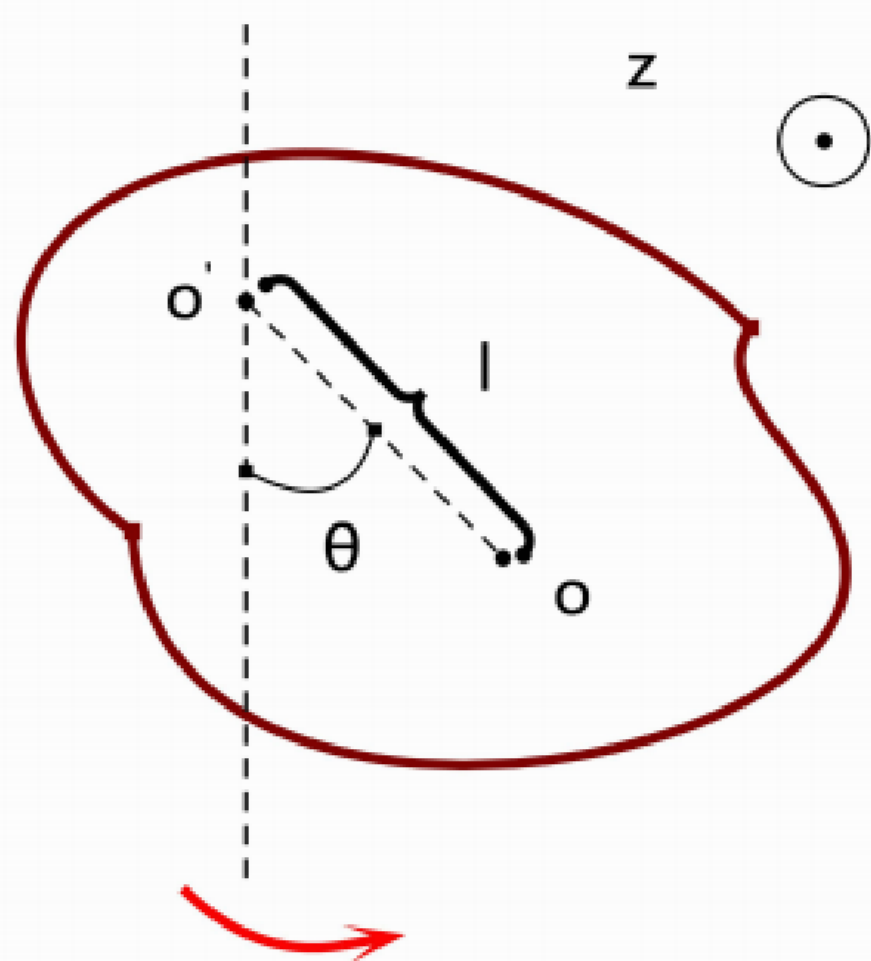
$$F_r = m a_x$$

$$\begin{cases} a_z = \frac{mg}{(m + \frac{1}{2}M)R} \\ a_y = \frac{m}{m + \frac{1}{2}M} g \\ T = \frac{Mmg}{2m + M} \end{cases}$$

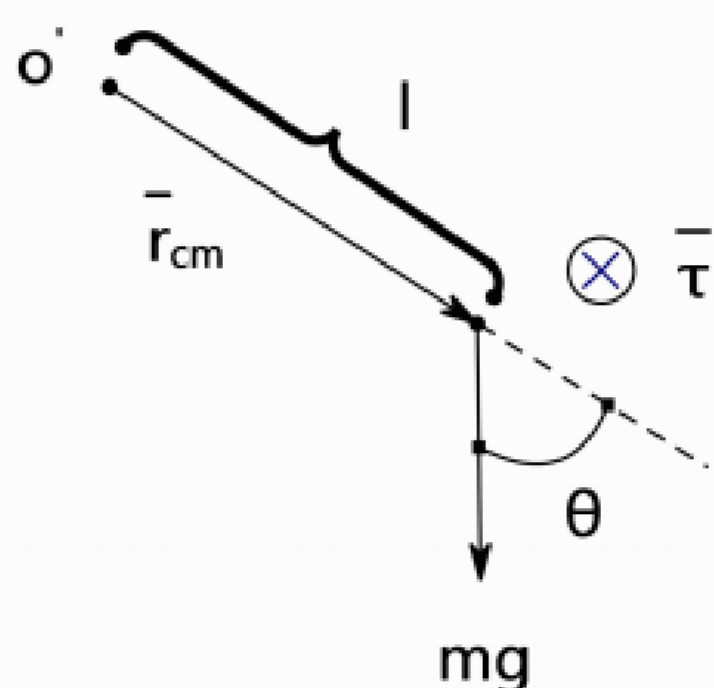
$$\tau = -mgL \sin \theta = I_{O'} \ddot{\theta}$$

$$\ddot{\theta} + \frac{mgL}{I_{O'}} \theta = 0$$

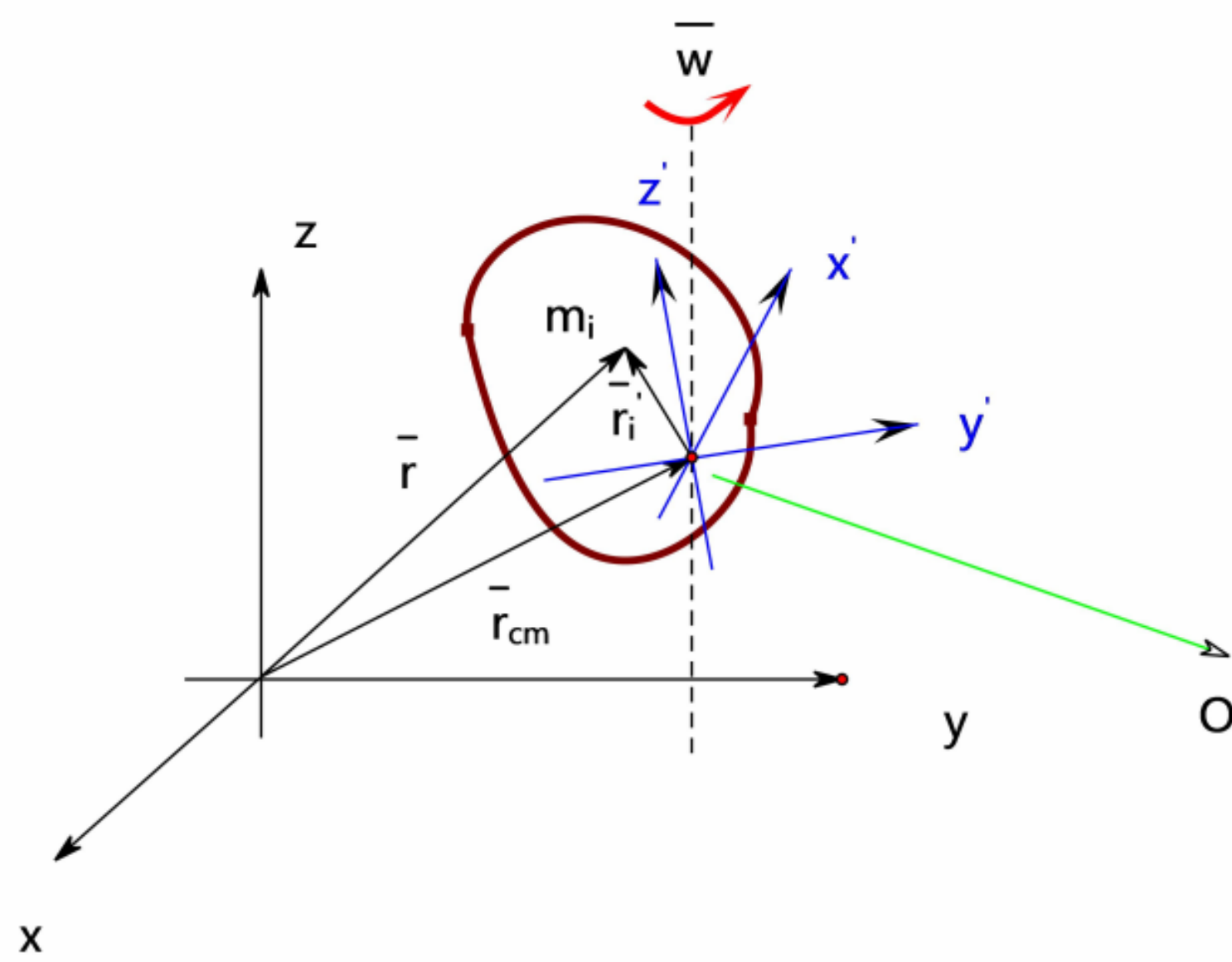
$$I_{O'} = I_O + mL^2$$



≡



Rigid Body Rotation About a Moving Axis

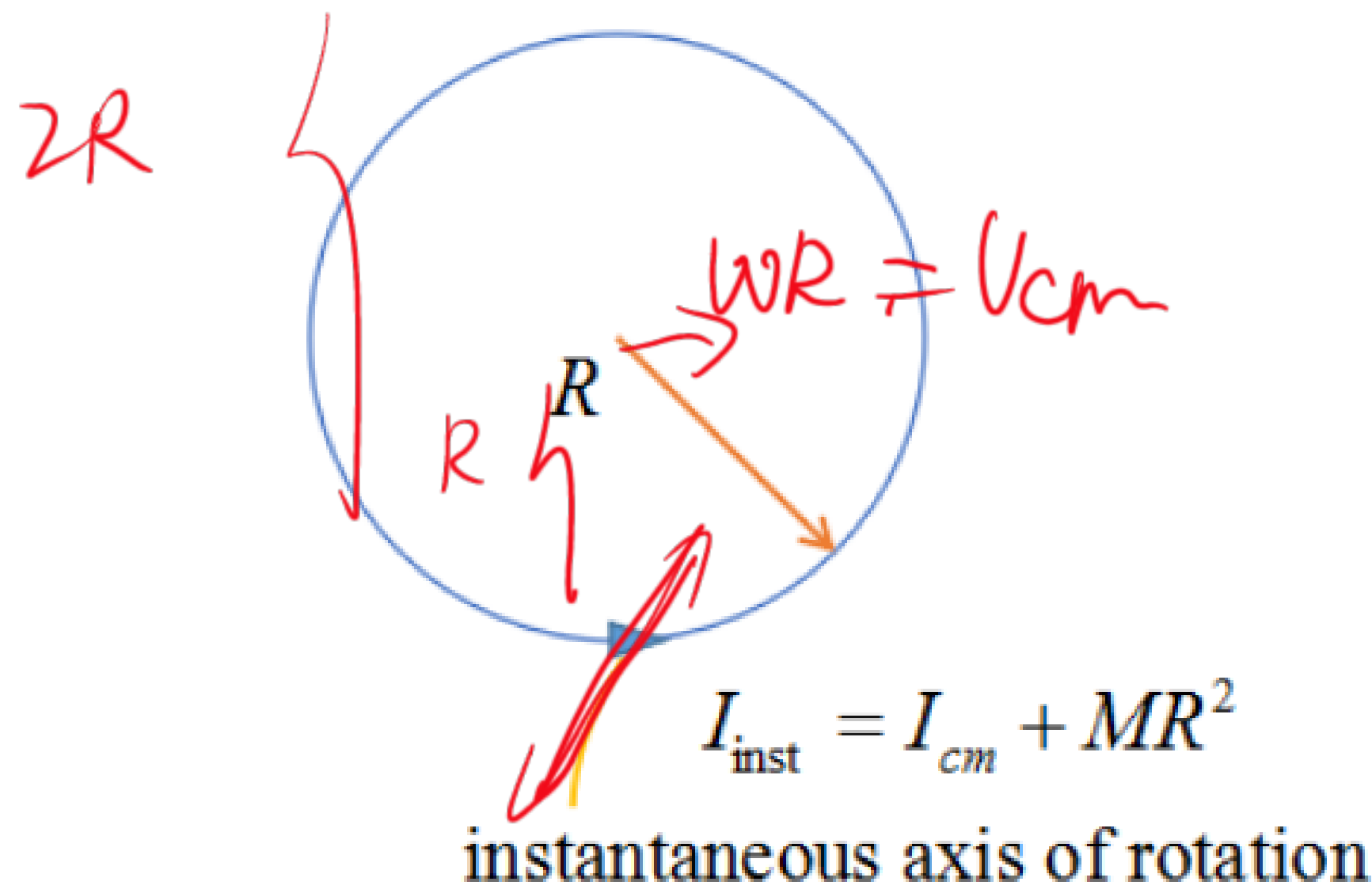


Velocity of the element of mass m_i

$$\begin{aligned}\bar{v}_i &= \bar{v}_{cm} + \bar{\omega} \times \bar{r}'_i \\ &= \bar{v}_{cm} + \bar{\omega} \times \bar{r}'_{i\perp}\end{aligned}$$

Origin placed at the center of mass

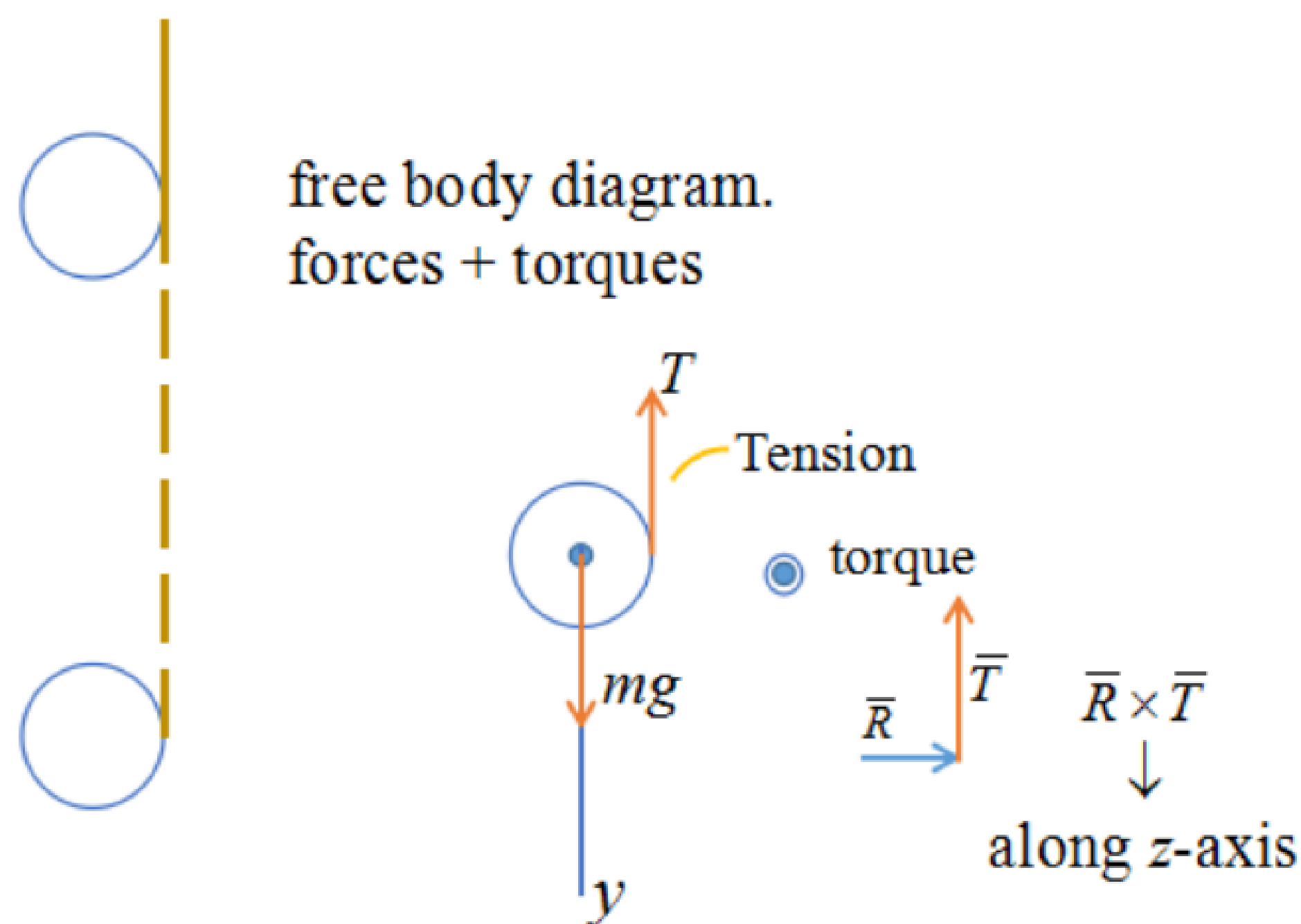
$$K = \frac{1}{2} M v_{cm}^2 + \frac{1}{2} I_{cm} \omega^2$$



$$\begin{aligned}K &= \frac{1}{2} I_{inst} \omega^2 = \frac{1}{2} (I_{cm} + MR^2) \omega^2 = \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} M (\omega R)^2 \\ &= \frac{1}{2} I_{cm} \omega^2 + \frac{1}{2} M v_{cm}^2\end{aligned}$$

$\omega R = v_{cm}$

Example (a): Thread Unwinding From Cylinder



$$(1) \quad mg - T = ma_{cm,y}$$

$$(2) \quad TR = I_{cm}\epsilon_z = \frac{1}{2}mR^2\epsilon_z$$

$$(3) \quad a_{cm,y} = R\epsilon_z \quad (\text{no slipping; follows from } v_{cm,y} = \omega_z R)$$

$$TR = \frac{1}{2}mR^2\epsilon$$

From (2) and (3): $T = \frac{1}{2}mR\epsilon_z = \frac{1}{2}ma_{cm,y}$
 From (1) and the above: $mg = \frac{3}{2}ma_{cm,y}$

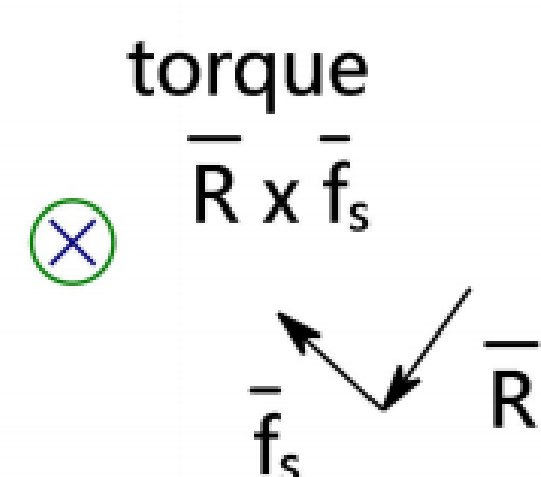
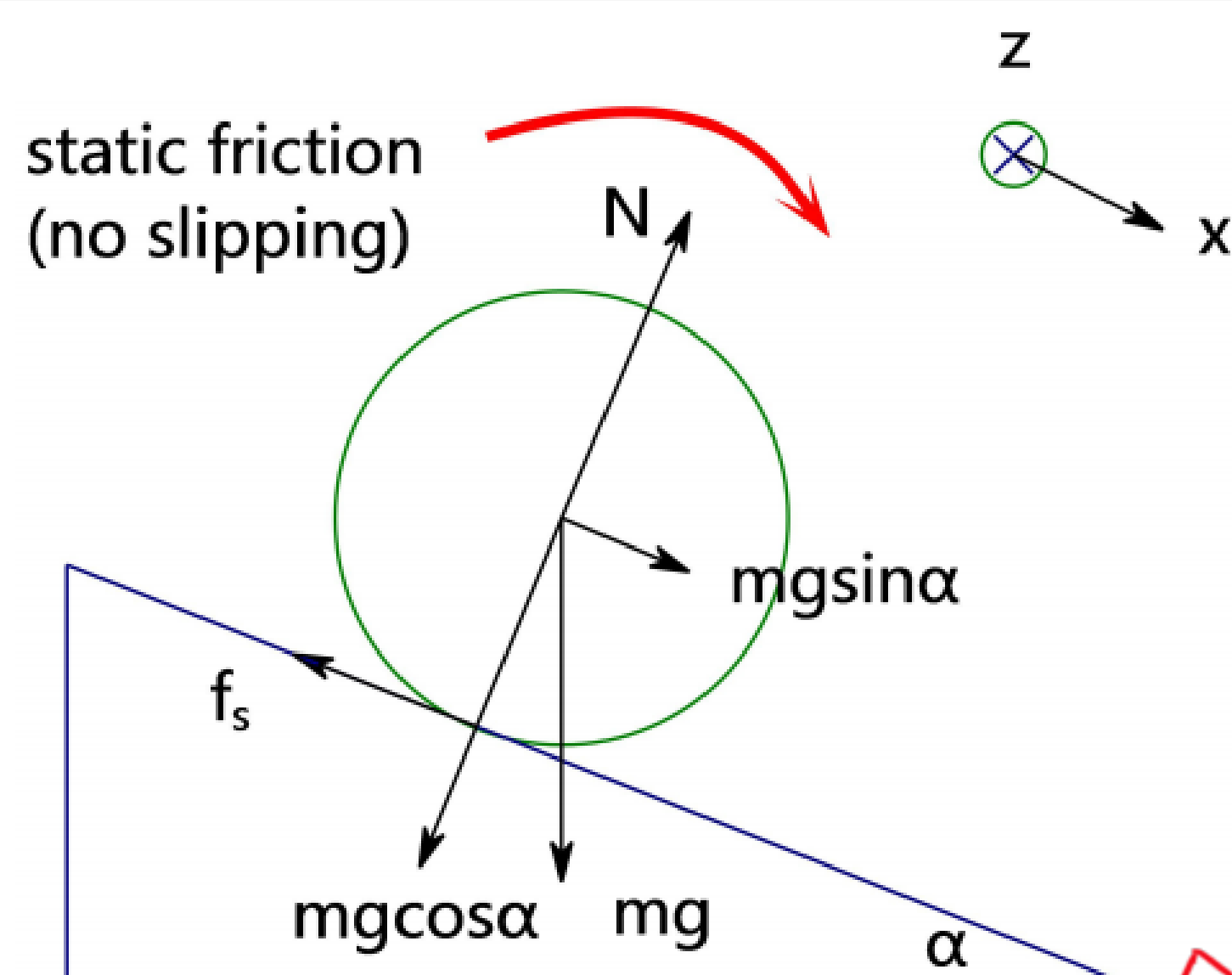
$$T = \frac{1}{3}ma_{cm,y}$$

These two equations yield

$$a_{cm,y} = \frac{2}{3}g$$

$$T = \frac{1}{3}mg$$

Example (b): Sphere Rolling Down an Incline (no slipping)



$$f_s = \frac{2}{5}ma_{cm,x}$$

$$f_s = Cma_{cm,x} \quad (m + Cm)a_{cm,x} = mg \sin \alpha$$

$$\begin{cases} ma_{cm,x} = mg \sin \alpha - f_s \\ f_s R = I_{cm}\epsilon_z = \frac{2}{5}mR^2\epsilon_z \\ a_{cm,x} = \epsilon_z R \end{cases}$$

斜面上球体下落加速度快于柱体

f_s 是否做功? $\times$ f_s 改变平动动能, 但增加了转动动能, 故不做功!!!

Solving the system, one can find

$$a_{cm,x} = \frac{5}{7}g \sin \alpha$$

$$f_s = \frac{2}{7}mg \sin \alpha$$

合外力做功改变平动动能!

Comments

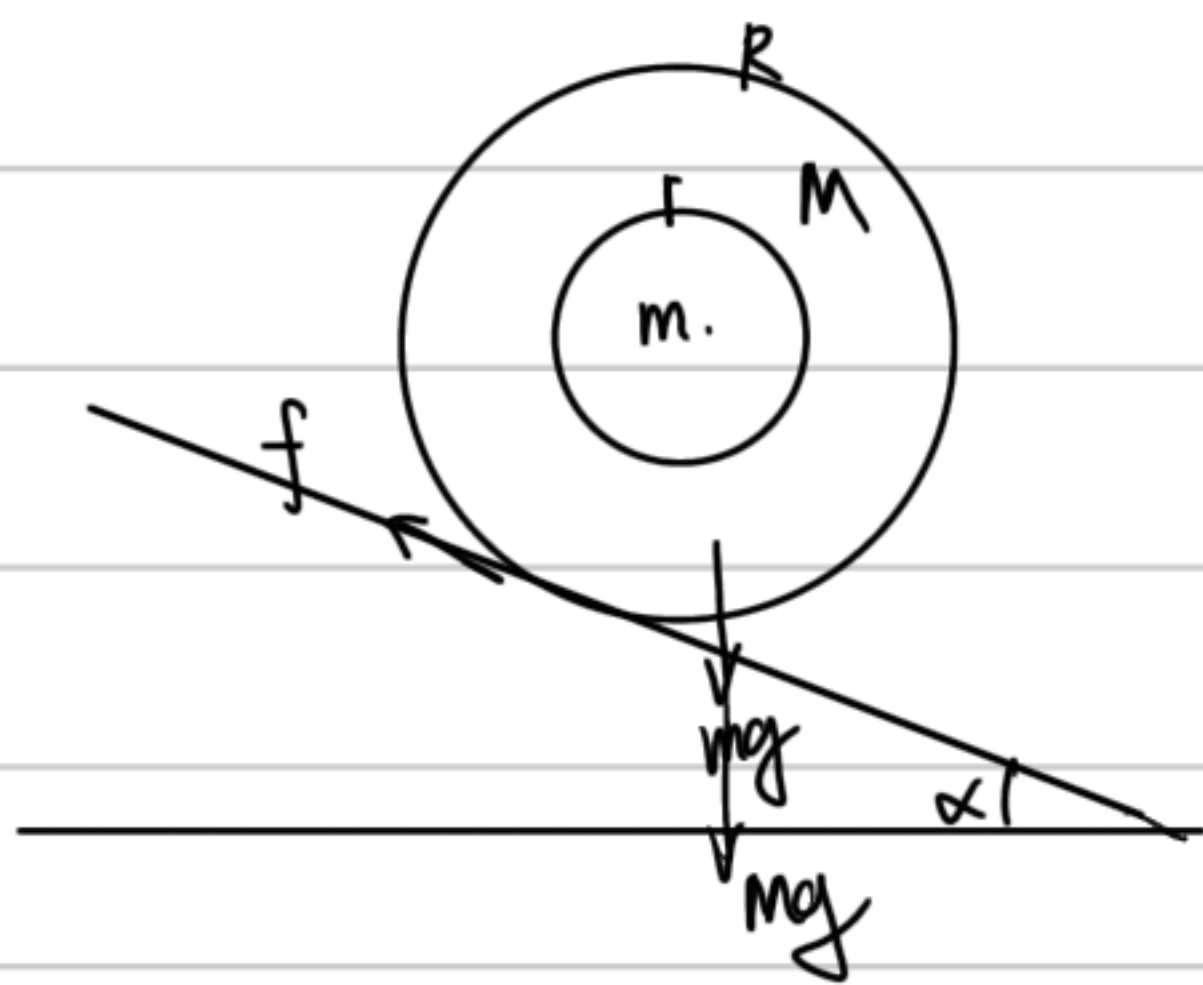
$$a_{cm,x} = \frac{g \sin \alpha}{1 + C}$$

for ball: $C = \frac{2}{5}$ for sphere $C = \frac{2}{3}$
 for cylinder $C = \frac{1}{2}$ $a_{cm,x}$

* Friction still in the same direction for a ball rolling uphill.

* In our examples we always neglect rolling friction (i.e. we assume perfect rigidity). Example: car tyre.





$$f = \frac{1}{2} M_{\text{total}} a.$$

$$\textcircled{1} a = \frac{2}{3} g \sin \alpha.$$

$$\textcircled{2} (mg + mg^{\text{cmo}}) - f = (M+m)a.$$

$$fR = \frac{a}{R} \cdot \frac{1}{2} M R^2 \Rightarrow f = \frac{1}{2} M a.$$

$$(m+M)g^{\text{cmo}} = (\frac{2}{3}M+m)a.$$

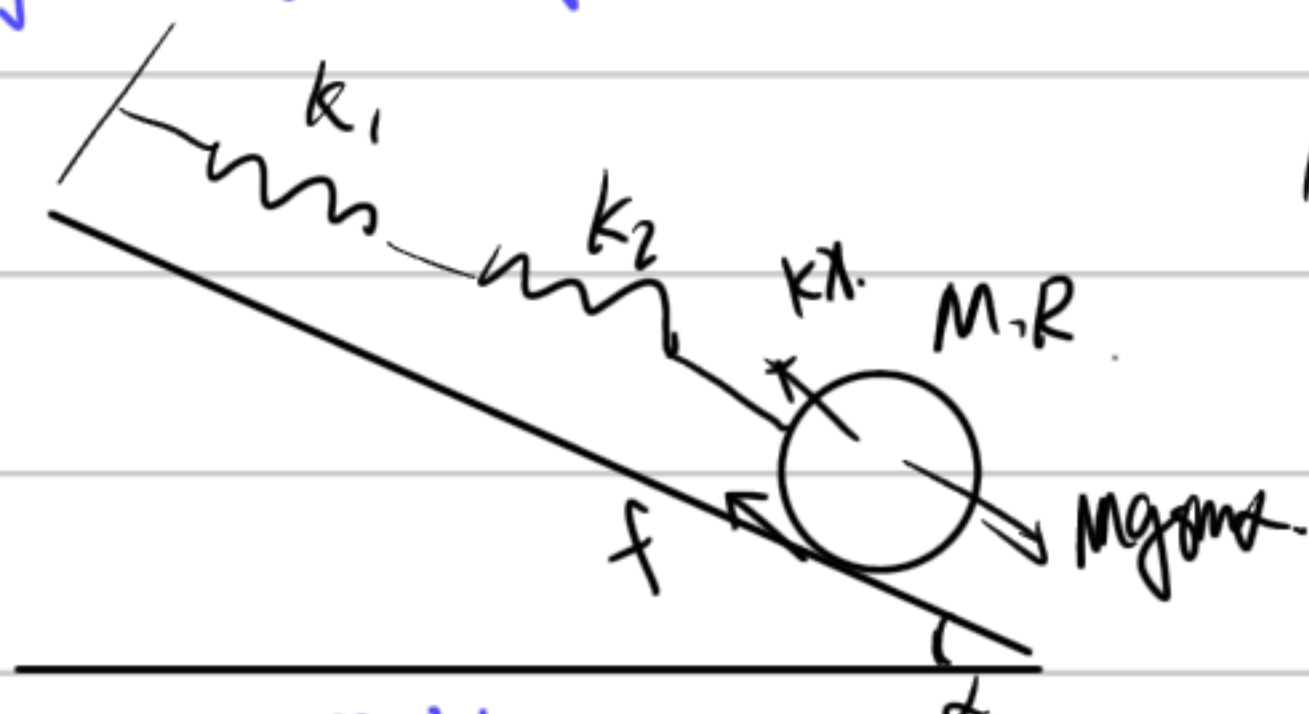
$$a = \frac{(m+M)g}{\frac{2}{3}M+m} \sin \theta$$

$$\textcircled{1} \frac{2}{3}$$

<

$$\textcircled{2} \frac{2M+2m}{3M+2m}$$

厘清滚筒还是实心球!



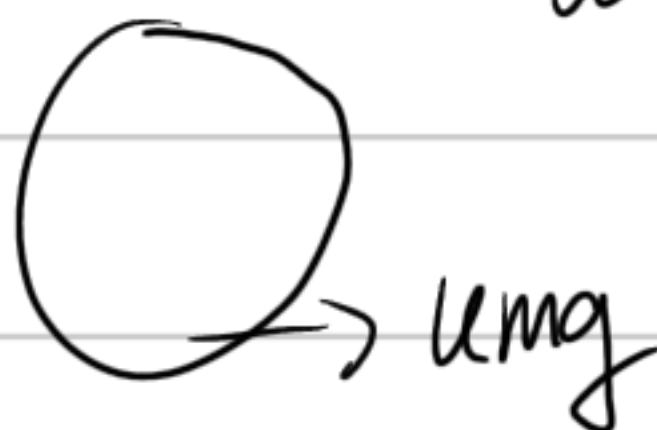
$$\begin{aligned} Mg \sin \alpha - f - kx &= Ma \\ f &= \frac{2}{5} M a \end{aligned} \quad \left| \begin{aligned} f &= ma \\ v_{\text{cm}}^{\uparrow} &= \end{aligned} \right.$$

在平面上的滚筒 f 近似为 0

a. ε 近似为 0

$$\begin{aligned} v: 0 &\rightarrow w'R \\ w: w_0 &\rightarrow w' \\ &\frac{1}{3} w_0 \end{aligned}$$

$$w'R = \mu g t.$$



$$\mu m g R = I \epsilon = \frac{1}{2} M R^2 \epsilon.$$

$$\epsilon = \frac{2\mu g}{R}$$

$$\epsilon = \frac{\mu g}{R}.$$

$$R w_0 - w'R = 2\mu g t = 2w'R$$

$$w' = \frac{1}{3} w_0$$

Chapter 13 – Rigid Body Mechanics (III)

Work and Conservation of Angular Momentum

Power (rate of work being done)

$$\delta W = \tau_z d\theta \quad \Rightarrow \quad \frac{\delta W}{dt} = \tau_z \frac{d\theta}{dt} = \tau_z \omega_z$$

$$P = \tau_z \omega_z$$

$$\vec{\tau} = \vec{r} \times \vec{F}$$

角动量.

$$\vec{L} = \vec{r} \times \vec{p}$$

SI unit: $\text{kg} \cdot \frac{\text{m}^2}{\text{s}}$

Rewrite $\vec{r} \times \vec{F} = \frac{d}{dt}(\vec{r} \times \vec{p})$

$$\vec{\tau} = \frac{d\vec{L}}{dt} \quad (\text{single particle})$$

Conclusion. The rate of change of the angular momentum of a particle equals the torque of the net force acting on it (both calculated about the same reference point!).

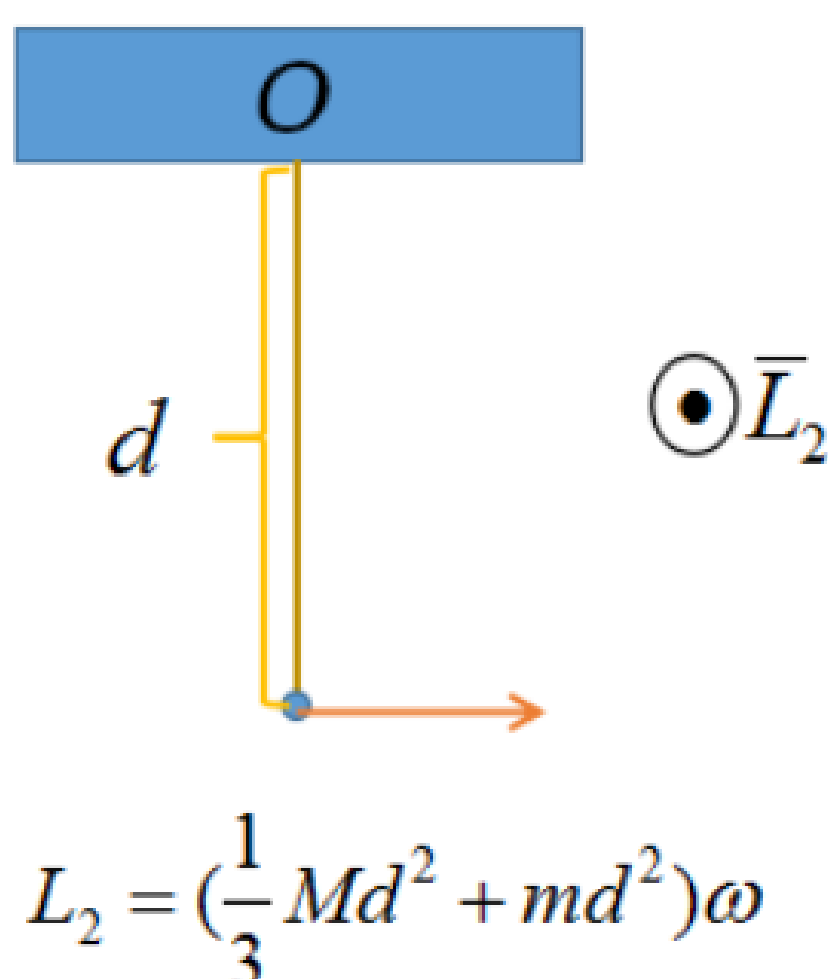
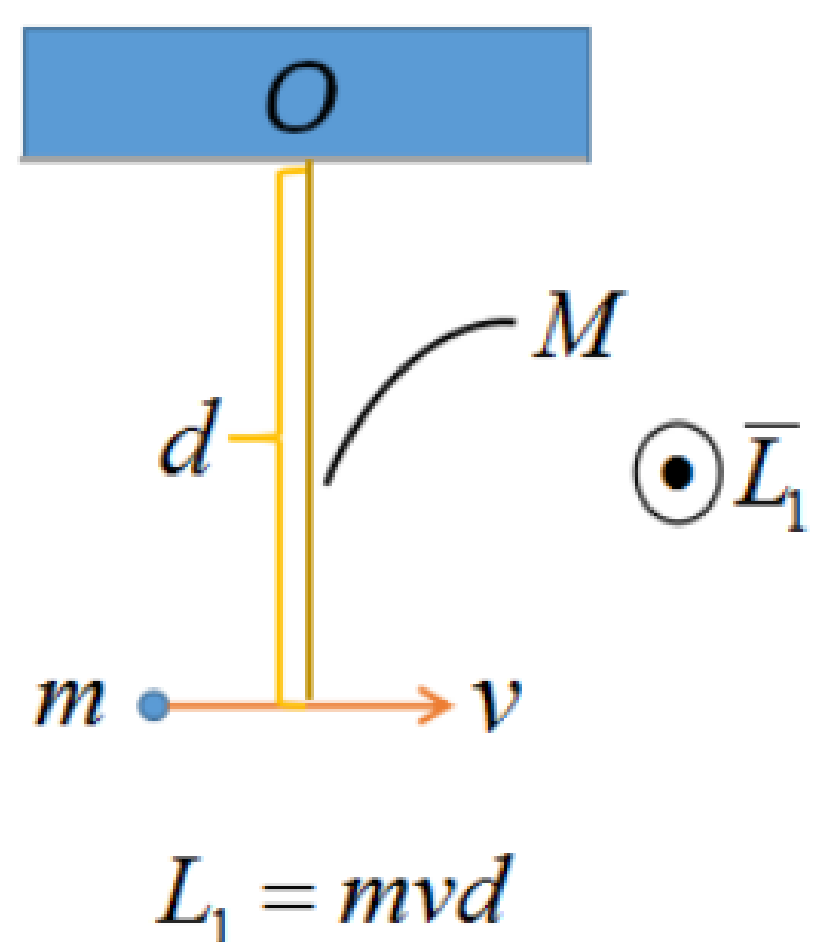
$$L = I_{\text{slice}} \omega.$$

Conservation of Angular Momentum.

1. $\frac{d\vec{L}}{dt} = \vec{\tau} \text{ ext.}$ 角动量守恒, 若 $\vec{\tau} = 0$

2. $\vec{L} = \vec{r} \times \vec{p} = I\omega$

3. Δ 哑铃收缩. 角动量守恒, 角速度 $\uparrow$
动能 $\uparrow$



$$I_{rod} = \frac{1}{3}MR^2$$

There are no external torques with respect to the suspension point O , hence the conservation of the total angular momentum implies

$$L_1 = L_2 \implies \omega = \frac{mvd}{\frac{1}{3}Md^2 + md^2}$$

Note. Here the rotation is not about a symmetry axis, but the rigid body (rod+bullet) is planar.

向上的
粗糙斜面滚筒，先滑动再滚动，停下来所需总时间一样
 $t_h = \frac{R\omega_0}{2g\sin\alpha}$

但是停止滑动所需时间不一样

The Gyroscopic Effect.

? not horizontal. ?

Calculate the work done by the friction force on the cylinder as it moves from where it was set down to where it begins to roll without slipping.

Express your answer in terms of the given quantities.

☐ $\sqrt{}$
☐ $\Delta\Sigma\Phi$

$W = \frac{\omega_0^2 R^2 M}{6}$

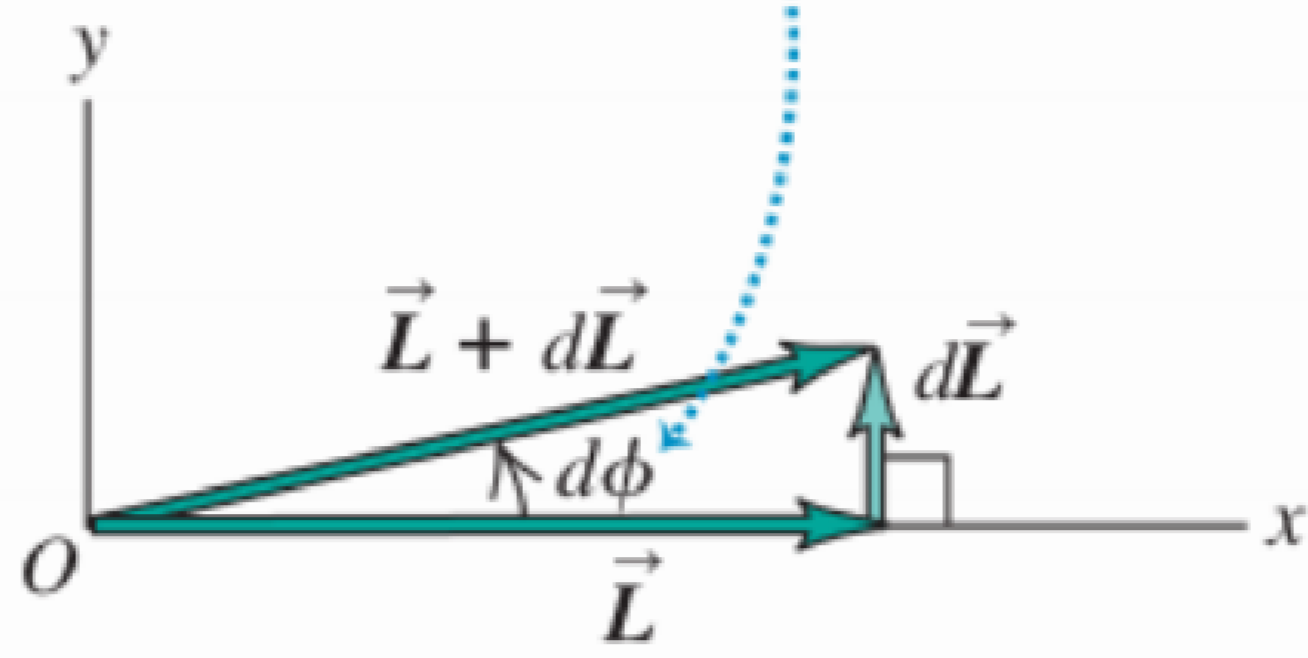
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斜面滚筒经典例题
① 动摩擦做负功
② 相对位移 = $V_{\text{轮}} - V_{\text{cm}}$

The Gyroscopic Effect.

In a time dt , the angular momentum vector and the flywheel axis (to which it is parallel) precess together through an angle $d\phi$.



$$d\mathbf{L} = \boldsymbol{\tau}^{\text{ext}} dt \perp \mathbf{L}$$

corresponding angular displacement

$$d\phi = \frac{|d\mathbf{L}|}{|\mathbf{L}|}$$

The rate at which the flywheel axis rotates about the z axis (*precession angular speed*)

$$\gamma = I\omega \approx$$

$$\Omega = \frac{d\phi}{dt} = \frac{\frac{|d\mathbf{L}|}{|\mathbf{L}|}}{dt} = \frac{\tau^{\text{ext}}}{L} = \frac{mgr}{I\omega}$$

Observation: the angular speed of precession is inversely proportional to the angular speed of rotation (spinning) of the flywheel
(in other words: a rapidly spinning gyroscope precesses slowly).

Note. We have ignored the contribution to the angular momentum due to precession (this component is along the z axis). Our simplified analysis is valid for a rapidly spinning flywheel.

Equilibrium and Elasticity

Chap 15

Conditions for Equilibrium $\left\{ \begin{array}{l} 1. F^{\text{ext}} = 0 \\ 2. \tau^{\text{ext}} = 0 \end{array} \right.$

Strain, Stress, and Elastic Modulus

Some definition:

$$\text{stress} = \frac{\text{force}}{\text{area}}$$

$$\text{strain} = \frac{\Delta l (\text{deformation})}{l_0}$$

$$\text{elastic modulus} = \frac{\text{stress}}{\text{strain}}$$

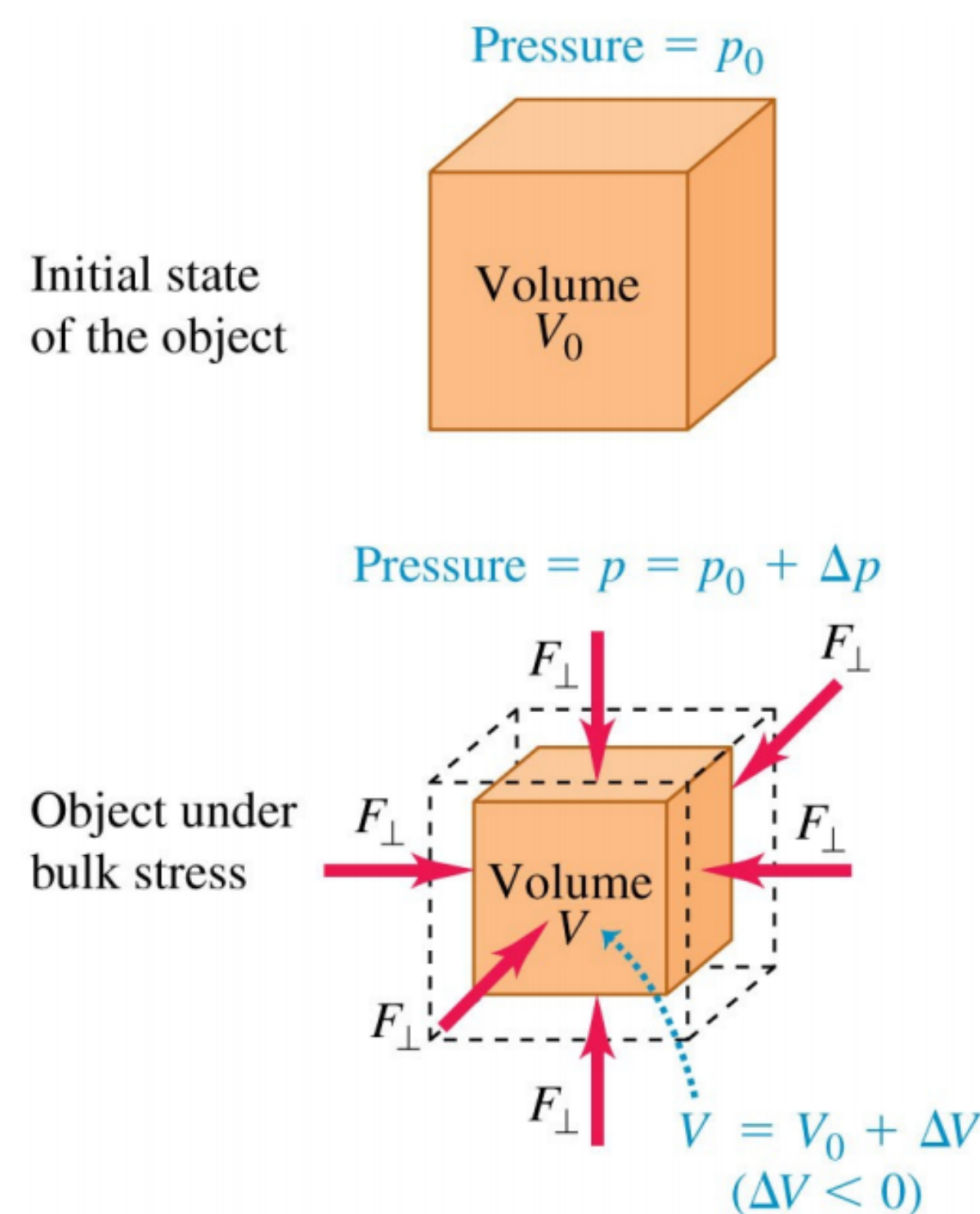
$$\text{tensile stress} = \frac{F_{\perp}}{A}$$

$$\text{tensile strain} = \frac{\Delta l}{l_0}$$

Young's modulus is tensile stress divided by tensile strain, and is given by

$$Y = \frac{\frac{F_{\perp}}{A}}{\frac{\Delta l}{l_0}}$$

tensile stress & strain



bulk stress and strain

pressure $P = \frac{F_{\perp}}{A}$

bulk stress : ΔP

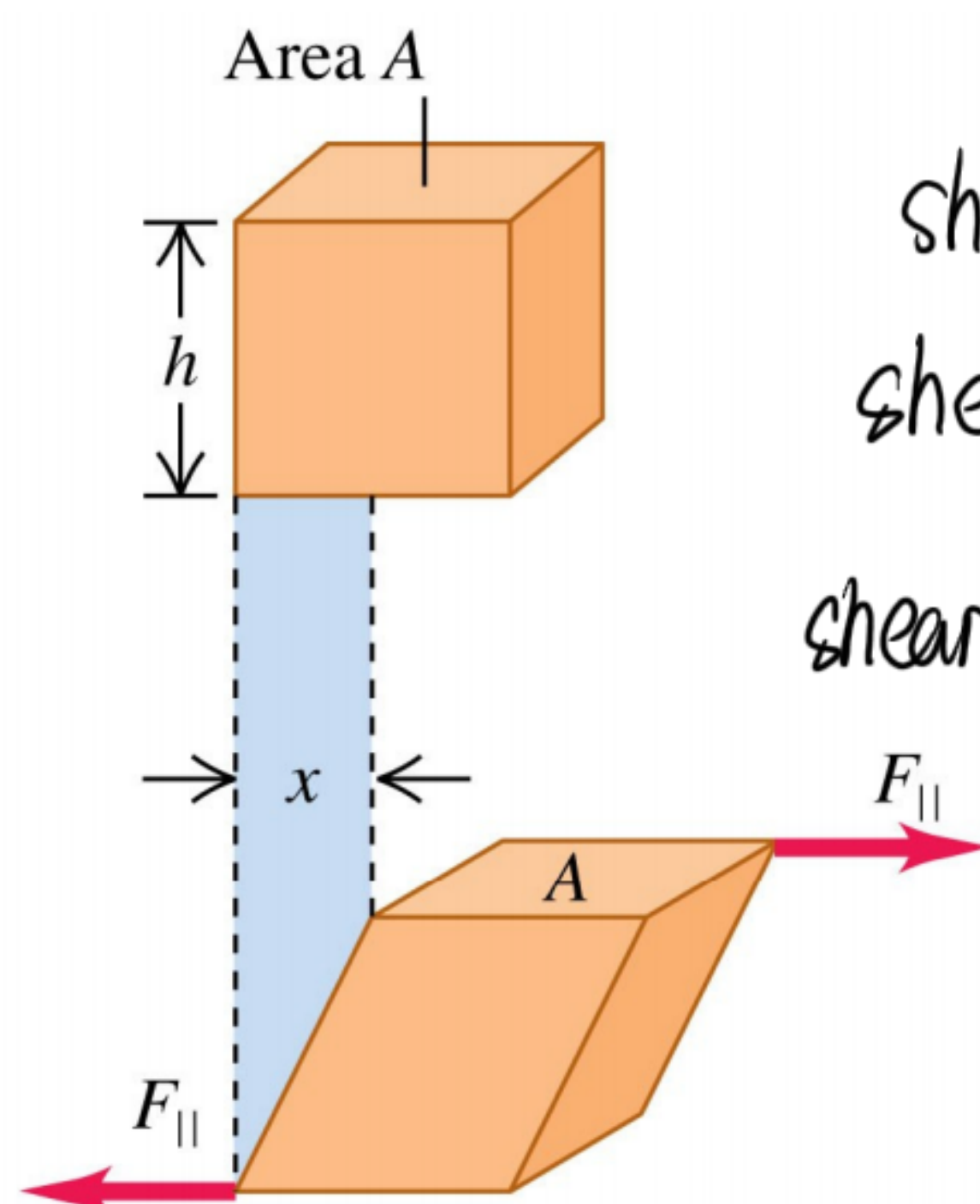
bulk strain : $\frac{\Delta V}{V_0}$

bulk modulus $B = - \frac{\Delta P}{\frac{\Delta V}{V_0}} \quad (\Delta V < 0)$

shear stress & strain

Initial state of the object

Object under shear stress



shear stress $\frac{F_{\parallel}}{A}$

shear strain $\frac{x}{h}$

shear modulus $S = \frac{\frac{F_{\parallel}}{A}}{\frac{x}{h}}$

Table 11.1 Approximate Elastic Moduli

Material	Young's Modulus, Y (Pa)	Bulk Modulus, B (Pa)	Shear Modulus, S (Pa)
Aluminum	7.0×10^{10}	7.5×10^{10}	2.5×10^{10}
Brass	9.0×10^{10}	6.0×10^{10}	3.5×10^{10}
Copper	11×10^{10}	14×10^{10}	4.4×10^{10}
Crown glass	6.0×10^{10}	5.0×10^{10}	2.5×10^{10}
Iron	21×10^{10}	16×10^{10}	7.7×10^{10}
Lead	1.6×10^{10}	4.1×10^{10}	0.6×10^{10}
Nickel	21×10^{10}	17×10^{10}	7.8×10^{10}
Steel	20×10^{10}	16×10^{10}	7.5×10^{10}

Elasticity and Plasticity

Elastic limit or yield point

Table 11.3 Approximate Breaking Stresses

Material	Breaking Stress (Pa or N/m ²)
Aluminum	2.2×10^8
Brass	4.7×10^8
Glass	10×10^8
Iron	3.0×10^8
Phosphor bronze	5.6×10^8
Steel	$5 - 20 \times 10^8$

Proportional limit

Elastic behavior

Plastic deformation

Plastic behavior

Permanent set

此时会突然拉得很紧

即使撤消F, 弹簧无法恢复原长

Fracture point

Strain

$$\frac{1920}{0.18} \div \frac{8.1 \times (10^{-2})^2}{18}$$

$$\frac{8.1 \times 1920}{0.18} = 10^7$$

$$d = \sqrt{\frac{3 \times 10^7}{6400}}$$

Elements of Fluid Dynamics

$$\rho = \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V}$$

Relative density ("specific gravity")

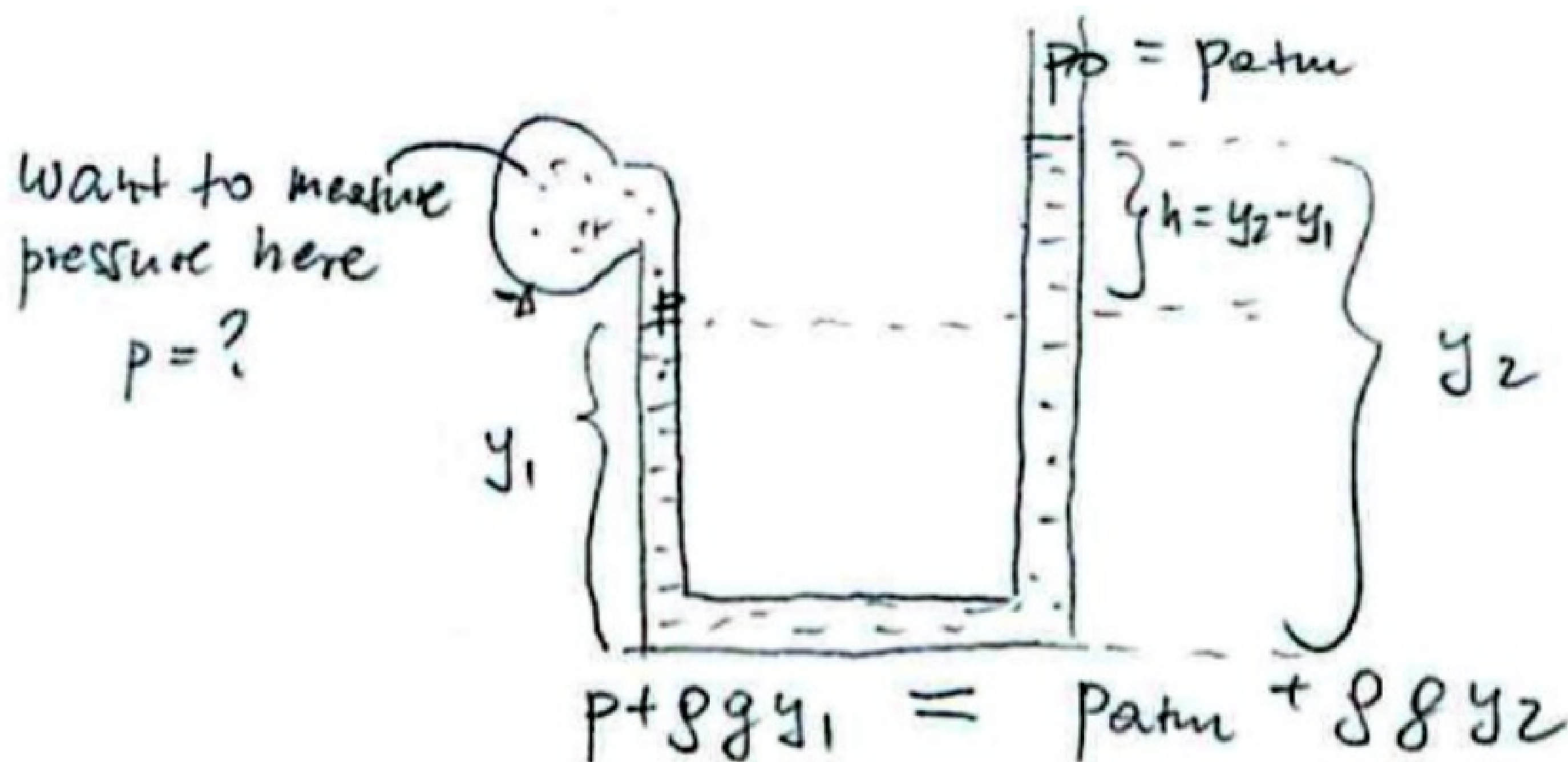
$$r = \frac{\rho}{\rho_{\text{H}_2\text{O}, 4^\circ\text{C}}} = \frac{\rho}{1000 \text{ kg/m}^3}$$

$$p \stackrel{\text{def}}{=} \frac{\Delta F \perp}{\Delta A} \quad [Pa = N/m^2] \quad \text{position - dependent}$$

Alternative units: 1 bar = 10^5 Pa

1 psi = 1 lb/in² = 6 895 Pa

normal atmospheric pressure $p_{\text{atm}} = 1.013 \times 10^5 \text{ Pa}$
 $= 14.70 \text{ lb/in}^2 = 1 \text{ atm}$

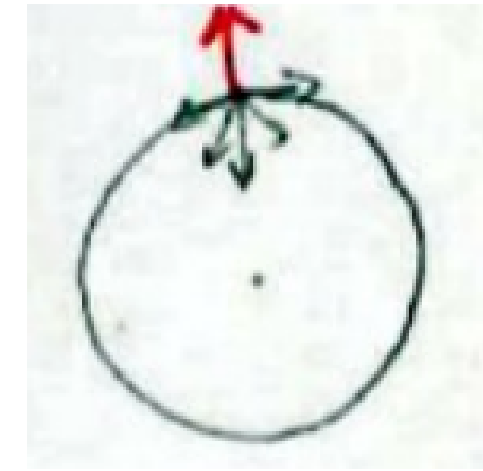


$$p_{\text{gauge}} = p - p_{\text{atm}} = \rho g (y_2 - y_1) = \rho g h$$

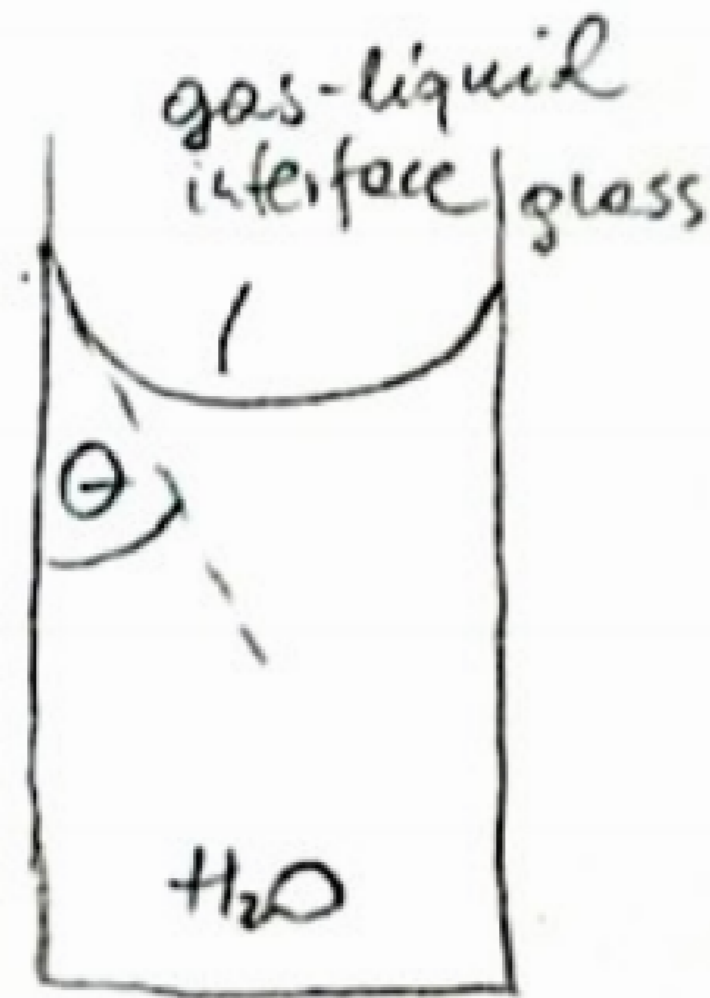
excess height of the liquid
in the right column

Surface Tension

In the weightless state, the liquid will form a ball – a solid that, with a given volume, has minimum surface area, hence minimizes the energy due to surface tension.

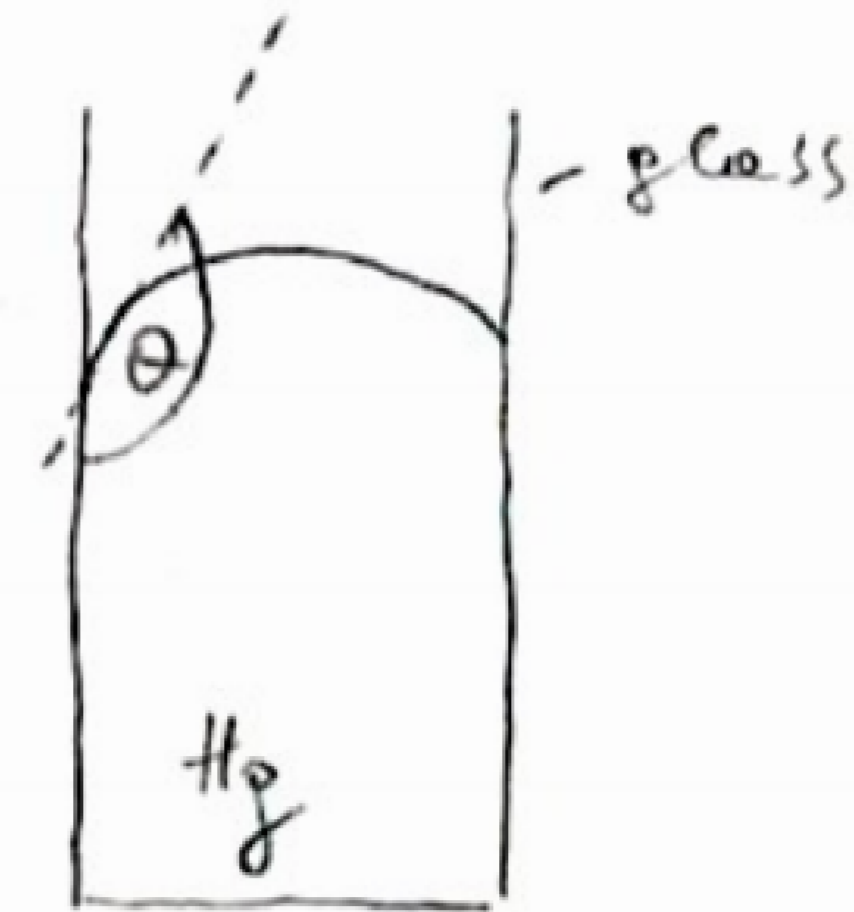


As a result of competition between cohesion forces (forces between the molecules of the liquid) and adhesion forces (forces between the liquid and the container), the surface of the liquid may take up various shapes.



wetting liquid $0 < \theta < \frac{\pi}{2}$

adhesion forces dominate



non-wetting liquid $\frac{\pi}{2} < \theta < \pi$

cohesion forces dominate

Capillarity (capilar action) –

ability of a liquid to flow in narrow tubes in opposition external forces (e.g. gravity).

Fluids in Motion

Ideal Fluid

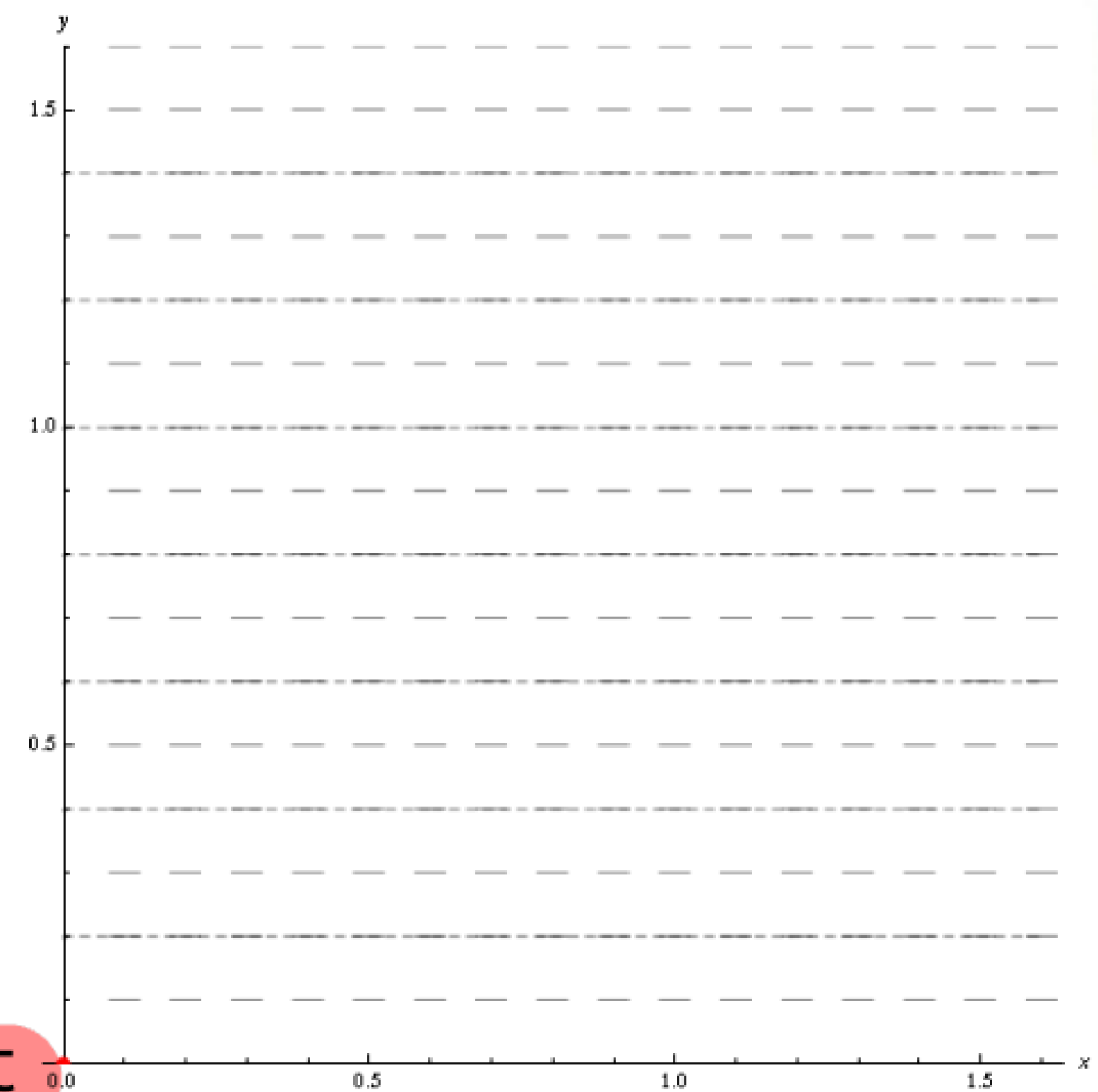
fluid density does not change (the fluid is incompressible);
experiences no internal friction (no viscosity)

Flow Lines (Path Lines)

trajectories of individual particles in a fluid

Stream Lines

family of curves such that at each point, the
tangent line to the curve coincides with
the direction of the fluid velocity at that point

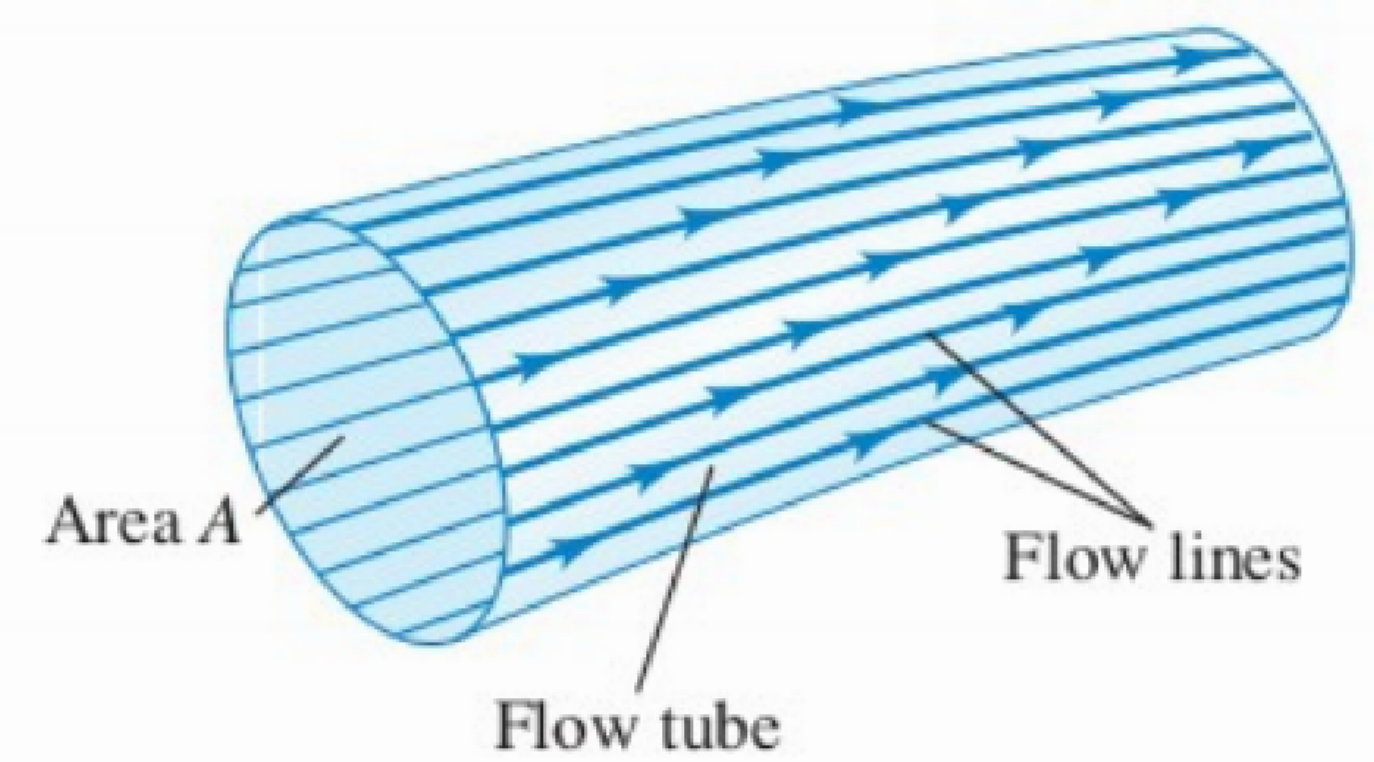


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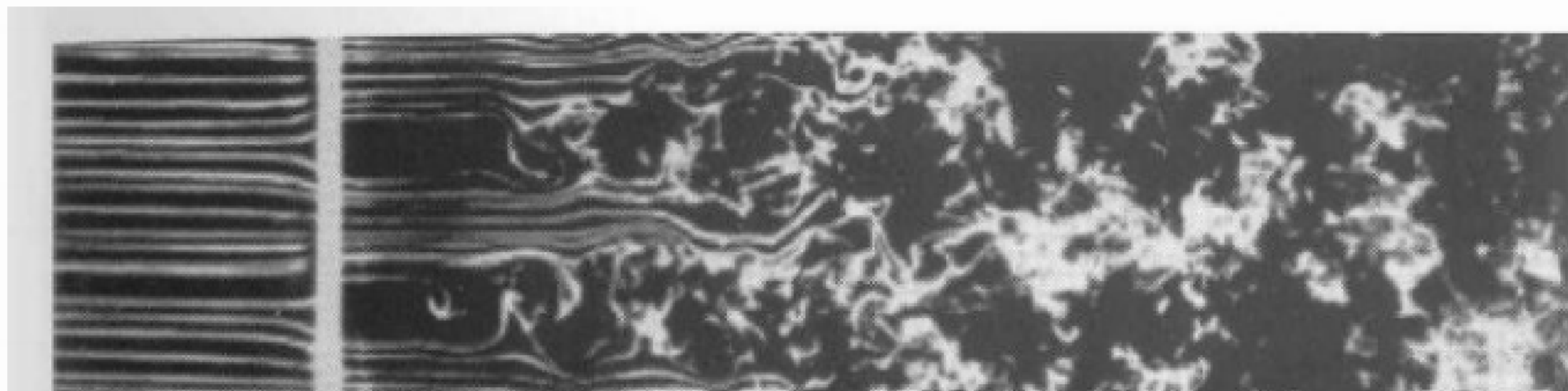
Note. When the flow pattern changes, flow lines do not coincide with stream lines.
We will discuss only steady-flow situation, when the two do coincide.

Flow Tube

a tube formed by flow lines passing through the edge of an imaginary
element of area. Note that in steady (laminar) flow no fluid can cross
the side walls of a flow tube; fluids in different flow tubes cannot mix.



Laminar vs. Turbulent Flow



Continuity equation: $A_1 v_1 = A_2 v_2$

$$vA = \text{const}$$

Bernoulli's Equation

$$\delta W = dE = d(K + U_{\text{grav}}) = dK + dU_{\text{grav}}$$

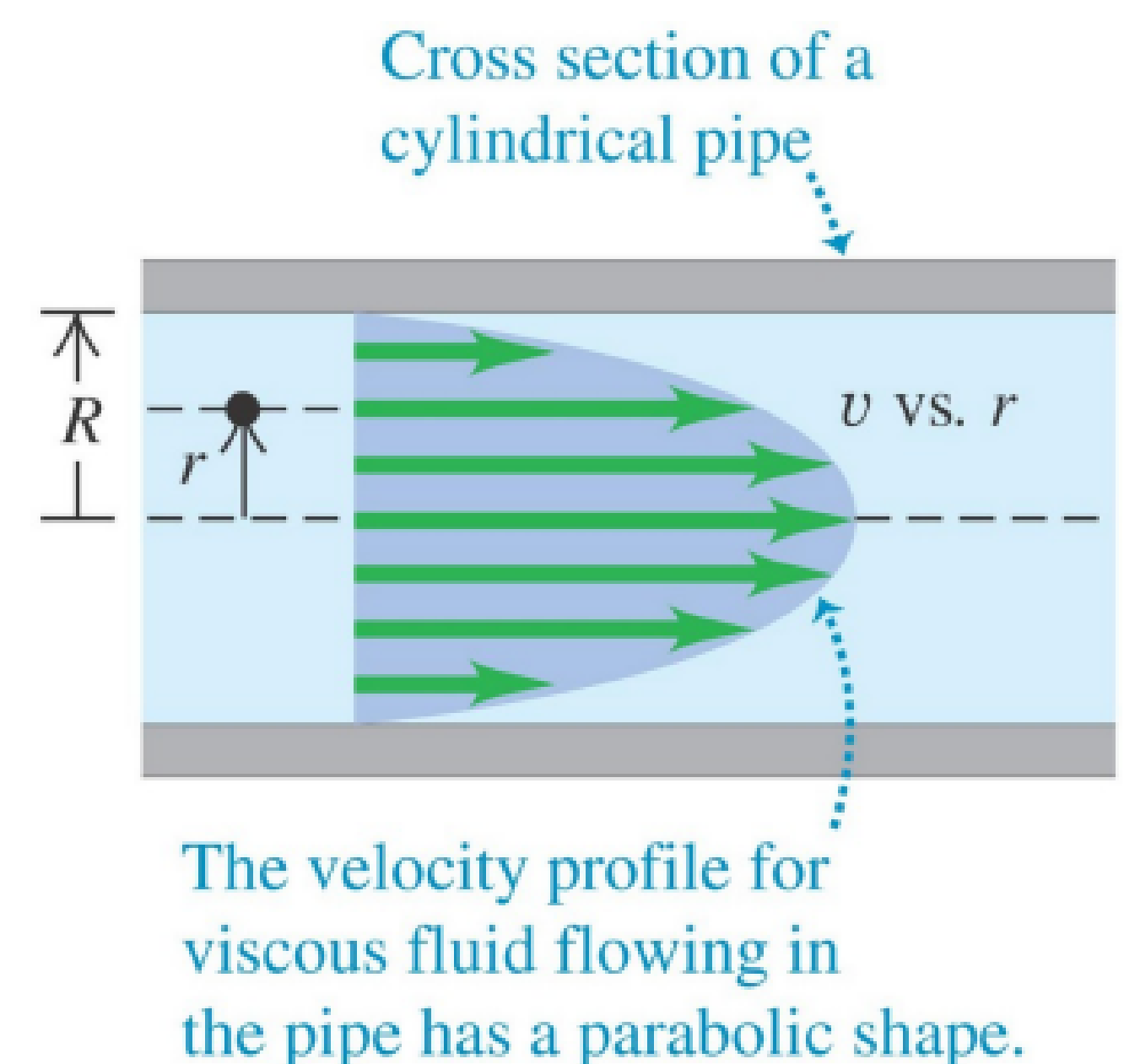
$$dK = \frac{1}{2} \rho (v_2^2 - v_1^2) dV$$

$$dU_{\text{grav}} = \rho g (y_2 - y_1) dV$$



$$p + \frac{1}{2} \rho v^2 + \rho g y = \text{const}$$

- *Viscosity* is internal friction in a fluid.
- *Turbulence* is irregular chaotic flow that is no longer laminar.



$$\frac{1}{1 + \frac{h}{R}} \quad \frac{1}{1 + x} \quad 1 + (-1)\frac{h}{R} + 1\left(\frac{h}{R}\right)^2 + (-1)\left(\frac{h}{R}\right)^3$$

Gravitation

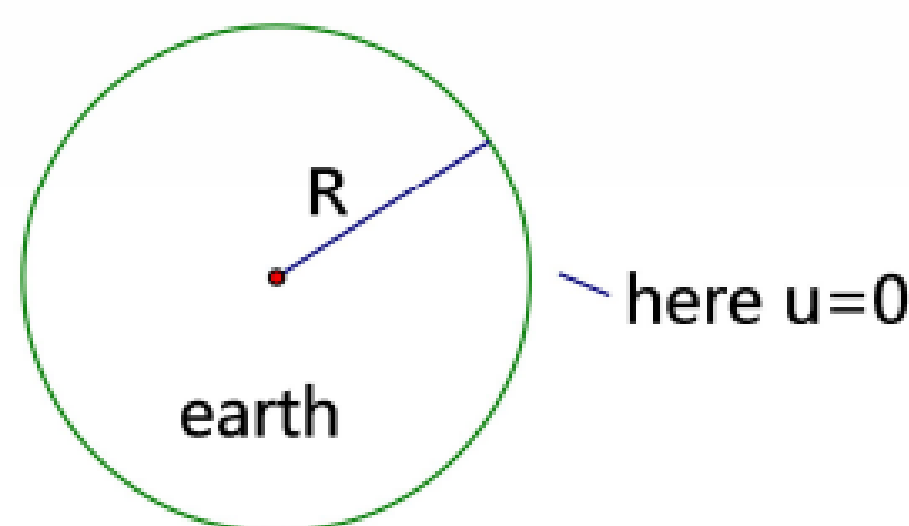
1. vector field $\vec{E}_G = \frac{\vec{F}_G}{m} = -G \frac{M}{r^2} \frac{\vec{r}}{r}$
2. $GM = gR^2$ close to Earth's surface
3. gravitational potential energy
 * central forces are conservative

4. $U(r) = -G \frac{Mm}{r} + C$ $\vec{F} = -\text{grad } U = -\nabla U$

① $U(\infty) = 0 \Rightarrow C = 0$ and $U(r) = -G \frac{Mm}{r}$

② $U(R) = 0 \Rightarrow -G \frac{Mm}{R} + C = 0 \Rightarrow C = G \frac{Mm}{R}$ and

$$U(r) = GMm \left(\frac{1}{R} - \frac{1}{r} \right)$$

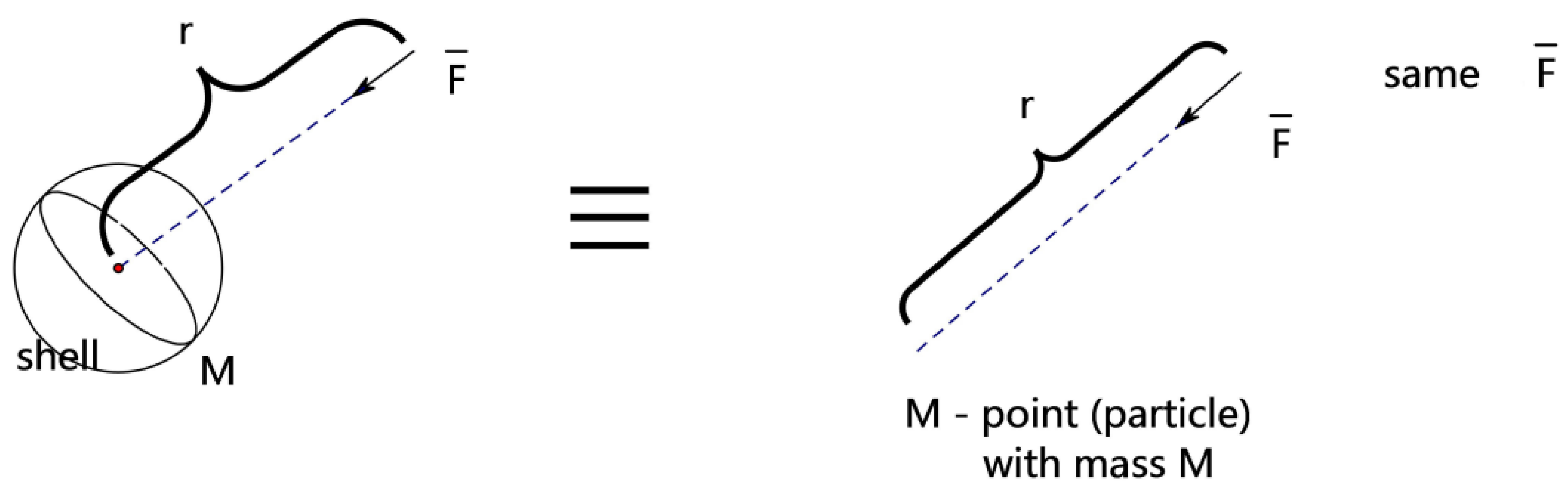


Let $r = R + h$ (with $h \ll R$), then

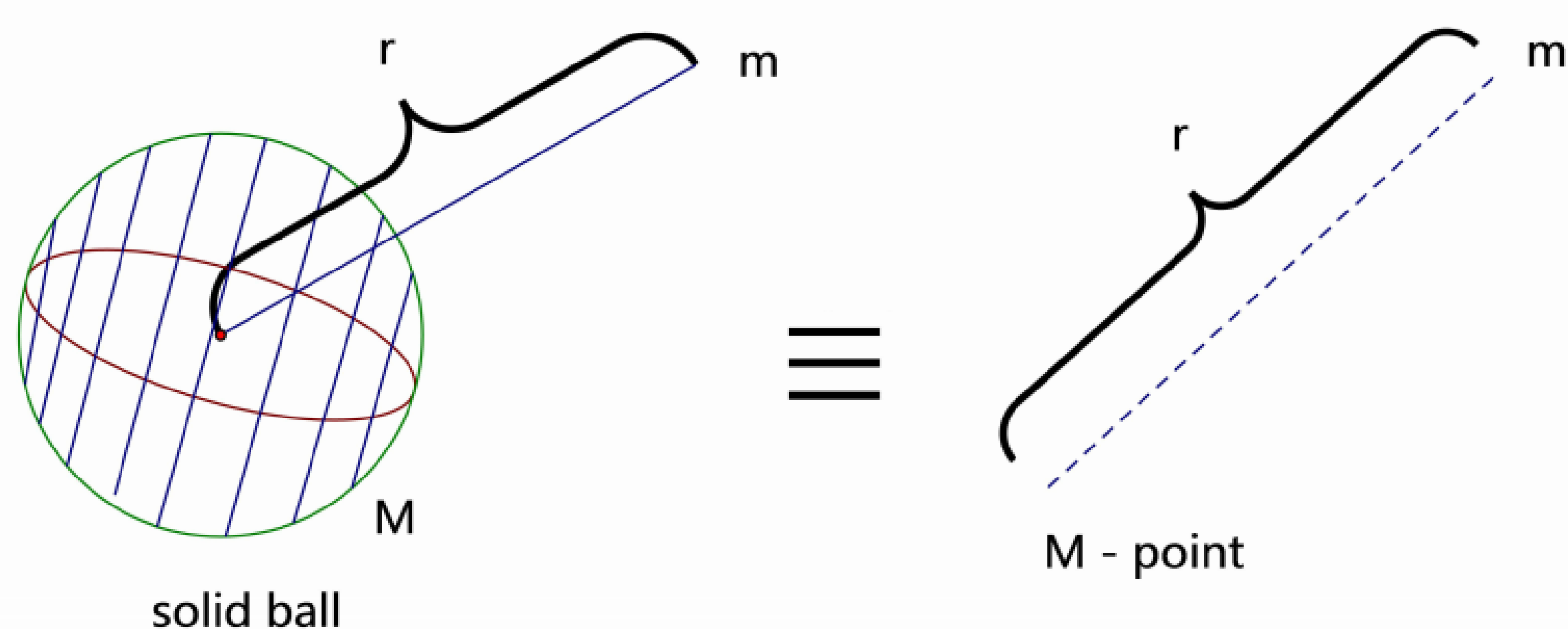
$$\begin{aligned} U(r) &= GMm \left(\frac{1}{R} - \frac{1}{R+h} \right) = G \frac{Mm}{R} \left(1 - \frac{1}{1 + \frac{h}{R}} \right) \\ &= G \frac{Mm}{R} \left(1 - 1 + \frac{h}{R} - \left(\frac{h}{R} \right)^2 + \dots \right) \quad [\text{for } h \ll R] \\ &\approx G \frac{Mm}{R} \left(1 - 1 + \frac{h}{R} \right) = G \frac{Mm}{R^2} h = mgh \end{aligned}$$

逃逸速率: $\frac{1}{2} m V_{esc}^2 - G \frac{M_{Earth} m}{R} = 0 \Rightarrow V_{esc} = \sqrt{\frac{2GM_{Earth}}{R}} \approx 11.2 \text{ km/s}$

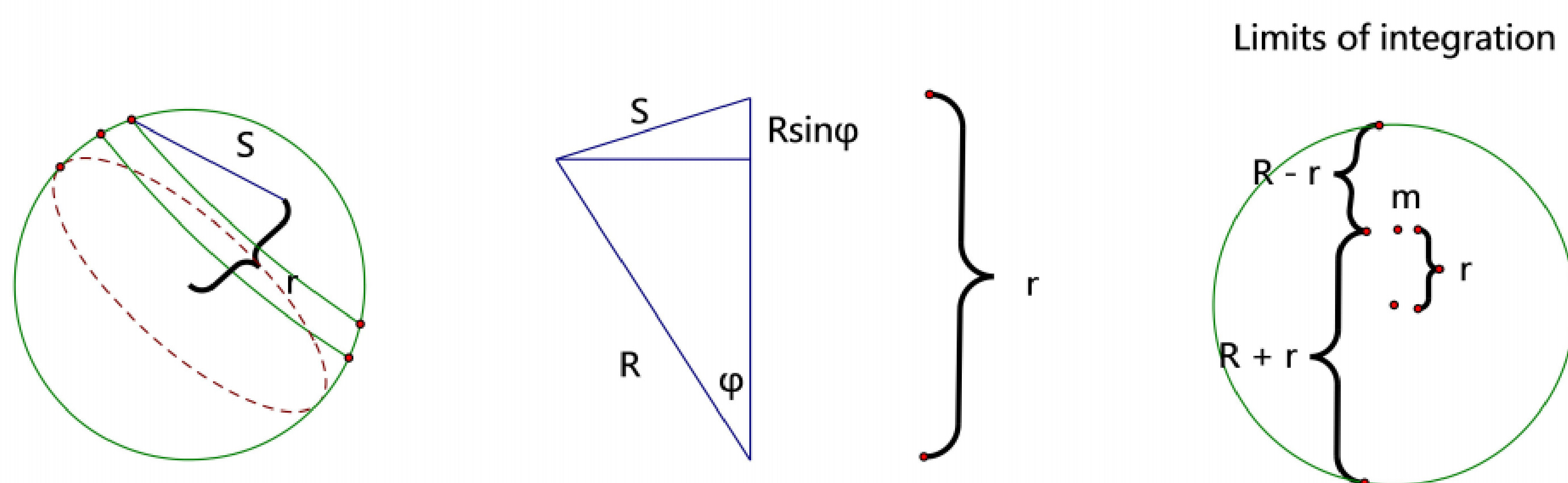
conclusion
(*)



(*) For a solid ball with a spherically symmetric distribution of mass, divide it into shells and use the above fact



(*) If m is inside the sphere

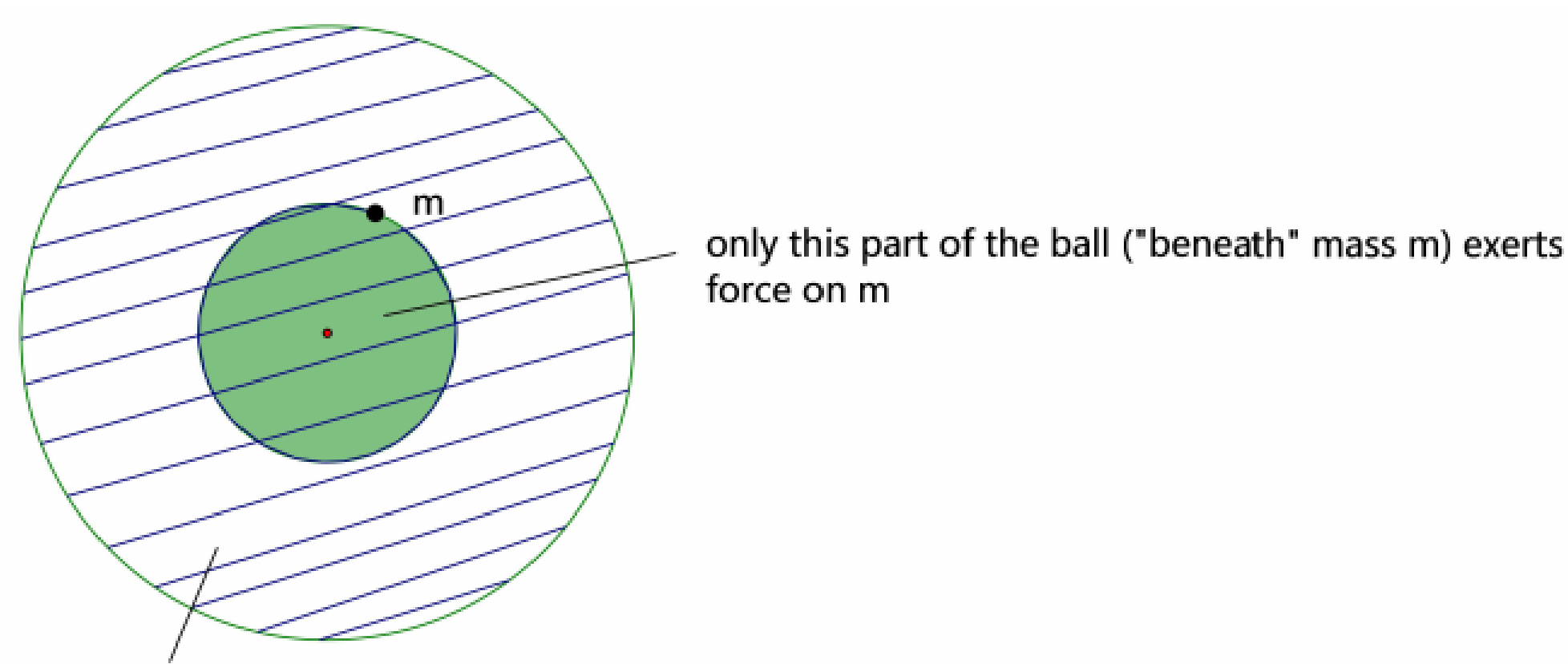


$$U(r) = -G \frac{mM}{2rR} \int_{R-r}^{R+r} ds = -G \frac{Mm}{R} = \text{const}$$

$$\Rightarrow \vec{F} = -\text{grad } U \equiv 0$$

Gravitational force on mass m inside the sphere is zero!

(*) Consequently



Satellites in Circular Orbit

$$v = \sqrt{\frac{GM}{r}} \quad T = \frac{2\pi r}{v} = 2\pi \frac{r^{\frac{3}{2}}}{\sqrt{GM}}$$

(a) geostationary satellite ($T_{\text{geost}} = 24h$)

$$r_{\text{geost}} \simeq 35786 \text{ km}$$

$$v_{\text{geost}} \simeq 3.1 \text{ km/s}$$

(b) ISS ($r = 6800 \text{ km}$)

$$v = 7.7 \text{ km/s}$$

$$T = 93 \text{ min}$$

(c) moon ($r \simeq 380000 \text{ km}$)

$$v = 1 \text{ km/s}$$

$$T \simeq 28 \text{ days}$$

Central Field : $\vec{F}(\vec{r}) = f(r)\vec{r}$

$$\vec{\tau} = \vec{r} \times \vec{F} = 0 = \frac{d\vec{L}}{dt}$$

同心力 $\Rightarrow$ 角动量守恒 $\left\{ \begin{array}{l} \text{大小} \\ \text{方向} \Rightarrow \text{粒子在同一平面运动} \end{array} \right.$

motion in a plane

Conclusion

For motion in the field of a central force, $\vec{L} = \text{const}$ (both magnitude and direction are constant).

Note. Constant direction of $\vec{L} \Rightarrow$ motion in a plane.

Aerial Velocity

$$\text{看扫过的面积} \quad \frac{dA}{dt} = \frac{1}{2} \left| \vec{r} \times \frac{d\vec{r}}{dt} \right| = \frac{1}{2} |\vec{r} \times \vec{v}|$$

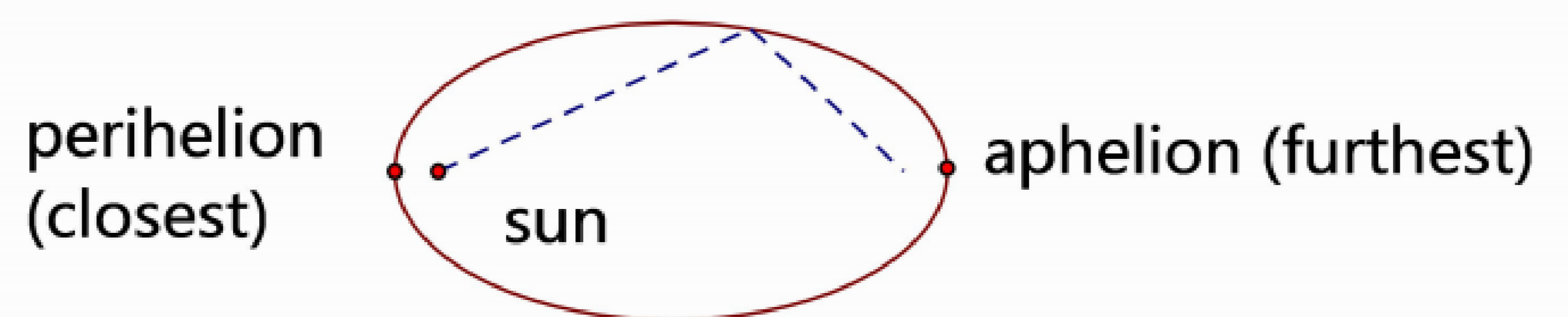
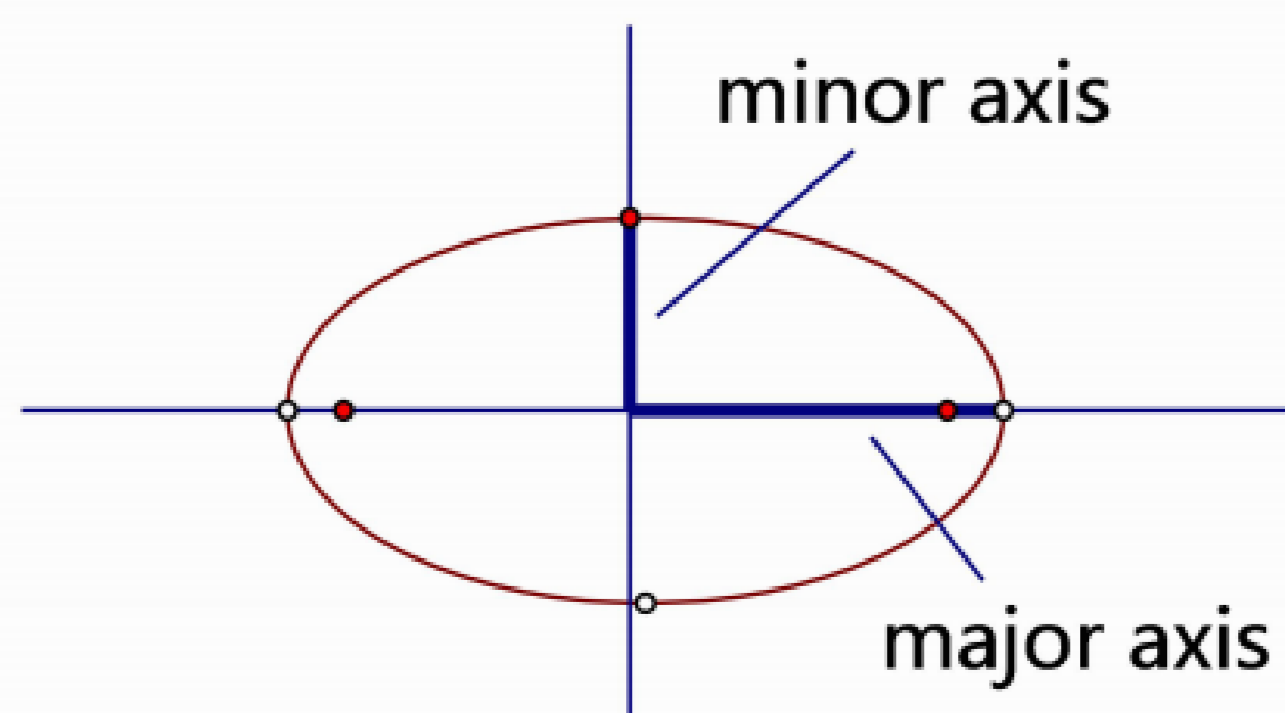
$$\vec{a} = \frac{1}{2}(\vec{r} \times \vec{v})$$

$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v} = \vec{a} \cdot 2m.$$

Kepler's Laws

- ① Each planet moves in an elliptical orbit with the Sun at one of the focal points.
- ② The line from the Sun to a given planet sweeps out equal areas in equal times.
- ③ The period of motion of a planet is proportional to the 3/2-power of the major axis length of its orbit

$$T = \frac{2\pi a^{3/2}}{\sqrt{GM_{\text{Sun}}}}.$$



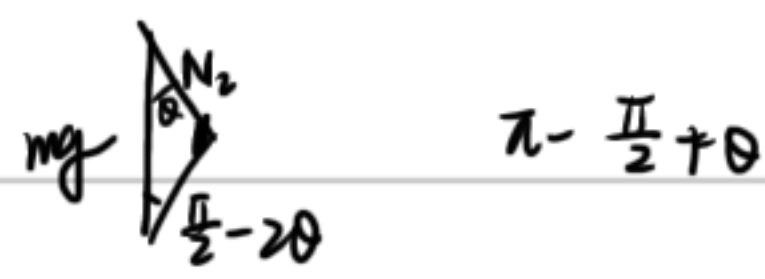
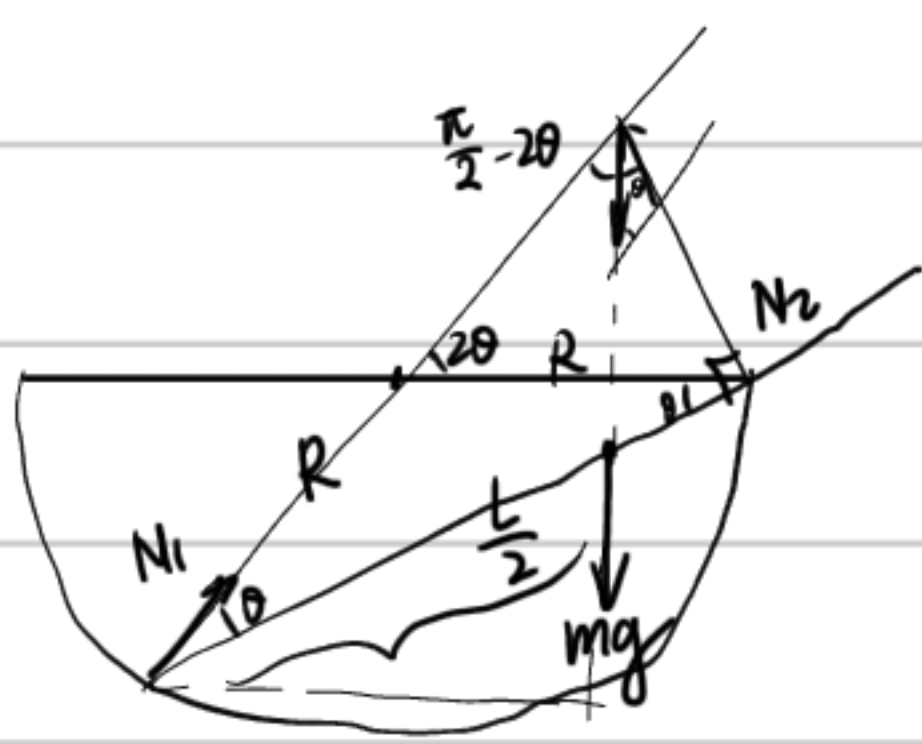
Note. Precisely, motion of both the Sun and the planet is about their center of mass (in practice, the center of mass is very close to the center of the Sun).

Mechanical Waves

1. $\xi(x, t) = \xi_0 \cos(kx - \omega t)$

★ $v_{ph} = \frac{\omega}{k}$

2. the classical wave equation in 1D*

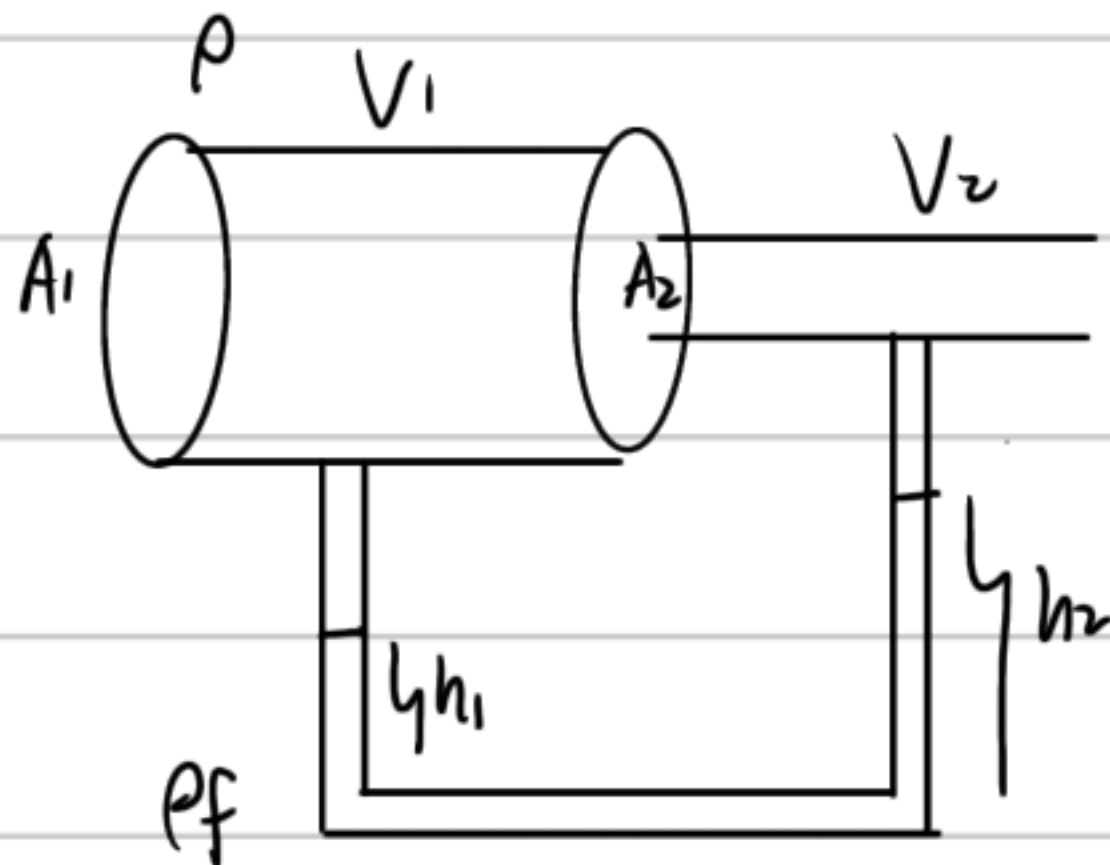


$$\frac{L}{2} \cos \theta \cdot mg = 2R \cos \theta \cdot N_2$$

$$\frac{N_2}{mg} = \frac{\sin(\frac{\pi}{2} - \theta)}{\sin(\frac{\pi}{2} + \theta)} = \frac{L}{4R}$$

$$\frac{2C^2 - 1}{\cos \theta} = \frac{L}{4R}$$

$$\Delta 2C - \frac{1}{C} = \frac{L}{4R}$$



$$P + \rho gh + \frac{1}{2} \rho V^2$$

$$V_2^2 = V_1^2 \frac{A_1^2}{A_2^2}$$

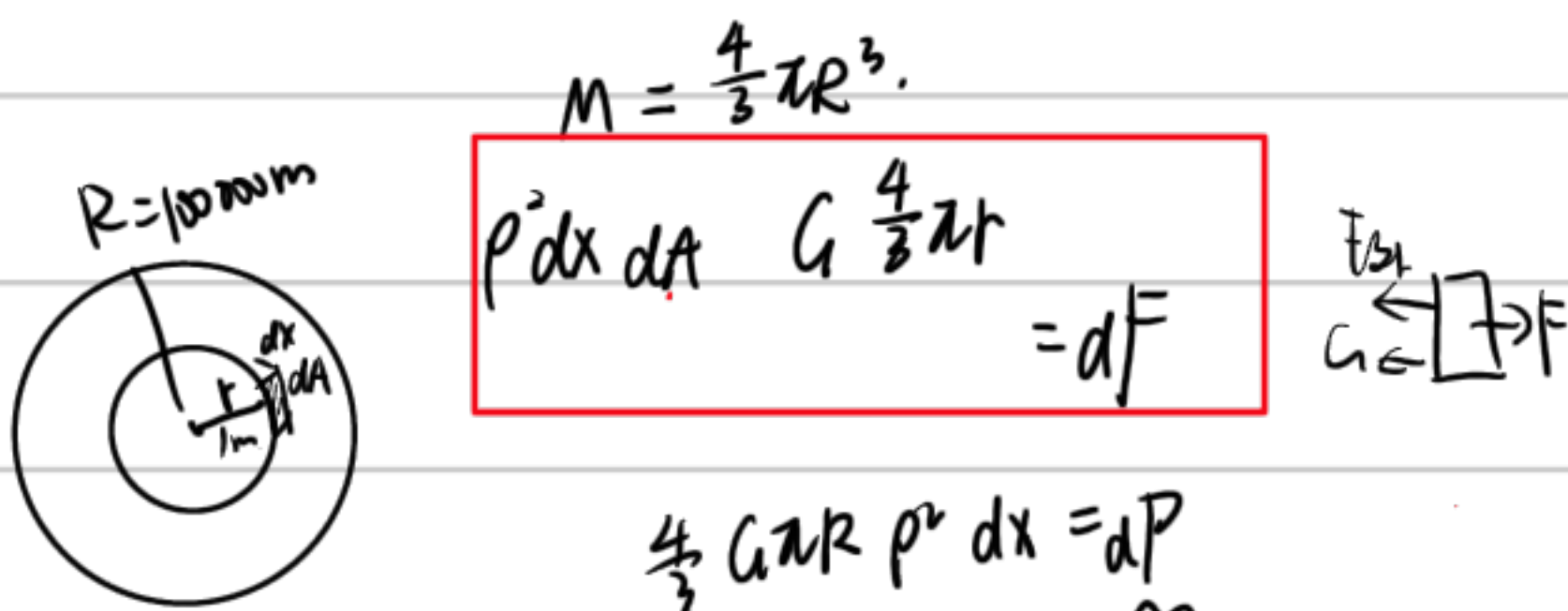
$$P_1 + \rho g(h - h_1) + \rho_f g h_1 = P_2 + \rho g(h - h_2) + \rho_f g h_2$$

$$\frac{V_1}{V_2} = \frac{A_2}{A_1}$$

$$P_1 + \frac{1}{2} \rho V_1^2 = P_2 + \frac{1}{2} \rho V_2^2$$

$$2\rho_f g h - 2\rho g h = V_2^2 - V_1^2 = V_1^2 \left(\frac{A_1^2}{A_2^2} - 1 \right)$$

$$\frac{1}{2} \rho V_2^2 - \frac{1}{2} \rho V_1^2 = -\rho g h_2 + \rho_f g h_2 + \rho g h_1 - \rho_f g h_1 = -\rho g h + \rho_f g h$$



$$M = \frac{4}{3} \pi R^3 \rho$$

$$\rho^2 dx dA \cdot G \frac{4}{3} \pi r^3 = dF$$

$$F_{\text{net}} = \int dF$$

$$\frac{4}{3} \pi G R \rho^2 dx = dP$$

$$F_{\text{gravitation}} = \frac{G \frac{4}{3} \pi R^3 \rho \cdot \rho dx dA}{r^2} = \frac{4}{3} \pi \rho^2 G R dx dA$$

$$P = \frac{F}{dA} = \frac{4}{3} \pi \rho^2 G R dx$$

这样只能算出来某一微小高度造成压强

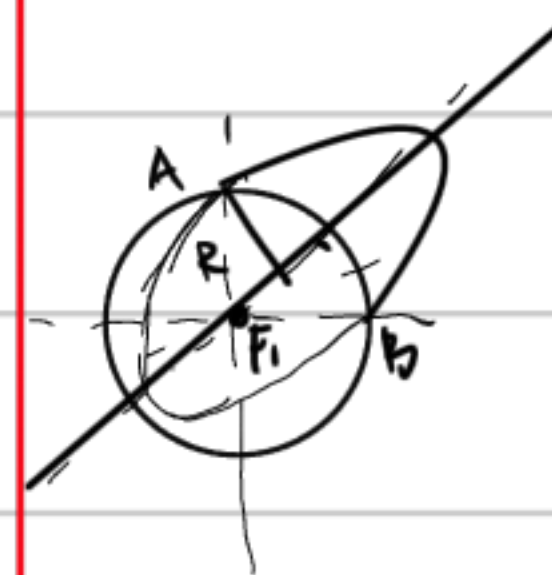
实际上



$$G + F_{\text{外}} = F$$

$$\int G = \int F - F_{\text{外}} = F$$

$$\frac{4}{3} \pi \rho^2 G R^2 \Big|_0^R = \frac{F}{dA} = P$$



$$L = mrv$$

$$\frac{1}{2} m v^2 - \frac{G M m}{r} = E$$

$$\Rightarrow r^2 + \frac{G M m}{E} r - \frac{L^2}{2 m E} = 0$$

$$r_1 + r_2 = 2a = -\frac{G M m}{E}$$

$$E = -G \frac{M m}{2a} \quad a \downarrow E \downarrow$$

$$r_1 r_2 = a^2 - e^2 = b^2 = -\frac{L^2}{2 m E}$$

$$2a = R + \frac{R}{\sqrt{2}}$$

$$a = \left(\frac{1}{2} + \frac{1}{\sqrt{2}} \right) R$$

$$V^2 = \frac{2 G M}{R} - \frac{2 E G M (\sqrt{2} - 1)}{R}$$

$$= \frac{(2\sqrt{2} - 2) G M}{R}$$